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# Multi-Person Human Mesh Recovery with the Anny Body Model

**Master's Thesis**

by

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## Eigenständigkeitserklärung

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Kaiserslautern, den 13.11.2026

Muhammad Saad Najib

## Use of AI tools

AI assistants (Claude, Anthropic, and Codex, OpenAI) were used under the author's direction for code development, debugging, analysis scripts and assistance with drafting and revising the report. Numerical provenance is recorded in the repository and in Appendix A.

# Abstract

This thesis addresses multi-person whole-body human mesh recovery from a single image in the Anny body model, whose named shape parameters include age, at reduced inference cost. No real images with Anny labels exist, and the large Vision Transformer encoders of accurate single-shot models are expensive.

For real-image supervision, 3DPW's SMPL labels are converted to Anny by fitting corresponding joints. A ViT-L teacher based on Multi-HMR is trained on synthetic AnnyOne images and converted 3DPW with corrected head-anchor targets and a fixed rotation correction of four output bones. A ViT-B student inherits the teacher's prediction heads through a learned projector and is distilled from the teacher's features and outputs. For video, temporal filtering and alignment to detected 2D landmarks run at inference time.

The student has 3.14 times fewer parameters and 3.19 times fewer FLOPs than the teacher and runs 2.10 times faster, but is less accurate: its per-vertex error (PVE) is 119.6 mm on held-out AnnyOne images and 110.1 mm on converted 3DPW, against 73.7 and 89.0 mm for the teacher. On 3DPW its PVE is lower than that of the released Multi-HMR-Anny checkpoint (the released baseline, 126.9 mm), which was not trained on the converted labels. Neither longer training nor another distillation temperature closes the gap. An ablation shows that the teacher lowers the student's 3DPW PVE by 6.6 mm through its transferred heads, with no measurable further effect of the distillation losses in the configuration tested. The teacher's detection recall on real photographs falls with crowd size (78.5 % with two, 45.3 % with nine or more). On a walking video, filtering and alignment visibly steady the output and improve its placement (qualitative, without ground truth).

The synthetic test set is a small holdout of 100 images without children, the real-image labels are converted rather than measured, and every model was trained once. Crowd recall and body shape on real photographs were analysed for the teacher only, which separates children from adults better than the initial model but not reliably.

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# 1. Introduction

## 1.1. Motivation

Human mesh recovery estimates the pose and shape parameters of a parametric body model for each person in an image; the body model turns them into a triangle mesh of the body surface. This thesis addresses the multi-person, whole-body variant: pose, shape and distance from the camera for every person, including people who overlap or are partly occluded.

Multi-HMR [Bar+24] solves this task in a single forward pass. A Vision Transformer (ViT) encodes the image, a detection head selects one keypoint per person, and a Human Perception Head regresses each body and its distance. The released Multi-HMR-Anny checkpoint (the released baseline) predicts bodies in the Anny body model [Bré+26; NAV26b]; multi-person Anny prediction is therefore not new, and this checkpoint serves as the external baseline.

SMPL [Lop+15] represents shape by statistical coefficients learned from scans of adults, usually the first 10, which carry no direct semantic meaning. Anny instead uses 11 named phenotype parameters, such as age and height, and 163 bones for the whole body (Section 2.1, Table 2.1). Anny can therefore represent a child directly, which does not imply that a network recognises a child in a photograph.

Two difficulties remain. The first is supervision: AnnyOne [Bré+26] provides synthetic images with exact Anny labels but an appearance unlike real photographs, while 3DPW [Mar+18] provides real video frames labelled in SMPL; no real images with Anny labels exist. The second is cost: large image encoders have many parameters and long inference times. Knowledge distillation, in which a large trained teacher guides the training of a smaller student, reduces the cost and keeps single-pass inference.

## 1.2. Problem statement and research questions

The input is an RGB image with its camera intrinsics (focal length and principal point), which define the viewing ray of each pixel. For each detected person the model predicts an Anny body: bone rotations, phenotype values and the camera-space position of one head joint, the anchor. The deployed model is a ViT-B student; its ViT-L teacher, trained in this thesis, is the accuracy reference, and the released baseline the external reference. Four research questions structure the thesis:

1. **Real-image supervision.** How can the SMPL labels of 3DPW be converted into Anny labels, and which aspects of pose and shape do they constrain?

2. **Compact reconstruction.** How much smaller, cheaper and faster is a ViT-B student distilled from a ViT-L teacher, how much accuracy does it lose against the teacher and the released baseline, and what does distillation contribute?

3. **Multi-person detection.** How does detection recall depend on the number of people in an image?

4. **Video.** Can the image-based student process a single-person video, and do inference-time temporal filtering and landmark alignment make its output steadier and better placed?

The student processes one frame at a time; no multi-frame network is trained. Reconstruction is evaluated on a held-out part of AnnyOne and on converted 3DPW, age and gender for the teacher on Relative Human photographs with age-group and gender tags. No new shape labels for real images are collected, and age in years is not estimated.

## 1.3. Contributions

I implemented the Anny output head and its target construction on top of the Multi-HMR code, the SMPL-to-Anny converter, the training and evaluation pipeline, the distillation set-up and the video filter and alignment; the pretrained backbone weights come from the released checkpoints.

1. **Anny labels for real images.** A joint-based fitting pipeline converts the standard 3DPW splits into Anny labels by fitting corresponding joints and virtual torso joints; the labels were checked with mesh overlays and a joint comparison (Chapter 3).

2. **A compact reconstruction system.** A ViT-L teacher is trained on AnnyOne and converted 3DPW with two corrections, head-anchor targets consistent with the decoder and a fixed rotation correction of four output bones, both identified by diagnostic experiments (Appendix F). A ViT-B student receives a copy of the teacher's prediction heads, which read its features through a learned projector, and imitates the teacher's features and outputs.

3. **An evaluation under one protocol.** All models are evaluated with the same labels and settings, with pose, placement and shape assessed separately, ablations of the teacher corrections, an ablation that separates the distillation losses from the transferred teacher heads (Section 5.3), sensitivity analyses of training length and temperature on validation data, and detection recall by the number of people per image.

4. **A video pipeline.** The pose, shape and position of each tracked person are filtered over time, and position and depth are aligned to detected 2D landmarks, without retraining (Section 4.9.2). The student and the released baseline are compared on the same frames.

## 1.4. Thesis structure

Chapter 2 introduces the body models and related work. Chapter 3 describes the datasets and the label conversion, on which the method of Chapter 4 depends. Chapter 5 reports the experiments. Chapter 6 answers the research questions and states the limitations; Chapter 7 concludes. The appendices give reproduction details, the joint correspondence, settings, additional figures, earlier models, the diagnostic experiments, the conversion history and additional teacher results.

# 2. Background and related work

The network of this thesis predicts bodies in the Anny body model, whereas its real images, 3DPW, are labelled in SMPL; both models are therefore described first.

## 2.1. The body models SMPL and Anny

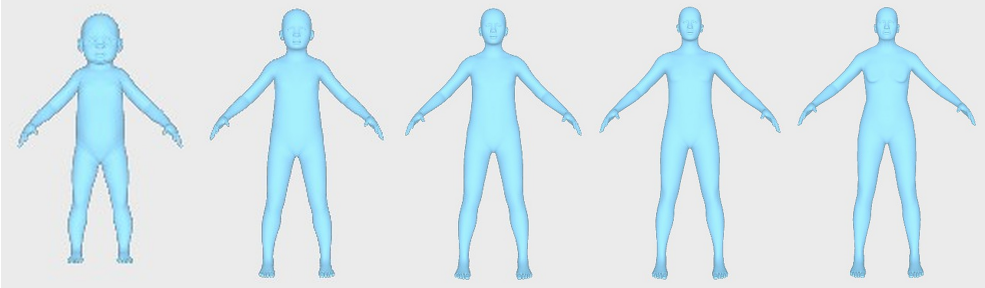
A parametric body model is a differentiable function $M(\boldsymbol{\theta}, \boldsymbol{\beta}) \rightarrow \mathbf{V} \in \mathbb{R}^{N\times 3}$ that maps pose parameters $\boldsymbol{\theta}$ and shape parameters $\boldsymbol{\beta}$ to the $N$ vertices of a triangle mesh, so that a network can be trained through it. The shape deforms a rest-pose template; the pose articulates a skeleton, and linear blend skinning (LBS) transfers its motion to the surface:

$$\mathbf{v}'_i \;=\; \sum_k w_{ik}\, \mathbf{G}_k(\boldsymbol{\theta}, \boldsymbol{\beta})\, \mathbf{v}_i(\boldsymbol{\beta}), \tag{2.1}$$

where $\mathbf{v}_i(\boldsymbol{\beta})$ is the shaped rest vertex, $w_{ik}$ the skinning weight of bone $k$ and $\mathbf{G}_k$ the world transform of bone $k$, obtained by composing the rotations along the kinematic chain from the root to $k$. A rotation error near the root therefore propagates to every bone below it.

**The SMPL family.** SMPL [Lop+15], the body model of most prior work and of the 3DPW labels, has 6,890 vertices and 24 joints. Its shape space consists of principal components of registered adult scans; most methods use the first 10 coefficients (betas), statistical directions without individual semantics. Child proportions are represented poorly. SMIL [Hes+18] is a separate infant model; AGORA [Pat+21] and BEV [Sun+22] blend the adult template towards an infant one with one extra coefficient. SMPL-H [RTB17] adds MANO hands with 15 joints each, and SMPL-X [Pav+19] also a face, for 55 joints on 10,475 vertices. SMPL has a research licence; assets, datasets and trained weights carry separate terms. Multi-HMR [Bar+24] was published for SMPL-X; a later checkpoint released by NAVER replaced its output layer with Anny (Section 2.3), and the teacher is initialised from its backbone weights with its own heads (Appendix A).

**Anny.** Anny [Bré+26], developed at NAVER LABS Europe, is derived from the open, artist-made MakeHuman model [Mak] rather than from scans; code and assets are licensed under Apache 2.0 and CC0 [NAV26a]. This thesis uses the full rig of 163 bones, for which the AnnyOne labels are written. The package loads a pruned rig of 104 bones by default, which matches neither the checkpoint nor the labels; package version, rig and mesh must be those used to



age 0 age 0.2 age 0.4 age 0.6 age 1

**Figure 2.1.:** *Anny at the mean AnnyOne shape in rest pose with only the age phenotype varied, each body scaled to the same displayed height so that its proportions can be compared. The skeletal height (head joint to mean ankle) grows from 0.53 m at age 0 to 1.54 m at age 1.*

create them. Pose is one rigid transform per bone, applied by Equation (2.1). 88 helper bones (toes, breasts, small face bones), whose rotation an image does not show, are not supervised, leaving 75 supervised bones (Section 4.2.2). The neck has the stacked bones `neck01` to `neck03`, followed by `head`.

Shape is set by 11 named *phenotype* values in $[0, 1]$, among them `gender`, `age`, `weight` and the blend weights `african`, `asian` and `caucasian` (Table 2.1). Each drives artist-made blend shapes (rest-vertex offsets), so predictions are directly interpretable (older, heavier). Gender 0 is the male end and 1 the female end, with rest statures of 1.753 m and 1.632 m. Age interpolates five anchor bodies spaced evenly from $-\frac{1}{3}$ to 1: age 0 is the baby, 0.333 the child, 0.667 the young adult and 1 the old adult. The newborn anchor lies below 0 and is unreachable for a network output bounded to $(0, 1)$ by a sigmoid.

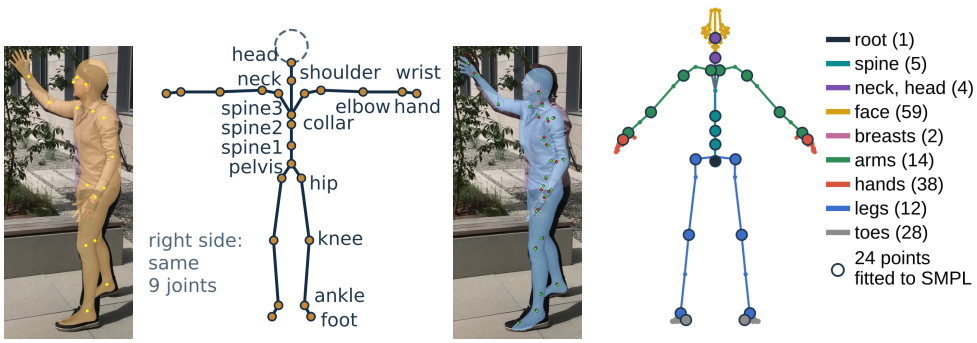
At the mean AnnyOne shape, increasing age from 0 to 1 raises the skeletal height from 0.53 m to 1.54 m (1.37 m at 0.5; Figure 2.1), so most growth occurs below about 0.5. A low height value alone does not yield child proportions: at age 0.6 the smallest reachable body is 1.22 m tall with adult proportions. Child proportions are marked by narrow shoulders relative to the head; the shoulder-to-head width ratio is 2.24 at age 0.1 and 3.00 at 0.6, and only a low age value produces the former. Ages are not converted to years, because three scales in the code disagree; a person is called *child-shaped* if the age value is below about 0.45.

**Comparison.** Table 2.1 and Figure 2.2 compare the two models. They differ most in the skeleton (24 joints against 163 bones) and in the shape parameterisation (10 unnamed principal-component coefficients against 11 named phenotypes). SMPL represents pose as one axis-angle vector per joint (direction: rotation axis; norm: angle); the network of this thesis predicts each Anny rotation in the continuous 6D form (6 numbers), mapped to a valid rotation matrix.

| | SMPL | Anny |
|---|---|---|
| Built from | registered scans of adults | the open MakeHuman project (artist-made) |
| Mesh | 6,890 vertices | one artist-made mesh; vertex count not stated[a] |
| Skeleton | 24 joints (23 + root): 6 on the centre line, 9 on each side | 163 bones: face 59, hands 38, toes 28, arms 14, legs 12, spine 5, neck and head 4, breasts 2, root 1 |
| Shape | 10 betas: principal components of the scans, without individual semantics | 11 named phenotypes in $[0, 1]$: gender, age, muscle, weight, height, proportions, cup size, firmness, African, Asian, Caucasian |
| Pose | one axis-angle rotation per joint, $24 \times 3 = 72$ numbers | one rigid transform per bone |
| Children | needs a separate infant model or a blend towards one | covered by the age phenotype (age 0: infant) |
| Licence | research licence terms | code Apache 2.0, assets CC0 |
| In this thesis | the 3DPW labels; refitted with Anny on 24 matched points (root and 18 bones move, 144 stay at rest) | predicted by the network: 163 rotations in 6D form ($163 \times 6 = 978$ numbers; 88 helper bones unsupervised, 75 supervised), 11 phenotypes and a 3D position (distance and image offset, 3 numbers) |

[a] The vertex count of the Anny mesh depends on the release and on whether unattached vertices are kept; it is not stated here.

**Table 2.1.:** *The two body models of this thesis: SMPL, the body model of the 3DPW labels and of most prior work, and Anny, the body model the network predicts. Bone counts per region refer to the full rest rig used by the AnnyOne labels and by the network (Figure 2.2).*




**(a)** *SMPL* **(b)** *SMPL, schematic* **(c)** *Anny* **(d)** *Anny rest rig*

**Figure 2.2.:** *(a) 3DPW SMPL label of one person (courtyard_basketball_01, image 570; yellow: OpenPose 2D points) and (c) the converted Anny label of the same person and frame (red: 3DPW SMPL joints; green: Anny joints); both are labels, not predictions. (b) Schematic of SMPL's 24 joints (positions illustrative). (d) Bone heads of Anny's 163-bone rest rig, coloured by region, with the 24 points fitted to the SMPL joints in the conversion enlarged.*

**AnnyOne.** AnnyOne is NAVER's synthetic multi-person dataset for Anny, with exact labels (all bone transforms and phenotypes of every person, and the camera). Its authors describe more than 780,000 images [Bra+26]; the release copy used here holds 579,960. It is the only source of complete Anny labels, shape included (Section 3.1).

## 2.2. Image features and attention

A Vision Transformer (ViT) [Dos+21] divides an image into square patches and embeds each as a token. Attention exchanges information between tokens through queries $\mathbf{Q}$, keys $\mathbf{K}_a$ and values $\mathbf{V}_a$:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}_a, \mathbf{V}_a) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}_a^\top}{\sqrt{d_k}}\right)\mathbf{V}_a, \tag{2.2}$$

where $d_k$ is the key dimension; each query returns a mean of the values weighted by its similarity to the keys. The subscript $a$ distinguishes these keys from the camera intrinsics $\mathbf{K}$ (Chapter 4). Multi-HMR uses self-attention among image patches and cross-attention from person queries to all image tokens. Both backbones are DINOv2 [ODM+23] ViTs as released by NAVER with Multi-HMR, a ViT-L for the teacher and a ViT-B for the student; the smaller token width of the ViT-B requires a projector (Section 2.6).

## 2.3. Single-person and multi-person mesh recovery

Human mesh recovery (HMR) estimates the parameters of a body model from an image.

**Regression and fitting.** HMR [Kan+18] regressed body-model parameters for one cropped person with a convolutional encoder, an iterative regressor and an adversarial pose prior, a discriminator separating plausible from implausible poses. HMR 2.0 [Goe+23] replaced the encoder with a ViT and showed that a large ViT trained on a broad dataset mixture performs well without further components. Multi-HMR and the models of this thesis follow this approach and regress the parameters directly, without test-time optimisation. SMPLify [Bog+16] instead fits SMPL to 2D keypoints by optimisation, and SPIN [Kol+19] places such a fit in the training loop. The 3DPW converter (Chapter 3) is a SMPLify-type optimisation, but in 3D, between two body models, run once offline.

**Single-shot multi-person methods.** Two-stage methods apply a person detector and then a single-person network to each crop, so their cost grows with the number of people. ROMP [Sun+21] performs both in one pass (single-shot): it predicts a heatmap of body centres and reads the body parameters there. Multi-HMR [Bar+24], the architecture of this thesis, retains this principle. Its ViT produces one token per 14×14-pixel patch, a 48×48 grid for

the 672×672-pixel input, and each token scores whether a person's primary keypoint, the head joint in the released code [NAV26b], lies within it. High-scoring tokens become queries of the *Human Perception Head* (HPH), which attends to all image tokens and predicts the body parameters and the camera distance of the head joint. This joint, the anchor, is central to the depth defect in Appendix F; Section 4.2.2 gives details.

The released SMPL-X checkpoints reach a PVE (mean vertex error, Section 4.8) on the 3DPW test set of 94.1 mm with ViT-L in 74 ms and 94.0 mm with ViT-B in 43 ms on a V100 GPU [NAV26b]. The ViT-B is equally accurate and faster, which motivates a ViT-B student; these SMPL-X errors are not comparable with the Anny-space errors of this thesis.

**BEV and Relative Human.** BEV [Sun+22] extends ROMP with each person's relative depth and a body model with an age parameter. Its authors collected Relative Human, real photographs annotating each person with an age group (adult, teenager, kid, baby), a depth layer and 2D keypoints, but no 3D body; it is the only available real dataset with age labels and is used for evaluation only (Section 3.6).

**Multi-HMR-Anny and Anny-Fit.** NAVER has released a Multi-HMR checkpoint that predicts Anny, trained on BEDLAM, AnnyOne and further real images [NAV26b]; multi-person Anny prediction is therefore prior work. This checkpoint, the released baseline, provides the teacher's backbone weights and serves as baseline. On the same backbone its decoder has 8 cross-attention blocks and 76.57 M parameters, against 2 blocks and 14.01 M in the models of this thesis; trained on AnnyOne, it may have seen the AnnyOne holdout (Section 5.2.1). Anny-Fit [Bra+26], by the same group, refines such predictions per image by optimisation and adapts them to children at test time. It was not used, since the models here remain single forward passes, but it is the first comparison for future work on children.

## 2.4. Synthetic training data

AGORA [Pat+21] renders scanned people with SMPL-X fits into realistic scenes, and BEDLAM [Bla+23] showed that synthetic images alone can train competitive networks; AnnyOne does the same for Anny. Synthetic labels are exact and complete, but the domain gap to real images can reduce accuracy on photographs. In this thesis, shape predictions followed the labels on the synthetic holdout (AnnyOne images excluded from training) but collapsed towards the training mean on real photographs (Section 5.7.6). The teacher therefore draws 30 % of every batch from real images, which require Anny labels.

## 2.5. Real-image labels for a new body model

3DPW [Mar+18] is the standard real-world benchmark for mesh recovery: 60 video sequences with SMPL labels for one or two actors each. As SMPL and Anny differ in skeleton and shape space, the labels must be refitted in Anny. The SMPL-X code transfers fits between SMPL-family models through a vertex correspondence [Pav+19], and the Anny package fits Anny to a target mesh [NAV26a]; neither converts a 3DPW label into an Anny label directly, so a dedicated converter was written (Section 3.3). It fits joint positions only and recovers only what they determine (Section 3.7). SHAPY [Cho+22] supervises shape on real images with body measurements and linguistic attributes; Anny's named phenotypes would suit such labels, but none were collected.

## 2.6. Knowledge distillation

Knowledge distillation [HVD15] trains a small *student* network to reproduce the outputs of a large, trained *teacher*, originally by matching its class probabilities softened by a temperature (output distillation). FitNets [Rom+15] also match an intermediate teacher representation through a small learned regressor (feature distillation). DINOv2 [ODM+23] distilled its smaller ViTs from its largest, making a DINOv2 ViT-B a suitable student initialisation. The student uses both forms (Section 4.6). It has 101.38 M parameters against the teacher's 318.38 M.

**The projector.** The student is initialised with a copy of the teacher's prediction heads. Its features have 768 channels, the teacher's features and heads 1024. A projector, one linear layer, maps one width to the other, so that student features can be compared with the teacher's and fed to the copied heads; it therefore remains in the deployed student.

**The temperature.** After the projector, each patch token is a vector of 1024 channel values. For teacher and student, the softmax over the channel values of each token is computed at temperature $T$ (after division by $T$), and the Kullback–Leibler (KL) divergence between the two distributions is the feature loss, zero when they coincide. A larger $T$ flattens the distribution, so that smaller channel responses also contribute (illustratively, weights [0.88, 0.12] at $T = 1$ become about [0.62, 0.38] at $T = 4$). The distributions describe feature activation patterns, not body-part probabilities. Because softening scales the gradient by about $1/T^2$, the loss, and with it the gradient, is scaled by $T^2$ to keep gradient magnitudes comparable across temperatures [HVD15].

Output distillation is an L1 loss between student and teacher rotations, phenotypes and distances, weighted as in the supervised loss and without temperature. The reported student uses $T = 4$; Section 5.4.2 examines other values.

| Approach | Relevant idea | Relation to this thesis |
|---|---|---|
| Multi-HMR and its released Anny model [Bar+24; NAV26b] | single-shot multi-person detection and body regression | framework; teacher's backbone weights; baseline |
| FitNets and knowledge distillation [Rom+15; HVD15] | feature and output transfer from a teacher | the student's projected feature matching and output distillation |
| BEV [Sun+22] | age-aware reconstruction and age-group labels on real photographs | reason to evaluate child proportions; not used in the student |
| VIBE, TCMR and SmoothNet [KAB20; Cho+21; Zen+22] | temporal features or output refinement | SmoothNet is the closest relative of the inference-time filter; learned temporal models are future work |

**Table 2.2.:** *Prior work and its relation to this thesis.*

## 2.7. Temporal models for video

Most video methods refine the per-frame features of an image network: HMMR [Kan+19] with a temporal convolution, VIBE [KAB20] with a recurrent network and an SMPL motion discriminator, and TCMR [Cho+21] with branches predicting the current frame from its neighbours to reduce jitter (frame-to-frame fluctuation). MPS-Net [Wei+22] and GLoT [She+23] attend over frames. SmoothNet [Zen+22] is a learned network that smooths the output poses of any estimator over a window. PHALP [Raj+22] and 4DHumans [Goe+23] track people through video, the first step of any multi-person video method, and WHAM [Shi+24] recovers motion in world coordinates.

The student is frame-wise, so its predicted body can jitter or change size between frames. Two inference-time steps are added without retraining (Section 4.9). An exponential moving average blends each tracked person's new bone rotations, phenotypes and position with the previous estimates; like SmoothNet it smooths outputs without image features, but it is a fixed formula, causal (only current and past frames contribute) and operates on body-model parameters. The body's position and depth are then adjusted to match 2D landmarks from a separate detector. Learned temporal models remain future work.

## 2.8. Relation to prior work

The method retains the single-shot design of Multi-HMR and reduces its cost through a smaller image encoder, connected to the teacher by a projector and copied heads; attention itself is unchanged (Table 2.2).

# 3. Datasets and Anny label generation

This chapter describes the datasets and the conversion of 3DPW's SMPL labels to Anny, against which all 3DPW errors are computed. Table 3.1 lists each split and its use. No test image is used for training, and results on earlier 3DPW label sets are kept apart from those on the corrected labels (Appendix G). Section 3.7 collects the limitations of the evaluation data.

| Split | Size (images / people) | Use | Chapters / appendices |
|---|---|---|---|
| AnnyOne training | 579,860 images | training of all models of this thesis; 70% of each mixed batch | 4, 5 |
| AnnyOne holdout | 100 / 222 | synthetic test set and in-training monitor; excluded from training | F, 5 |
| Converted 3DPW training | 5,882 / 7,584 | 30% of each mixed batch | 4, 5 |
| Converted 3DPW validation | 2,622 / 3,311 | 300 images scored during training (teacher every epoch, student every 5 epochs); monitoring only | 5 |
| Converted 3DPW test | 8,190 / 11,854 | reported 3DPW results; never used for training | 5 |
| Relative Human val. | 635 / 2,130 | age and gender on real photographs; evaluation only | F, 5 |
| Demo images | 12 images | visual comparison; no metric | F, 5 |
| Walking video | 356 frames | the walking clip; video comparison, no labels | 5 |

**Table 3.1.:** *Data splits used in this thesis, their sizes and their use. Converted 3DPW sizes are images / labelled people after the drop rules, for the corrected labels.*

## 3.1. AnnyOne

AnnyOne [Bré+26] is the synthetic dataset released with Anny. It contains 579,960 rendered images of 1280×1280 pixels with on average 2.73 people per image. Every visible person has an exact label: the Anny pose (163 bone transforms), the 11 phenotypes (Section 2.1) and the camera intrinsics (focal length and principal point). The network receives each image resized to 672 pixels. The field of view varies between images (51° to 110° on a sample of five images), so training and evaluation use each image's own intrinsics.

**Split.** AnnyOne has no official split. The last 100 images, with 222 people, form the holdout (indices 579,860 to 579,959; limitations in Section 3.7), which no model trained in this thesis has seen; all earlier images form the training set. Unless stated otherwise, holdout errors use the holdout protocol (Section 4.8).

**Age distribution.** Anny produces a child shape below an age value of about 0.4 to 0.5 (Section 2.1). Only 2.53% of AnnyOne's people are younger than 0.3 (1.06% below 0.2, 4.45% below 0.4). The median age is 0.601, and 41% of all people lie between 0.5 and 0.6. In a training batch, 2.1% to 2.3% of the supervised people are younger than 0.3. Children are evaluated on Relative Human (Section 3.6).

## 3.2. 3DPW

3DPW [Mar+18] is a real video dataset of 60 sequences recorded with a handheld phone camera at 1080×1920 pixels (portrait) and 30 frames per second. Inertial sensors on the actors provide SMPL ground truth. The standard release has 24 training, 12 validation and 24 test sequences with one or two labelled actors each, in total 87 person-sequences (62 men, 25 women). Bystanders are not labelled. 3DPW's splits are kept. One validation sequence, `outdoors_crosscountry_00`, yields no labels in any label set (Section G.2), which leaves 11 validation sequences.

**Need for conversion.** As 3DPW's labels are SMPL bodies (Table 2.1), real-image training and testing require an Anny body fitted to every labelled 3DPW person and stored in AnnyOne's label format.

## 3.3. Converting SMPL to Anny

Anny's skinning has no analytic inverse, so pose and shape are recovered from joint positions by optimisation. Appendix G documents earlier versions.

**Input and correspondence.** For each person and frame the targets are the 24 joints $\hat{\mathbf{J}}^S$ of 3DPW's per-frame `jointPositions`, together with gender, camera and frame index. The joints are not recomputed with the neutral SMPL model, because 3DPW's shape coefficients belong to the gendered models and the neutral model moves the joints by 37 mm on average. A correspondence $\pi$ assigns each SMPL joint $k$ to an Anny bone $\pi(k)$ (Section 3.4).

**Objective.** The optimisation variables are the 163 bone rotations $\mathbf{r}$ in 6D form (two columns of each rotation matrix), the phenotype logits $\boldsymbol{\ell}$, which the

sigmoid $\sigma$ maps into $(0, 1)$, and a translation $\mathbf{t}$:

$$\min_{\mathbf{r},\boldsymbol{\ell},\mathbf{t}} \frac{1}{24}\sum_{k=1}^{24}\left\|\mathbf{p}_{\pi(k)}(\mathbf{r},\sigma(\boldsymbol{\ell}))+\mathbf{t}-\hat{\mathbf{J}}_k^S\right\|_2^2 + \lambda_r \sum_{b\neq\text{root}} \|\mathbf{R}_b-\mathbf{I}\|_F^2 + \lambda_\phi \sum_{j\in\text{free}} \left(\sigma(\ell_j)-\tfrac{1}{2}\right)^2, \tag{3.1}$$

where $\mathbf{p}_b$ is the point attached to bone $b$ (its head or a virtual joint, Section 3.4) and $\mathbf{R}_b$ the bone's rotation matrix. The first regulariser ($\lambda_r = 10^{-4}$) pulls every non-root rotation towards identity, because a twist about a bone's own axis moves no joint but deforms the mesh. The second ($\lambda_\phi = 10^{-3}$) pulls the free phenotypes towards 0.5, so that a residual joint gap cannot drive a phenotype to an extreme value.

**Free parameters.** Only the root and 18 of the 23 mapped bones may rotate; 144 of the 163 bones stay at rest, among them `neck02`, `neck03`, `head` and all finger bones. Weight, muscle, cup size, firmness and the three ethnicity values move no joint, and adult age barely does. Gender is therefore taken from 3DPW and written in Anny's convention (men 0, women 1), age is fixed at 0.55 (an adult value) and the seven other phenotypes at 0.5; only height and proportions are fitted.

**Staged fitting.** Adam (learning rate 0.05) optimises Equation 3.1 in three stages: first the root, then the bone rotations, and finally the free phenotypes, which carry the weakest signal. Each frame starts from the previous frame's solution; a person's first frame uses 150 / 400 / 250 iterations per stage, later frames 20 / 80 / 60.

**One shape per person and sequence.** After the per-frame fits, the converter takes the median of the fitted phenotypes over the sequence as the person's shape, so that body proportions are constant through the video. With this shape fixed, the pose of every frame is refitted (100 iterations per frame) and the body is transformed into camera coordinates.

**Label files.** Every third valid frame is fitted (10 frames per second), on one frame grid per sequence (frame numbers divisible by 3), so that all labelled actors of a sequence appear in the same image label, as multi-person detection requires. The label builder writes one file per image in the AnnyOne format and verifies that the image exists. It drops a person whose fit error exceeds 80 mm, of whose mesh less than 30% lies inside the image, or whose box is narrower or shorter than 40 px. The data loader pads the portrait images to a square and shifts the principal point accordingly.

## 3.4. Geometric correspondence and virtual joints

SMPL and Anny place several joints, the hips among them, at different anatomical points, so the correspondence is geometric rather than by name, which would force the fitter to distort the body: both skeletons are placed in their rest pose and aligned at the hip midpoint and along the body axes. Each Anny hip bone is then chosen by distance and each spine bone by its normalised height along the torso. Appendix B gives the full correspondence.

For each of the nine torso targets $k$ attached to Anny bone $b = \pi(k)$, the fitter uses a virtual joint

$$\mathbf{p}_b(\mathbf{R}, \boldsymbol{\phi}) = \mathbf{h}_b(\mathbf{R}, \boldsymbol{\phi}) + \mathbf{Q}_b(\mathbf{R}, \boldsymbol{\phi})\boldsymbol{\delta}_k, \tag{3.2}$$

where $\mathbf{h}_b$ is the posed bone head, $\mathbf{Q}_b$ its orientation and $\boldsymbol{\delta}_k$ a constant offset in the bone's rest frame, measured on the aligned rest skeletons. Both skeletons are thus compared at the same body location while Anny's rig remains unchanged. Limb targets use the bone head itself. The offsets reduce the difference in joint placement; the two models remain anatomically different.

## 3.5. Label construction and validation

The corrected labels comprise 16,694 images and 22,749 people. They use Anny's gender convention and one frame grid per sequence, and every label has its image. The two earlier label sets, their defects and the re-fit that removed them are documented in Appendix G (Tables G.2 and G.3).

The fitting loss does not show whether the finished labels are correct. A separate check therefore compares four properties of the finished label files with 3DPW's source data: body-joint positions, projected keypoints, the presence of every actor, and the gender convention. Because it samples images, the check cannot prove that every label is free of error. On the corrected labels the median body-joint discrepancy is 14.1 mm, and the check passes with two remaining warnings (Table 3.2). These come from limb proportions and from joints that SMPL and Anny define differently, the hip among them.

In Figure 3.1, the converted Anny mesh covers the same two people as 3DPW's SMPL mesh, and the two joint sets lie close together.

## 3.6. Relative Human

Neither the AnnyOne holdout nor 3DPW contains children. Relative Human, introduced with BEV [Sun+22], contains 8,741 real photographs, many of them crowded and with children. Each person has a box, an age group (adult, teen, kid, baby) and a gender coded 0 for male, as in Anny; there are no 3D labels. The licence terms of the copy used were not verified. Only the validation split is used: 635 images with 2,130 labelled people (1,680 adults, 101 teens, 209 kids, 117 babies, 23 of unknown age).

The model runs once on each whole photo with a field of view of 55°, and detections are matched to labelled boxes as in Section 4.8. A predicted age

| Check | Corrected labels |
|---|---|
| Body joints vs 3DPW, median / p95 (mm) | 14.1 / 21.1 |
| Surface, Anny→SMPL / SMPL→Anny (mm) | 18.9 / 16.1 |
| 2D error over raw 3DPW (px) | +0.4 |
| Gender labels | men 0, women 1 |
| Two-person images, an actor unlabelled | 0 of 1,080 |
| Fitted frames; median frame error (mm) | 22,912; 14.0 |
| Head turn vs chest, labels / SMPL (°) | 19.6 / 19.3 |
| *Remaining warnings* | |
| Limb ratio Anny / SMPL, forearm / thigh | 0.948 / 1.064 |
| People with a joint more than 100 mm off | 3.2% |

**Table 3.2.:** *Independent check of the corrected labels of converted 3DPW on images sampled from every sequence, compared with 3DPW's joints, surface and 2D keypoints. The check passes with the two warnings of the last block; Appendix G compares the earlier label sets.*



**Figure 3.1.:** *Converted label (not a network prediction) for 3DPW test frame 771 of* `downtown_walking_00`*, corrected labels. Top: image; bottom left: 3DPW's SMPL label (orange) with OpenPose 2D keypoints (yellow); bottom right: converted Anny label (blue) with 3DPW's SMPL joints (red) and Anny joints (green).*

below 0.45 counts as child-shaped, a predicted gender below 0.5 as male. Per age group, recall and the mean predicted age are reported; overall, gender accuracy and the child-versus-adult AUC of the predicted age (the probability that a kid or baby receives a lower predicted age than an adult; 0.5 is chance). Shape numbers cover detected people only and are reported next to recall.

## 3.7. Limitations of the evaluation data

The evaluation data limit the conclusions of this thesis in three ways.

**AnnyOne holdout.** The synthetic test set consists of only 100 images (222 people). They were taken from the end of the dataset rather than sampled at random, the youngest person has age 0.513 (no children), and the same images serve as in-training monitor and as reported test set. Holdout results therefore do not support strong generalisation claims; these would require a larger, randomly drawn and age-balanced holdout.

**Converted 3DPW labels.** The labels are converted, not ground truth, and test three phenotypes: gender (from metadata), height and proportions (fitted; medians 0.47 and 0.51). The other 8 phenotypes are identical for every person: age 0.55, a full adult (Figure 2.1), and seven values at 0.5. The shape loss and mesh terms pull every real person towards these constants; 3DPW people are an estimated 16% of the supervised people in a 70/30 mixed batch. The teacher with full shape mask removes the shape loss on the constants (Section 4.4). The 3DPW error, measured against these neutral-body meshes, may favour a neutral shape prediction (not tested). The labels can neither teach nor test children. The bones `neck02`, `neck03` and `head` stay at identity, so the head turns only through `neck01` and the labels cannot test the neck bones of Appendix F. SMPL has no finger joints, so every label has rest-pose hands, and training masks their rotation terms for 3DPW people (Section 4.4). The labels differ from 3DPW's source joints by 14.1 mm at the median and 21.1 mm at the 95th percentile. Networks are compared with the fitted targets, not the source joints, so this residual is no error floor and is not subtracted from reported PVE values. The targets are Anny meshes rather than the original SMPL meshes, they carry this residual, and MPJPE averages all 163 Anny joints (Section 4.8). The 3DPW results are therefore not comparable with published 3DPW results. Because bystanders are unlabelled, a correct detection of a bystander counts as false, and the measured 3DPW precision is only a lower bound.

**Relative Human.** The labels are 2D boxes with age-group and gender tags, without 3D ground truth, so pose and shape error cannot be measured.

## 3.8. Demo images and the diagnostic manifest

The diagnostic checks use a fixed sample of 55 AnnyOne (image, person) entries, which all runs after the initial model exclude from sampling, so that their exposure is identical for all later models (Appendix F). Visual comparisons use 12 demo images: 4 real photographs and the middle frame of each of 8 3DPW test sequences. No model is trained on them and no metric is computed on them. The photo comparison with the released baseline (Section 5.6.1) and the walking video (Section 5.6) are likewise unlabelled and permit no reconstruction error.

# 4. Methodology

## 4.1. System overview

The deployed model is a ViT-B student distilled from a ViT-L teacher. The teacher is trained on AnnyOne and on 3DPW training images whose SMPL labels the converter of Chapter 3 turns into Anny labels. The student is trained on the same labels and on feature and output targets of the frozen teacher, which also provides the initial weights of the student's heads. Only the student runs at deployment (Figure 4.1).

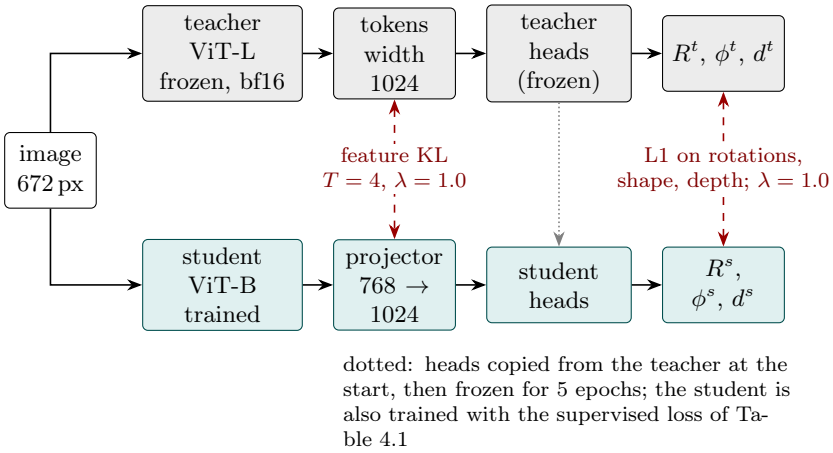



**Figure 4.1.:** *Teacher–student training (schematic). Student and frozen teacher receive the same image; the student is trained with the supervised and two distillation losses. At inference only the student runs.*

The student's encoder $f_B$ maps an image $I$ to $N_p$ tokens $\mathbf{Z}^s = f_B(I)$ of $C_s = 768$ channels, and a trainable linear projector maps them to the teacher's width:

$$\widetilde{\mathbf{Z}}^s = \mathbf{Z}^s\mathbf{W}^\top + \mathbf{b}, \qquad \mathbf{W} \in \mathbb{R}^{C_t \times C_s}, \quad C_t = 1024. \tag{4.1}$$

The projector adds no image information; the savings come mainly from the smaller encoder.

## 4.2. Multi-HMR for Anny

### 4.2.1. The task

The input is one RGB image with intrinsics $\mathbf{K}$ showing an unknown number of people. For each person $i$ the network predicts the rotations $\mathbf{R}_i \in SO(3)^{163}$ of

Anny's bones, the phenotypes $\boldsymbol{\phi}_i \in [0,1]^{11}$, and the image position $(u_i, v_i)$ and depth $z_i$ of one anchor joint $a$, the head. The Anny body model $M$ [Bré+26] maps $(\mathbf{R}_i, \boldsymbol{\phi}_i)$ to vertices $M_V$ and joints, and the mesh is translated so that its anchor joint $M_{J,a}$ lies at the back-projected anchor pixel:

$$\mathbf{V}_i = M_V(\mathbf{R}_i, \boldsymbol{\phi}_i) - M_{J,a}(\mathbf{R}_i, \boldsymbol{\phi}_i) + \mathbf{t}_i, \qquad \mathbf{t}_i = z_i \, \mathbf{K}^{-1} \, (u_i, v_i, 1)^\top. \tag{4.2}$$

### 4.2.2. Architecture

The teacher follows Multi-HMR [Bar+24] with outputs for Anny (Figure 4.2). A DINOv2 ViT-L [ODM+23] taken from the released baseline encodes the image into tokens; one MLP scores each token for the head anchor, a second predicts the anchor's offset inside the token. Each detected token becomes a query of the Human Perception Head (HPH, Section 2.3), which attends to the other people and to all image tokens.

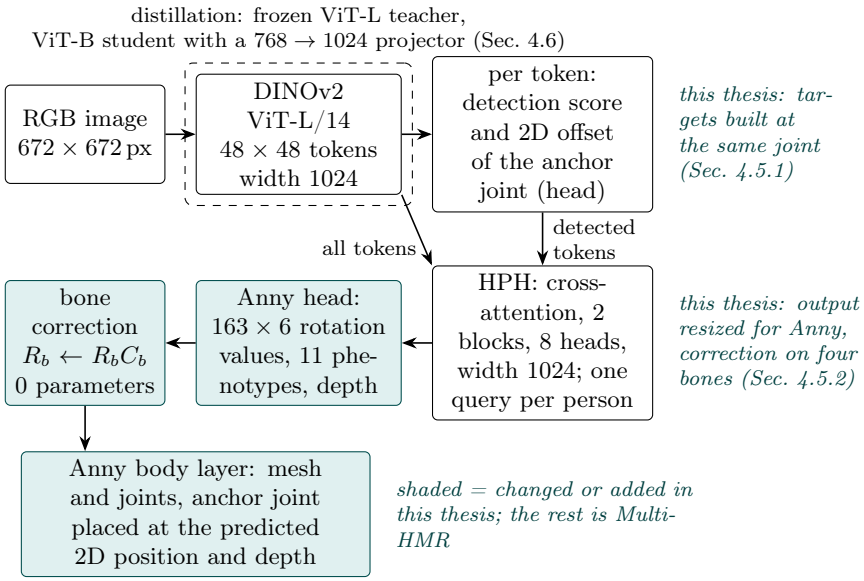



**Figure 4.2.:** *Teacher architecture (schematic). Shaded boxes are new in this thesis; the dashed box marks the part that distillation replaces.*

In the Anny head, which replaces the SMPL-X head, each rotation is predicted as 6 values and orthogonalised by Gram–Schmidt [Zho+19]; a sigmoid bounds the phenotypes to $[0,1]$, and the depth is Multi-HMR's focal-normalised log-distance. Four bones are then multiplied by one constant rotation each (Section 4.5.2). During training a fixed mask sets 88 helper bones (toes, breasts, face and subdivision bones) to the identity in prediction and label, so only 75 bones are supervised; the network still outputs helper-bone rotations, which are unsupervised and are set to rest pose at evaluation only in the holdout protocol (Section 4.8). The Anny layer, with the same full rig as the target builder, is part of the network, so the 3D and 2D losses are computed on the placed mesh and their gradients propagate through it. The

student keeps everything after the encoder, replaces the encoder by ViT-B and adds the projector (Table 4.2).

### 4.2.3. Outputs at inference

At inference the model receives only the image and $\mathbf{K}$. When the calibration is unknown, as for the demo images and the video, $\mathbf{K}$ is built from an assumed field of view; an incorrect assumption changes the recovered depth.

## 4.3. Supervised training objective

Target vertices and joints are decoded from each person's labelled parameters; the anchor targets refer to the head joint (Section 4.5.1). The total loss is

$$\begin{aligned}\mathcal{L} = {} & \alpha_{\text{det}}\mathcal{L}_{\text{det}} + \alpha_{\text{off}}\mathcal{L}_{\text{off}} + \alpha_{\text{rot}}\mathcal{L}_{\text{rot}} + \alpha_{\text{shape}}\mathcal{L}_{\text{shape}} \\ & + \alpha_{\text{dist}}\mathcal{L}_{\text{dist}} + \alpha_{\text{t}}\mathcal{L}_{\text{t}} + \alpha_{\text{j3d}}\mathcal{L}_{\text{j3d}} + \alpha_{\text{v3d}}\mathcal{L}_{\text{v3d}} \\ & + \mathbb{1}[e \geq e_{\text{2D}}]\,(\alpha_{\text{j2d}}\mathcal{L}_{\text{j2d}} + \alpha_{\text{v2d}}\mathcal{L}_{\text{v2d}}) + \lambda_{\text{kd}}\mathcal{L}_{\text{kd}} + \lambda_{\text{kd-out}}\mathcal{L}_{\text{kd-out}},\end{aligned} \tag{4.3}$$

with $e$ the epoch, $e_{\text{2D}}$ the first epoch of the 2D terms and weights $\alpha$ and $\lambda$; the two distillation terms exist only for the student. Table 4.1 lists the weights, which were not tuned in this thesis.

### 4.3.1. Detection and anchor localisation

Let $s_q$ be the detection probability at token $q$, $y_q \in \{0, 1\}$ its target and $N_+ = \sum_q y_q$. The detection term is the focal loss of CornerNet [LD18]:

$$\mathcal{L}_{\text{det}} = -\frac{1}{\max(1, N_+)} \sum_q \begin{cases} (1 - s_q)^2 \log(s_q + \epsilon), & y_q = 1, \\ s_q^2 \log(1 - s_q + \epsilon), & y_q = 0. \end{cases} \tag{4.4}$$

The sum runs over all tokens of the batch; positive tokens contain a head anchor, and the squared factors down-weight well-classified tokens. For $N_h$ supervised people with sub-token offsets $\mathbf{o}_i$,

$$\mathcal{L}_{\text{off}} = \frac{1}{N_h} \sum_i \|\widehat{\mathbf{o}}_i - \mathbf{o}_i\|_1 \tag{4.5}$$

penalises the localisation error of the anchor within its token.

### 4.3.2. Pose, shape and placement

With rotation matrices $\widehat{\mathbf{R}}_{ib}$, $\mathbf{R}_{ib}$ of bone $b$ of person $i$, pose and shape use L1 losses:

$$\mathcal{L}_{\text{rot}} = \frac{1}{N_h} \sum_i \sum_b m_{ib} w_b \|\widehat{\mathbf{R}}_{ib} - \mathbf{R}_{ib}\|_1, \qquad \mathcal{L}_{\text{shape}} = \frac{1}{N_h} \sum_i \sum_k |\widehat{\phi}_{ik} - \phi_{ik}|. \tag{4.6}$$

| Term | Supervised quantity | Teacher | Uncorr. teacher | Student |
|---|---|---|---|---|
| Detection | focal loss on the per-token anchor score | 10 | 10 | 10 |
| Offset | L1 on the anchor's position inside its token | 1 | 1 | 1 |
| Rotations | L1 over the 9 entries of each predicted/target bone rotation | 0.1 | 0.1 | 0.1 |
| neck and hand boost | renormalised: $\times$2.41 neck, head, hands; $\times$0.60 rest | 4 | 4 | 4 |
| Shape | L1 on the 11 phenotypes after the sigmoid | 20 | 20 | 20 |
| Distance | L1 on the anchor's focal-normalised log-depth | 5 | 1 | 5 |
| Translation | L1 on the anchor's 3D position in the camera frame | 5 | 1 | 5 |
| 3D joints | L1 on the 163 joints, each at its own pelvis | 100 | 100 | 100 |
| 3D vertices | the same for the mesh vertices | 100 | 100 | 100 |
| 2D joints | L1 in pixels on the joints projected from anchor and depth | 0.1 | 0.1 | 0.1 |
| 2D vertices | the same for the projected vertices | 0.1 | 0.1 | 0.1 |
| first 2D epoch | epoch $e_{2D}$ at which the 2D terms start | 0 | 0 | 0 |
| Feature distillation | KL of softened student and teacher tokens, $T = 4$ | – | – | 1.0 |
| Output distillation | L1 to the teacher's rotations, phenotypes, depth | – | – | 1.0 |
| Target joint | joint of the four anchor targets | head | pelvis | head |
| Bone correction | neck02, head, metacarpal1.L, finger4-1.R | on | none | on |

**Table 4.1.:** *Loss terms and their weights for the teacher, the uncorrected teacher and the student (Table 4.3). The total loss is the weighted sum of all rows above the rule.*

Here $m_{ib}$ is 0 for unlabelled 3DPW hand bones (Section 4.4), $w_b$ the normalised bone weight. The teacher supervises all phenotypes, including those the converter leaves constant; the shape-mask variants (Section 4.4) test this choice. Both terms sum over a person's bones (or phenotypes) before averaging over people, which sets the scale of each weight. An L1 loss is minimised by the conditional median, so weak image cues pull the predicted shape towards the training median (Appendix F).

For predicted and target log-depth $d_i$ and camera-space anchor translation $\mathbf{t}_i$,

$$\mathcal{L}_{\text{dist}} = \frac{1}{N_h} \sum_i |\widehat{d}_i - d_i|, \qquad \mathcal{L}_{\text{t}} = \frac{1}{N_h} \sum_i \|\widehat{\mathbf{t}}_i - \mathbf{t}_i\|_1. \tag{4.7}$$

Only these terms supervise the absolute placement directly; the 2D terms constrain it indirectly.

### 4.3.3. Geometric and reprojection supervision

For joints or vertices $\mathbf{X}_i$, each body's own pelvis is subtracted first, $\overline{\mathbf{X}}_{i\ell} = \mathbf{X}_{i\ell} - \mathbf{J}_{i,\text{root}}$. With $K_X$ points and weights $w_\ell$ ($w_\ell = 1$ for vertices, the configured joint weights for joints),

$$\mathcal{L}_{X3D} = \frac{1}{N_h K_X} \sum_i \sum_{\ell=1}^{K_X} w_\ell \|\widehat{\overline{\mathbf{X}}}_{i\ell} - \overline{\mathbf{X}}_{i\ell}\|_1. \tag{4.8}$$

This term is independent of the global position. The 2D terms project the uncentred camera-space points:

$$\mathcal{L}_{X2D} = \frac{1}{|\Omega_X|} \sum_{(i,\ell)\in\Omega_X} \|\pi_{\mathbf{K}}(\widehat{\mathbf{X}}_{i\ell}) - \pi_{\mathbf{K}}(\mathbf{X}_{i\ell})\|_1, \tag{4.9}$$

where $\pi_{\mathbf{K}}$ is the perspective projection and $\Omega_X$ the target points inside the image; a term that evaluates to NaN or infinity is skipped.

Unlike these L1 losses, PVE (Section 4.8) averages Euclidean distances. Training uses 32-bit floats with TF32 matrix products, because mixed precision produced NaN values in the Anny layer; steps with NaN or Inf gradients are skipped (16 of 10,000 for the teacher).

## 4.4. Training with real images

A weighted sampler draws on average 30% of each batch from converted 3DPW-train (corrected labels, 5,882 images) and 70% from AnnyOne (share of real people: Section 3.7). The 3DPW test split is never used for training.

**Hand mask.** Fingers, thumbs and metacarpals of every converted 3DPW label are at rest (Section 3.7). In all models trained on 3DPW, the rotation term skips these 38 bones for 3DPW people; finger rotations are supervised by AnnyOne only. The mesh terms still contain the rest-pose hands, and the output distillation uses the teacher's hand predictions.

**Shape mask.** The converter leaves 8 phenotypes constant, among them age (Section 3.7). The teacher with full shape mask excludes all of them from the shape term for 3DPW people, the teacher with partial shape mask the 7 other than age; the mesh terms still contain the neutral body. The mask is off in all other models. The model without shape mask was retained as the teacher because both masked variants lost, on real photographs, the improvement in predicted gender that the teacher shows over the initial model (Section 5.7.7).

## 4.5. Teacher training and its two corrections

The initial model combines the released baseline's ViT-L backbone with Anny heads trained on AnnyOne and is the starting point of every fine-tune (earlier

stages: Appendix E). The teacher is fine-tuned from it for 10 epochs of 1,000 iterations at batch size 4 with a constant learning rate of $10^{-5}$; only the last 4 backbone blocks and the heads are trained. Three changes are introduced together: head-anchor targets with distance and translation weight 5 (Section 4.5.1), the bone correction (Section 4.5.2), and 30% real images per batch with the hand mask. The 55 diagnostic-manifest images (Chapter 3) are never sampled; training took 4,292 s on one A100. The teacher without corrections (uncorrected teacher) differs only by pelvis targets with weight 1 and no bone correction. Its difference from the teacher is due to the two corrections, its difference from the initial model to the real images together with the cost of restarting training (Section F.7); each correction was tested in isolation only in short pilots (Appendix F).

### 4.5.1. Consistent head-anchor supervision

The body is placed by its head joint (Equation (4.2)), so all four anchor targets must refer to that joint. For the teacher, the target builder decodes the labelled body and takes the head's camera-space position and projection, also used for visibility checks and one-person-per-token assignment. The initial model and the uncorrected teacher build these targets at the pelvis, so targets and decoder disagree about the body's position, which the pelvis-centred losses cannot detect (Figure 4.3; measurements in Section F.3). The anchor and the placement weight (5 against 1) change together.

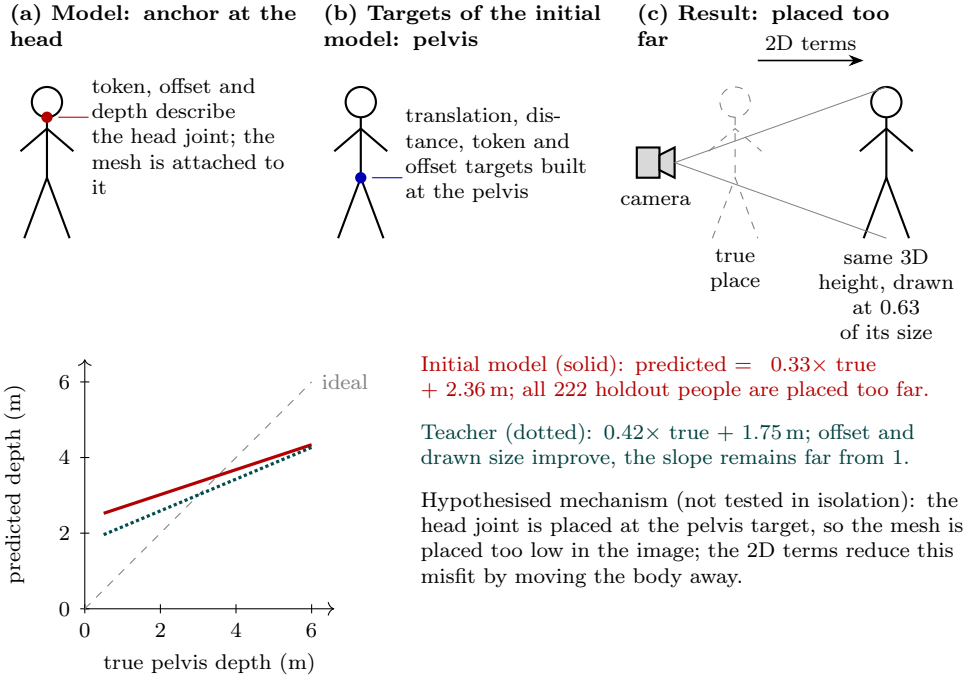



**Figure 4.3.:** *Placement anchor of the model and of the training targets (schematic). Measured depth scatter: Figure 5.7.*

### 4.5.2. Fixed output rotation correction

In the initial model, neck02 and the head bone are each rotated by approximately 180° (Section F.4); the origin of this flip was not identified. It is corrected by one constant rotation per affected bone, applied to the pose output before decoding:

$$\mathbf{R}_{ib}^{\text{corr}} = \widehat{\mathbf{R}}_{ib}\mathbf{C}_b, \qquad \mathbf{C}_b = \text{Proj}_{SO(3)}\left(\frac{1}{N_{\text{cal}}}\sum_i \widehat{\mathbf{R}}_{ib}^{\top}\mathbf{R}_{ib}^{\text{target}}\right), \tag{4.10}$$

with $\text{Proj}_{SO(3)}$ the nearest rotation to the mean residual over $N_{\text{cal}}$ people, estimated on 200 random training images, never on evaluation images, and only for bones whose mean offset exceeds 90°: for the teacher, neck02, head, metacarpal1.L and finger4-1.R (Figure 4.4). It acts before helper masking, decoding and the loss and has no parameters. Because a constant cannot remove an error that varies per image, the teacher was fine-tuned with the correction in place (student: Section 4.7).

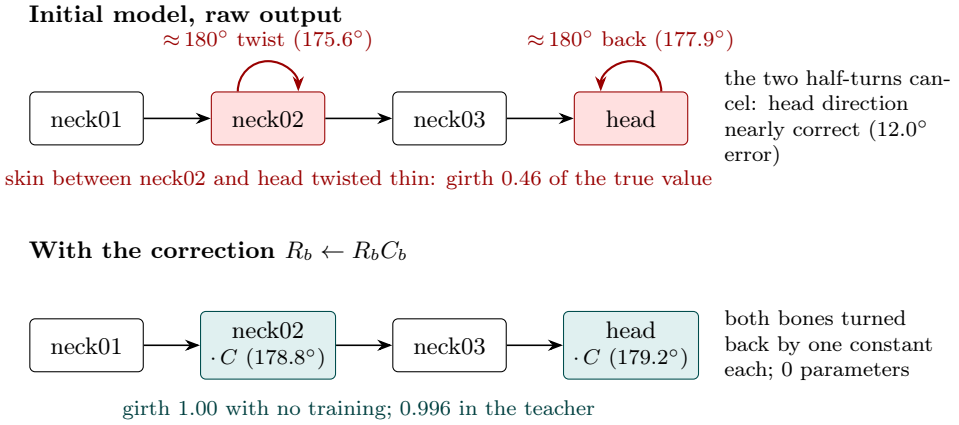



**Figure 4.4.:** *The neck chain of the initial model (top) and with the constant correction of neck02 and head (bottom), schematic. Girth ratios are measured on the AnnyOne holdout.*

## 4.6. Knowledge distillation

**Feature distillation.** For image $u$ and token $n$, the teacher token and the projected student token are converted into distributions over the $C_t$ channels, $p_{unc}^t = \text{softmax}_c(\mathbf{z}_{un}^t/T)$ and $p_{unc}^s = \text{softmax}_c(\tilde{\mathbf{z}}_{un}^s/T)$. With batch size $B$, the loss is their Kullback–Leibler (KL) divergence, summed over tokens and averaged over images:

$$\mathcal{L}_{\text{kd}} = \frac{T^2}{B}\sum_{u=1}^{B}\sum_{n=1}^{N_p}\sum_{c=1}^{C_t} p_{unc}^t \log\frac{p_{unc}^t}{p_{unc}^s}. \tag{4.11}$$

A higher $T$ flattens both distributions, so small channel responses contribute (Section 2.6); $T^2$ keeps the gradient magnitude comparable across temperatures [HVD15]. Following FitNets [Rom+15], the student side is the projected tokens that the heads read; the teacher side is detached.

**Head transfer and output distillation.** The student is initialised with a copy of every teacher head tensor whose name and shape match. Both models are queried at the true person locations, so person $i$ corresponds in both, and

$$\mathcal{L}_{\text{kd-out}} = \frac{1}{N_h} \sum_i \Big[ \alpha_{\text{rot}} \sum_b \|\mathbf{R}^s_{ib} - \mathbf{R}^t_{ib}\|_1 + \alpha_{\text{shape}} \sum_k |\phi^s_{ik} - \phi^t_{ik}| + \alpha_{\text{dist}} |d^s_i - d^t_i| \Big] \tag{4.12}$$

penalises the L1 deviation of the student's outputs from the detached teacher's. It uses no temperature, and its rotation term has neither the per-bone boost nor the hand mask, because the teacher predicts hands where a converted label has none. The supervised terms are retained because the teacher can be wrong. The weights are $\lambda_{\text{kd}} = 1.0$ and $\lambda_{\text{kd-out}} = 1.0$.

**Warm-up freeze.** Initially the copied heads receive features unlike the teacher's, and the gradients of their wrong outputs can corrupt the copied weights. The heads are therefore frozen for the first 5 epochs while backbone and projector adapt; afterwards all parameters are trained. Every head is built at the teacher's width, so all head tensors transfer (earlier designs: Section E.2).

**Choice of temperature and schedule.** Before training, $T = 4$ and the distillation weights were taken over from the earlier ViT-S distillation training configuration, and the 30-epoch schedule was fixed by the compute budget; neither was tuned on test data (selection protocol: Section 5.1). Section 5.4.2 varies $T$ over 5 values up to 16 as a sensitivity analysis.

## 4.7. The ViT-B student

### 4.7.1. Training configuration

The student's ViT-B backbone (Table 4.2) is initialised from NAVER's Multi-HMR ViT-B, already trained for mesh recovery, and fully trained. The supervised targets are built as for the teacher, and no data augmentation is applied. The copied heads contain the half-turn that the teacher's correction undoes (Section 4.5.2), so the student trains with that correction; its own correction, estimated after training on 200 training images, is used in every evaluation and in the video pipeline. The reported student is the 30-epoch run; a 60-epoch run of the same training configuration (learning rate halved every 24 epochs, at the same fraction of the schedule) serves only the training-length analysis (Section 5.4.1).

| | Teacher | Student |
|---|---|---|
| Backbone | ViT-L/14 (width 1024) from the initial model; last 4 of 24 blocks trained; frozen in bf16 when teaching | ViT-B/14 (width 768) from NAVER's HMR-trained Multi-HMR ViT-B; fully trained |
| Heads | HPH at width 1024 (14.01 M) | copied from the teacher, frozen 5 epochs, fed by a projector 768→1024 (0.79 M) |
| Params / GFLOPs / ms | 318.38 M / 1,927.4 / 59.7 | 101.38 M / 603.3 / 28.5 (3.14× / 3.19× / 2.10× less) |
| Schedule | 10 epochs × 1,000 iterations × batch 4 = 40,000 images | 30 epochs × 1,000 iterations × batch 8 = 240,000 images |
| Learning rate | $10^{-5}$, constant | $3 \times 10^{-5}$, halved every 12 epochs |
| Targets | head anchor, depth weight 5, 30% 3DPW (corrected labels), hand mask | identical |
| Distillation | – | feature KL $\lambda = 1.0$, $T = 4$; output L1 $\lambda = 1.0$ |
| Bone correction | four bones, estimated on the initial model | the teacher's during training; its own, estimated after training, at evaluation |

**Table 4.2.:** *Training configuration and cost of the student next to its teacher. Cost is measured at 672 px with one person on one L40S at batch size 1, both models in the same job. The student's errors and the 95 % retention thresholds are reported in Table 5.3.*

### 4.7.2. Retention threshold

The 95 % retention threshold was fixed before training: $E_s \leq E_t/0.95$ for an error metric $E$ with teacher error $E_t$, together with fewer parameters, fewer FLOPs and a faster forward pass. Read as "error at most 1.05 times the teacher's", the thresholds would be 77.4 and 93.4 mm. The evaluation job recomputes the thresholds from the teacher as re-measured in that job (Table 5.3). The rule fixed before training covers PVE and MPJPE on both test sets and the three cost measures; the PA-PVE and recall thresholds shown in Table 5.3 apply the same factor for reference. The same thresholds are applied separately against the released baseline (comparison not training-matched; Section 5.2.1).

### 4.7.3. Models compared

Table 4.3 defines the compared models; their run identifiers are listed in Table A.1 (Appendix A). Earlier models are in Appendix E, the two ablation models in Section 5.3.

| Name in the text | Backbone | Training data | Distinguishing feature |
|---|---|---|---|
| Initial model | ViT-L | AnnyOne | start of every fine-tune; pelvis targets, no bone correction |
| Uncorrected teacher | ViT-L | AnnyOne + 3DPW | initial model fine-tuned with hand mask; otherwise as above |
| Teacher | ViT-L | AnnyOne + 3DPW | head-anchor targets and bone correction |
| Teacher with full / partial shape mask | ViT-L | AnnyOne + 3DPW | teacher with the shape mask of Section 4.4 |
| Student | ViT-B | AnnyOne + 3DPW | distilled from the teacher, 30 epochs; the deployed model |
| 60-epoch student | ViT-B | AnnyOne + 3DPW | same training configuration, 60 epochs; training-length analysis only |
| Released baseline (Multi-HMR-Anny) | ViT-L | AnnyOne, BEDLAM, COCO, MPII (no 3DPW) | released by NAVER, own decoder; external reference, source of the teacher backbone |

**Table 4.3.:** *Models compared in this thesis. Each name denotes one training run; 3DPW is the converted training split (corrected labels, 30% of each batch).*

## 4.8. Evaluation protocol

A greedy 2D matcher pairs each labelled person with at most one detection: candidate pairs are taken in order of increasing distance between their projected 2D joints, and a pair is accepted, as a true positive, only if neither member is matched yet and the boxes enclosing the two joint sets overlap with an IoU of at least 0.05. The same matcher is used on the AnnyOne holdout and on 3DPW; errors are means over matched people (Table 4.4).

For one person with predicted and target vertices $\hat{\mathbf{v}}_k, \mathbf{v}_k$ ($N_V$ of them), joints $\hat{\mathbf{J}}_j, \mathbf{J}_j$ ($J = 163$) and pelvis joints $\hat{\mathbf{p}}, \mathbf{p}$,

$$\text{PVE} = \tfrac{1}{N_V} \textstyle\sum_k \left\| (\hat{\mathbf{v}}_k - \hat{\mathbf{p}}) - (\mathbf{v}_k - \mathbf{p}) \right\|_2, \tag{4.13}$$

$$\text{PA-PVE} = \tfrac{1}{N_V} \textstyle\sum_k \left\| s^* \mathbf{R}^* \hat{\mathbf{v}}_k + \mathbf{t}^* - \mathbf{v}_k \right\|_2, \tag{4.14}$$

$$\text{MPJPE} = \tfrac{1}{J} \textstyle\sum_j \left\| (\hat{\mathbf{J}}_j - \hat{\mathbf{p}}) - (\mathbf{J}_j - \mathbf{p}) \right\|_2, \tag{4.15}$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}), \tag{4.16}$$

with $s^*, \mathbf{R}^*, \mathbf{t}^*$ the least-squares (Procrustes) similarity fit of prediction to target, and TP and FN the matched and missed labelled people. PVE, after pelvis alignment, includes differences in body size; PA-PVE is the residual after the similarity fit. Cost is reported as parameters, GFLOPs (one multiply-accumulate counted as two operations) and the median forward time on one GPU at batch size 1 with one person.

**Two protocols.** In the *holdout protocol* the helper bones are set to rest pose in prediction and target before decoding, as in the demo images and qualitative results, so that the error covers the 75 supervised bones. In the *3DPW*

| Metric | What it measures | Blind to |
|---|---|---|
| PVE (mm) | vertex error after both pelvises are put at the same point | absolute placement and depth |
| PA-PVE (mm) | PVE after the best rotation, scale and shift (Procrustes) | also orientation and scale |
| MPJPE (mm) | PVE on the 163 Anny joints | as PVE |
| Recall, precision (%) | labelled people matched; detections matched | false detections; bystanders |
| Root error (m) | 3D pelvis distance, no alignment | pose and shape |
| Size ratio | drawn head-to-pelvis distance, predicted over true (1 is right) | 3D body size |
| Depth slope / intercept | line fitted to predicted against true pelvis depth (1 / 0 m) | errors around the line |
| Bone angle error (deg) | angle of $\hat{R}_b^\top R_b$ for a parent-relative rotation; ≈180 = flipped | compensating bones |
| Head direction (deg) | camera-frame head orientation error | the chain below the head |
| Neck girth ratio | neck-skin radius, predicted over true (1 is right) | neck length |
| $\mathrm{PVE}_{\mathrm{neck}}$ (mm) | vertex error over the neck skin, centred at neck01 | the rest of the body |
| Child AUC | chance that a child gets a lower predicted age than an adult (0.5 = chance) | undetected people |
| Gender accuracy (%) | predicted label using a threshold of 0.5 | undetected people |

**Table 4.4.:** *Metrics used in this thesis. Errors are means over matched people; child AUC and gender accuracy are computed on Relative Human.*

*protocol* the helper bones are scored as predicted, with the unsupervised rotations the network outputs. Both exist because the evaluation routine of the training code, which produces the in-training validation values and all 3DPW test results, does not apply the mask, whereas the holdout analysis was run with the mask used for rendering. On the same holdout people the 3DPW protocol gives a PVE 3.8 mm higher for the initial model.

**AnnyOne holdout.** The 100 holdout images (Section 3.1) are excluded from all training in this thesis and are scored in the holdout protocol unless stated otherwise; limitations: Section 3.7. Placement and per-bone metrics are computed on the same matched people.

**3DPW test.** All 8,190 test images (11,854 people) are evaluated against the corrected labels in the 3DPW protocol, with the same evaluation code for every model; MPJPE averages all 163 Anny joints (Section 3.7).

**Relative Human and demo images.** The Relative Human validation split (635 images, 2,130 people) is used for evaluation only, with boxes from the projected vertices and matching at IoU $\geq$ 0.3; a prediction counts as child-shaped when its predicted age is below 0.45. The 4 demo images and 8 3DPW

test frames place each body from its detection box, which hides depth errors, and are assessed visually.

**Run-to-run variation.** Each model is a single training run without a fixed seed; no confidence intervals were computed. Two identical short training runs differed by 7.6 mm PVE and 1.9° head direction (Section F.7). The two students trained with the same setting ($T = 4$; the reported run and the repeat within the temperature sweep) differ by 2.5 mm PVE on the AnnyOne holdout, 0.0 mm on 3DPW and 1.5 mm on the 3DPW validation images. Differences of a few millimetres between single runs are therefore not interpreted.

## 4.9. Applying the student to video

### 4.9.1. Frame-wise inference and speed

Each selected video frame is letterboxed (resized and padded without distortion), with an assumed field of view of 55° along the longer side for every model. The raw outputs keep the model's camera-space placement. A geometric tracker follows one person by box overlap and normalised centre distance; a frame without a prediction is left empty, and no frame-to-frame difference is computed across a track break. It records pelvis-relative joint velocity and acceleration ($\mathbf{q}_{tj} = \mathbf{J}_{tj} - \mathbf{J}_{t,\text{root}}$, frame spacing $\Delta t$)

$$\mathbf{v}_{tj} = \frac{\mathbf{q}_{tj} - \mathbf{q}_{t-1,j}}{\Delta t}, \qquad \mathbf{a}_{tj} = \frac{\mathbf{q}_{tj} - 2\mathbf{q}_{t-1,j} + \mathbf{q}_{t-2,j}}{\Delta t^2} \tag{4.17}$$

and the phenotype spread and change; these describe predicted motion, not error.

Input, student and released baseline are shown side by side. Playback FPS is the source rate divided by the frame stride. Inference FPS is measured at batch size 1 with a GPU-synchronised clock around the model call:

$$\text{FPS}_{\text{infer}} = \frac{N_{\text{timed}}}{\sum_{t \in \mathcal{T}} s_t}, \qquad s_t = \text{forward seconds of frame } t, \tag{4.18}$$

where $\mathcal{T}$ holds the $N_{\text{timed}}$ frames after a warm-up of 10 frames (which remain in the videos and measurements). The time $s_t$ includes detection and Anny decoding and excludes preprocessing, transfers outside the model, rendering (timed separately), the offline landmark pass and video decoding and encoding; $\text{FPS}_{\text{infer}}$ is therefore an inference speed, not an end-to-end speed. Values: Section 5.6.

### 4.9.2. Temporal filtering and landmark alignment

Per-frame predictions vary between frames, and the raw student mesh sometimes extends beyond the person (Section 5.6). Two inference-time steps, neither a learned temporal model, address this without retraining.

**Temporal filter.** A causal exponential moving average updates a state $\bar{x}$ of each smoothed quantity with each new prediction $x_t$:

$$\bar{x}_t = (1 - \alpha)\,\bar{x}_{t-1} + \alpha\,x_t, \qquad \alpha = 1 - \exp(-\Delta t/\tau_c),$$

with $\tau_c$ the time constant of quantity $c$; a longer $\tau_c$ reduces jitter at the cost of lag. The time constants are 0.08 s for the bone rotations, 0.12 s for the camera-space head anchor and 1.0 s for the phenotypes. Rotations are blended by shortest-arc quaternion interpolation, which keeps them valid near $\pm 180°$. The filtered parameters are decoded again, so vertices are never averaged. The state resets after a missed detection, a new track or a gap above 0.25 s; a missed frame is never filled with an earlier body. Raw and filtered outputs share one forward pass. The constants are pipeline defaults, inspected visually and not tuned.

**Landmark alignment.** To correct a misplaced or oversized body, the 2D landmark detector MediaPipe Pose [Goo] runs as a separate offline pass on the CPU over the original video. Its shoulders, elbows, wrists, hips, knees and ankles correspond to the heads of the Anny bones upperarm01, lowerarm01, wrist, upperleg01, lowerleg01 and foot. A robust least-squares fit estimates one 3D translation of the body that aligns the projected Anny points with the landmarks; this corrects an oversized or shifted mesh, but not a wrongly articulated limb. A fit is accepted only with at least 4 landmarks of confidence $\geq 0.5$ covering a shoulder, a hip and a knee or ankle, and a depth within 0.5–2 times the current median joint depth; otherwise the frame is left unchanged and the depth history resets. The accepted depth is smoothed in log space (0.35 s, at most 0.5 log-units per second), moving the body along the current pelvis camera ray; a reset can still cause a visible change in size. The alignment acts on the filtered body when the filter is on. The projection error is recomputed after smoothing, which trades some current-frame alignment for less size fluctuation.

**Scope of the two steps.** A lower landmark error indicates agreement with the landmark detector, not a more accurate 3D body; the clip has no 3D labels, and a stable body can still be wrong. Any improvement of the aligned output stems from this refinement, not from the network. The video panels report different timings (raw: Equation (4.18); filtered: plus filtering and decoding; aligned: plus the fit, without landmark extraction; input: playback FPS), which are not comparable (Table 5.8, Figure 5.5).

# 5. Experiments and results

This chapter evaluates the ViT-B student, the model intended for deployment, against its ViT-L teacher and the released baseline. It then measures what distillation contributes with an ablation, judges the sensitivity to training length and distillation temperature on validation data, and reports detection recall in crowded images, and the student on a photograph and a video. The last section analyses the teacher.

## 5.1. Experimental setup

The evaluation sets are described in Chapter 3 and scored as in Section 4.8; model names follow Table 4.3.

- *AnnyOne holdout*: the last 100 AnnyOne images (222 people, no children), excluded from training in this thesis and scored with the holdout protocol. Its size and composition limit what it can show (Section 3.7). Placement metrics are computed on the same matched people, with each body at the depth its model predicts.
- *3DPW test*: all 8,190 test images with the corrected converted labels, scored with the 3DPW protocol. The numbers hold only against these labels (Section 3.7).
- *3DPW validation*: 300 images with corrected labels, evaluated during training.
- *Relative Human* validation split: real photographs with an age group, a gender and a box per person, but no 3D ground truth.
- *Demo images*: 4 real photographs and 8 3DPW test frames without labels, judged by visual inspection, with each body placed from its detection box.

Every model is a single training run without a fixed seed, so PVE differences of a few millimetres between two runs can be run-to-run variation (Section 4.8).

**Model-selection protocol.** The student's training configuration was fixed before it was trained. The distillation temperature $T = 4$ and the loss weights were taken over from the earlier ViT-S distillation runs (Appendix E), and the length of 30 epochs was set by the compute budget. Every evaluation uses the last checkpoint of a run; there is no early stopping, and no checkpoint or hyperparameter of the student was selected on the 3DPW test set. The in-training

| | AnnyOne holdout | | | 3DPW test | | | |
|---|---|---|---|---|---|---|---|
| Model | PVE | PA-PVE | Recall | PVE | PA-PVE | Recall | 3DPW training |
| Released baseline | 60.3[a] | 44.6[a] | 89.0[a] | 126.9 | 68.8 | 97 | no |
| Teacher | 73.7 | 50.3 | 87.0 | 89.0 | 57.6 | 99 | yes |
| Student | 119.6 | 80.8 | 85.4 | 110.1 | 71.8 | 99 | yes |
| Uncorrected teacher | 77.4 | 56.4 | 89.0 | 91.5 | 59.7 | 99 | yes |
| Initial model | 70.5 | 53.3 | 90.2 | 121.8 | 72.0 | 100 | no |

[a]Potentially in-sample (Section 5.2.1). 3DPW errors are measured against the converted Anny labels and are not comparable with published SMPL-space results (Section 3.7).

**Table 5.1.:** *Reconstruction of the student, its teacher and the released baseline on the AnnyOne holdout and the 3DPW test set (errors in mm, recall in %; one run per model). The models differ in their training data, and the released baseline may have seen the holdout, so a lower PVE does not show a better architecture (Section 5.2.1).*

| Model | Params (M) | GFLOPs | Latency L40S (ms) | Latency A100 (ms) |
|---|---|---|---|---|
| Released baseline | 380.94 | $\geq$1917.2[b] | n/m | n/m |
| Teacher | 318.38 | 1927.4 | 59.7 | 77.9 |
| Student | 101.38 | 603.3 | 28.5 | n/m |

One multiply-add counts as two FLOPs. Latency: median batch-one forward time in fp32 with TF32; the student and teacher L40S times come from the same job. n/m: not measured. [b]Backbone only; the total is higher. The initial model and the uncorrected teacher share the teacher architecture, and the fixed rotation corrections add no trainable parameters.

**Table 5.2.:** *Model cost at 672 px input with one person. Latencies are comparable only within a GPU column.*

validation set is 3DPW validation; the reported student ends at 87.7 mm PVE there. The 60-epoch run and the temperature sweep were trained afterwards as sensitivity analyses, judged on the 3DPW validation images (Section 5.4), and changed no setting. Two limitations remain. The AnnyOne holdout serves both as in-training monitor and as the reported synthetic test set, and the two teacher corrections were adopted on diagnostics measured on it (Appendix F); the choice of the teacher over its two shape-masked variants was made on Relative Human (Section 5.7.7). Neither set is therefore independent of the development process in the way a sealed test set would be.

## 5.2. Student, teacher and released baseline

Table 5.1 lists the student, its teacher and the released baseline on both test sets, with the teacher without corrections (uncorrected teacher) and the initial model for reference. The teacher has the lowest errors on the converted 3DPW labels, 89.0 mm PVE, against 110.1 mm for the student and 126.9 mm for the released baseline. Against the released baseline the student has a lower PVE

on 3DPW (by 16.8 mm) and a lower MPJPE (112.0 against 129.4 mm), but a slightly higher PA-PVE (71.8 against 68.8 mm). Since PA-PVE first removes scale, rotation and translation, the student's advantage lies in the overall size and orientation of its bodies.

### 5.2.1. Fairness of the comparison

The comparison with the released baseline is not a controlled comparison of architectures, and Table 5.1 compares complete trained systems. On 3DPW the bias favours the models of this thesis: they were trained on 3DPW training sequences labelled by the same converter, with its constant phenotypes and rest-pose hands, and the released baseline was not. The teacher's advantage of 37.9 mm over the released baseline therefore includes the effect of training on converted 3DPW labels. The only like-for-like pair is the released baseline and the initial model, neither of which saw 3DPW (126.9 against 121.8 mm PVE). On the AnnyOne holdout the bias is in the opposite direction: the released baseline was trained on the AnnyOne release, which contains the 100 holdout images, so its 60.3 mm PVE is potentially in-sample. The two systems are also scored differently on 3DPW: the released baseline sets the helper bones (face, toes, breasts) to rest pose by design, whereas the models of this thesis are scored with predicted helper bones (3DPW protocol), a difference that does not favour them. A lower PVE of the student or the teacher therefore does not show a better architecture, and a baseline trained on the same data does not exist (Section 6.3).

### 5.2.2. Retention against the teacher

The student follows the training configuration of Table 4.2 (Section 4.7). Its PVE is 1.62 times the teacher's on the holdout (119.6 against 73.7 mm) and 1.24 times on 3DPW. The block "AnnyOne holdout, 3DPW protocol" of Table 5.3 is the standalone value compared with the in-training monitor of Table 5.5. Against the 95 % retention thresholds set in that section (Table 5.3), the student misses every error threshold on both sets, for example 110.1 mm PVE on 3DPW against a threshold of 93.7 mm, and meets the thresholds only on recall and cost. Against the released baseline the 3DPW PVE threshold (133.6 mm) is met and the AnnyOne threshold (63.5 mm) is not, which is not interpreted (possible in-sample evaluation). No labelled real photographs were available to test the student's hands and head.

### 5.2.3. Model size and speed

The student has 101.38 M parameters, 3.14 times fewer than the teacher (318.38 M) and 3.76 times fewer than the released baseline (380.94 M; Table 5.2). It needs 3.19 times fewer FLOPs than the teacher. On one L40S, in the same job, a student forward pass took 28.5 ms and a teacher pass 59.7 ms, a speed-up of 2.10; latencies are compared only within one GPU. Figure 5.1 shows the student with lower cost and higher 3DPW error than the teacher.

| Metric | Teacher | Student | Ratio | Threshold | Met? |
|---|---|---|---|---|---|
| *AnnyOne holdout* | | | | | |
| PVE (mm) | 73.7 | 119.6 | 1.62 | $\leq$ 77.6 | no |
| PA-PVE (mm) | 50.3 | 80.8 | 1.61 | $\leq$ 53.0 | no |
| MPJPE (mm) | 77.9 | 127.2 | 1.63 | $\leq$ 82.0 | no |
| Recall (%) | 87.0 | 85.4 | 0.98 | $\geq$ 82.7 | yes |
| *AnnyOne holdout, 3DPW protocol* | | | | | |
| PVE (mm) | 77.5 | 122.7 | 1.58 | $\leq$ 81.6 | no |
| PA-PVE (mm) | 54.4 | 84.2 | 1.55 | $\leq$ 57.3 | no |
| Recall (%) | 87 | 85 | 0.98 | $\geq$ 82.7 | yes |
| *3DPW test* | | | | | |
| PVE (mm) | 89.0 | 110.1 | 1.24 | $\leq$ 93.7 | no |
| PA-PVE (mm) | 57.6 | 71.8 | 1.25 | $\leq$ 60.6 | no |
| MPJPE (mm) | 88.3 | 112.0 | 1.27 | $\leq$ 92.9 | no |
| Recall (%) | 99 | 99 | 1.00 | $\geq$ 94.1 | yes |
| *Cost (L40S, batch 1, one person)* | | | | | |
| Params (M) | 318.38 | 101.38 | 3.14× | fewer | yes |
| GFLOPs | 1927.4 | 603.3 | 3.19× | fewer | yes |
| ms per image | 59.7 | 28.5 | 2.10× | faster | yes |

**Table 5.3.:** *The student against its teacher and the 95 % retention thresholds: each error at most the teacher's divided by 0.95, each recall at least 0.95 of the teacher's, and lower cost; the rule fixed before training covers PVE, MPJPE and cost, and the PA-PVE and recall thresholds are given for reference (Section 4.7.2). The student misses every error threshold and meets those on recall and cost (both models timed in the same job).*

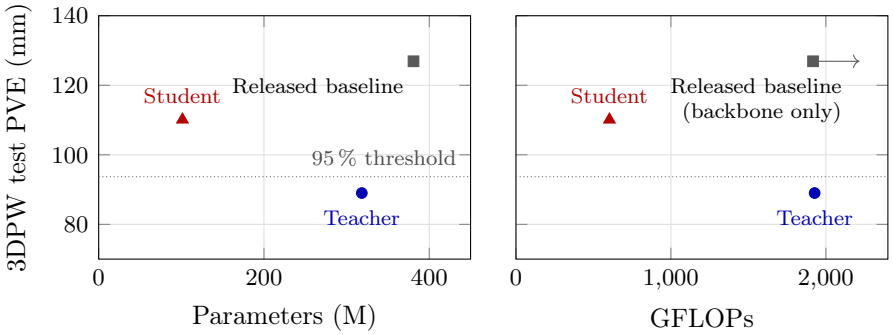



**Figure 5.1.:** *Parameters and GFLOPs against 3DPW-test PVE (corrected labels, all 8,190 images). The released baseline's GFLOPs count its backbone only (arrow: the total is higher); the dotted line marks the 95 % retention threshold (93.7 mm).*

| Model | Distillation losses | Teacher heads | AnnyOne holdout PVE | AnnyOne holdout PA-PVE | AnnyOne holdout MPJPE | 3DPW test PVE | 3DPW test PA-PVE | 3DPW test MPJPE |
|---|---|---|---|---|---|---|---|---|
| Student (reported) | yes | yes | 119.6 | 80.8 | 127.2 | 110.1 | 71.8 | 112.0 |
| Student, repeat run | yes | yes | 117.1 | 81.4 | 124.2 | 110.1 | 72.0 | 111.5 |
| Without distillation losses | no | yes | 118.2 | 79.7 | 125.3 | 110.0 | 71.6 | 111.6 |
| Without teacher | no | no | 122.6 | 81.0 | 130.0 | 116.7 | 79.1 | 123.1 |
| Teacher (ViT-L) | – | – | 73.7 | 50.3 | 77.9 | 89.0 | 57.6 | 88.3 |

**Table 5.4.:** *Distillation ablation: ViT-B models with the student's data, schedule and backbone initialisation, with and without the distillation losses and the transferred teacher heads (errors in mm, last checkpoint, one run each). The repeat run is the $T = 4$ run of the temperature sweep and indicates the difference between two runs of one configuration.*

This benchmark did not time the released baseline; the only speed comparison with it on the same GPU is the video of Section 5.6.

## 5.3. Contribution of distillation

The teacher reaches the student by two routes: its output heads are copied into the student, and two distillation losses pull the student's features and outputs towards the teacher's (Section 4.6). Two ablation models with the student's data, schedule, architecture and backbone initialisation separate these routes (Table 5.4); each was trained once and is evaluated like the student. A repeat run of the student indicates how far two runs of one configuration differ: 2.5 mm PVE on the holdout, 0.0 mm PVE and 0.5 mm MPJPE on 3DPW.

The teacher contributes clearly. A ViT-B model trained without it is 6.6 mm worse in PVE, 7.1 mm in PA-PVE and 11.6 mm in MPJPE on the 3DPW test set than the repeat run, and 5.5 mm worse in PVE on the holdout, several times the run-to-run difference. This gain is already obtained by the model that keeps the transferred heads, with their warm-up freeze and the teacher's bone correction in training, but switches both distillation losses off: its errors lie within the run-to-run difference of the student's on both datasets (3DPW PVE 110.0 against 110.1 mm, holdout PVE 118.2 against 117.1 mm), and the 3DPW validation images agree (86.1 against 86.2 mm). In this configuration the distillation losses therefore show no measurable additional effect on accuracy beyond head transfer, although the feature loss does align the features as intended (its term on the validation images is 13.0 with the loss and 306.0 without). The finding holds for single runs with one loss weighting, one feature layer and one schedule and is not a statement about distillation in general; these limits and the unseparated parts of head transfer are discussed in Section 6.3.

| | | 3DPW val., last epoch | | | Holdout monitor[a] | 3DPW val. PVE | |
|---|---|---|---|---|---|---|---|
| Run | Epochs | PVE | PA-PVE | MPJPE | PVE | best (epoch) | last − best |
| *Temperature sweep* | | | | | | | |
| $T = 1$ | 30 | 86.7 | 61.2 | 85.6 | 119.4 | 85.4 (29) | 1.2 |
| $T = 2$ | 30 | 87.7 | 61.4 | 86.7 | 121.2 | 86.3 (29) | 1.4 |
| $T = 4$ (repeat run) | 30 | 86.2 | 60.7 | 85.2 | 118.4 | 85.0 (29) | 1.3 |
| $T = 8$ | 30 | 85.2 | 60.8 | 84.0 | 122.8 | 84.9 (29) | 0.3 |
| $T = 16$ | 30 | 85.5 | 60.4 | 84.1 | 123.4 | 85.1 (29) | 0.4 |
| *Training length* ($T = 4$, evaluated every 5 epochs) | | | | | | | |
| Student (reported) | 30 | 87.7 | 60.6 | 86.9 | 122.8 | – | – |
| 60-epoch student | 60 | 86.0 | 60.2 | 85.0 | 113.0 | 84.8 (55) | 1.2 |

[a]The holdout is the synthetic test set; its monitored value is given for reference and is not a validation value (in-training evaluation; the standalone evaluation of Table 5.3 gives 122.7 mm for the reported student). Differences quoted in the text are computed from the unrounded values.

**Table 5.5.:** *Temperature sweep and training length judged on validation data: errors in mm at the last epoch on the 300 3DPW validation images and on the AnnyOne holdout as monitored during training (3DPW protocol), with the lowest 3DPW validation PVE of each run and its epoch. All runs use the student's configuration and differ only in $T$ or in the number of epochs.*

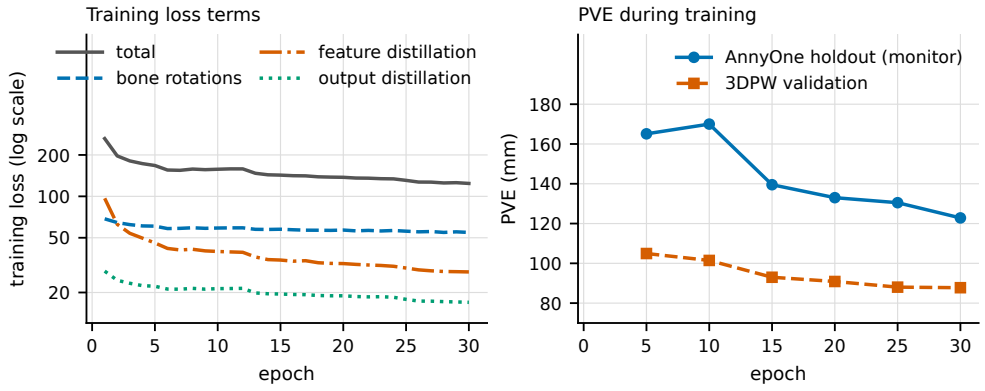
## 5.4. Sensitivity to training length and temperature

Both sensitivity analyses are judged on the 300 3DPW validation images, at the last epoch of each run (Table 5.5); the test-set evaluation of each run follows as its final evaluation and is not used for selection. $T$ and the training length were fixed before training (Section 5.1), and neither analysis changed a setting.

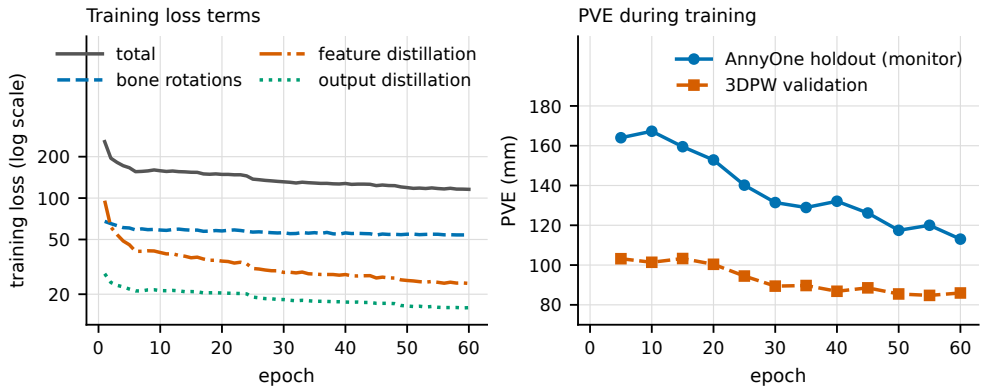
### 5.4.1. Training length

After 30 epochs the student's holdout PVE monitored during training (Figure 5.2) was still falling, from 139.5 mm at epoch 15 to 122.8 mm at epoch 30, while its 3DPW validation PVE had changed little since about epoch 25 (88.0 to 87.7 mm). To test whether longer training helps, the same training configuration was trained for 60 epochs. No validation loss was logged for these runs; the validation PVE, measured every 5 epochs, is used instead. The holdout PVE rises between epochs 5 and 10, when the copied heads are unfrozen; the large drops of the training loss follow the learning-rate halvings (every 12 and 24 epochs).

Over the last 20 % of the epochs the training loss (−0.87 and −0.29 % per epoch) and the holdout PVE (−1.21 and −0.38 % per epoch) still fall in both runs, but the 3DPW validation PVE is flat (−0.06 and +0.06 % per epoch). At the last epoch it is 87.7 mm after 30 and 86.0 mm after 60 epochs (Table 5.5), a difference comparable to the 1.5 mm by which two runs of the 30-epoch configuration differ on the same images; the 60-epoch run ends 1.2 mm above




**(a)** *Student, 30 epochs (reported)*




**(b)** *60-epoch student*

**Figure 5.2.:** *Training curves of the student and the 60-epoch student. Left: the training loss terms per epoch (log scale); right: the PVE evaluated every 5 epochs during training on the AnnyOne holdout monitor (the synthetic test set, not a validation set) and on the 300 3DPW validation images. The holdout monitor still falls over the last 20 % of the epochs, while the 3DPW validation PVE is flat.*

| | Teacher | Student 30 epochs | Student 60 epochs | Threshold |
|---|---|---|---|---|
| *AnnyOne holdout* | | | | |
| PVE (mm) | 73.7 | 119.6 | 107.7 | $\leq$ 77.6 |
| PA-PVE (mm) | 50.3 | 80.8 | 74.6 | |
| MPJPE (mm) | 77.9 | 127.2 | 114.5 | $\leq$ 82.0 |
| Recall (%) | 87.0 | 85.4 | 85.0 | |
| *3DPW test (8,190 images)* | | | | |
| PVE (mm) | 89.0 | 110.1 | 110.9 | $\leq$ 93.7 |
| PA-PVE (mm) | 57.6 | 71.8 | 72.5 | |
| MPJPE (mm) | 88.3 | 112.0 | 112.9 | $\leq$ 92.9 |
| Recall (%) | 99 | 99 | 99 | |
| *Cost, batch 1, one person* | | | | |
| ms per image[a] | 59.7 / 60.6 | 28.5 | 28.9 | faster |

Thresholds: teacher error divided by 0.95, set on PVE and MPJPE. [a]Each student job timed itself and the teacher on its own GPU (teacher: 30-epoch job first). Both students have 101.38 M parameters and 603.3 GFLOPs.

**Table 5.6.:** *Test-set evaluation of the student after 30 and 60 epochs against the teacher and the 95 % thresholds (errors in mm, recall in %). These test-set values were not used to choose the training length, which was fixed in advance (Section 5.1); the sensitivity analysis is read on validation data (Table 5.5).*

its best evaluation. Longer training thus clearly improves the synthetic monitor and changes the real validation images only slightly; the 30 epochs fixed in advance were kept, since the longer run was a sensitivity analysis and not a selection step.

The final evaluation (Table 5.6) is consistent with this. Doubling the epochs lowered the holdout PVE by 11.9 mm, to 107.7 mm, and the holdout root error from 830.7 to 697.9 mm (teacher: 728.5 mm). The 3DPW PVE changed from 110.1 to 110.9 mm, a change of 0.8 mm that is not interpreted (Section 4.8). The 60-epoch student misses all four retention thresholds at the same cost, so training duration alone does not explain the student's real-image gap. No other training length was evaluated.

### 5.4.2. Distillation temperature

The temperature $T$ sets how strongly the feature distillation softens the channel distributions before the student is compared with the teacher (Section 2.6). Since $T = 4$ was taken over without a search, 5 runs with $T \in \{1, 2, 4, 8, 16\}$ and the reported training configuration otherwise test how much the result depends on it; the output distillation has no temperature. Because the sweep trains $T = 4$ again, it also provides a second run of the reported setting.

On the validation images (Table 5.5) the last-epoch 3DPW PVE spans 2.6 mm over the five temperatures, from 85.2 mm at $T = 8$ to 87.7 mm at $T = 2$, while the two runs with $T = 4$ differ by 1.5 mm on the same images. Selection on validation data would have favoured $T = 8$, by 1.0 mm over the repeat run with $T = 4$; this difference is not interpreted, and the value fixed

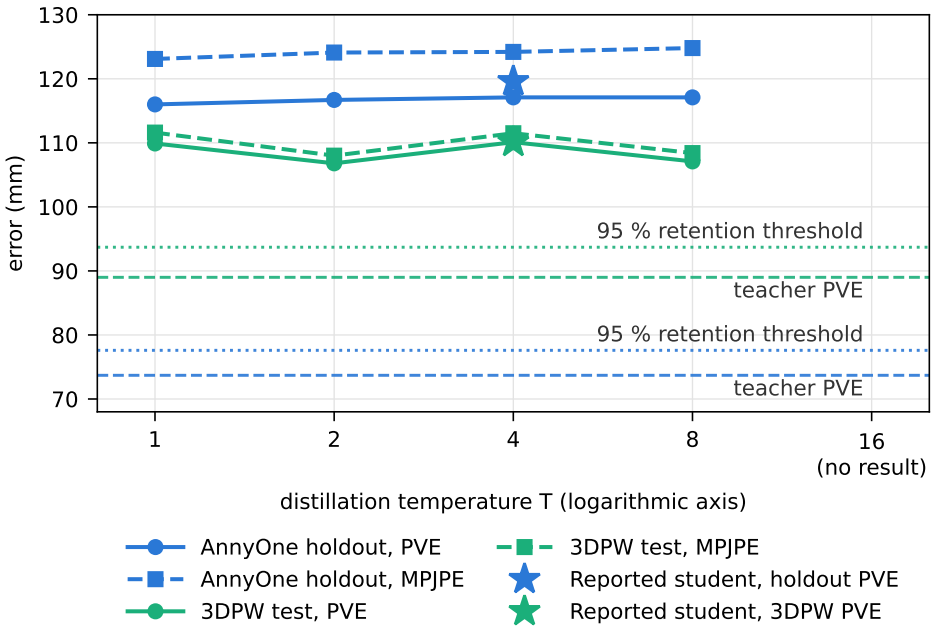



**Figure 5.3.:** *Test-set evaluation (secondary view) of the student against the feature-distillation temperature $T$: PVE (solid) and MPJPE (dashed) on the AnnyOne holdout and the 3DPW test set, with the teacher's PVE (dashed lines) and the 95 % retention thresholds (dotted). Stars mark the reported $T = 4$ run; $T = 16$ has no test evaluation (validation values: Table 5.5).*

| Labelled people per image | Images | People | Initial model recall (%) | Teacher recall (%) |
|---|---|---|---|---|
| 1 | 21 | 21 | 33.3 | 76.2 |
| 2 | 314 | 628 | 51.0 | 78.5 |
| 3–4 | 190 | 630 | 47.6 | 67.8 |
| 5–8 | 82 | 509 | 44.4 | 56.6 |
| 9 or more | 28 | 342 | 43.9 | 45.3 |
| All | 635 | 2,130 | 47.1 | 64.7 |

**Table 5.7.:** *Detection recall on the Relative Human validation split by the number of labelled people per image, with a detection counted at IoU $\geq$ 0.3 with a labelled box. The teacher's recall falls from 78.5 % with two people to 45.3 % with nine or more (teacher only; Section 5.5).*

beforehand is kept. The holdout monitor does not rank the temperatures in the same order, and recall is 100 % in every run. In every run the last epoch is at most 1.4 mm above the best epoch, so using the last checkpoint without early stopping costs little.

The test-set evaluation (Figure 5.3; values in Table H.4, Appendix H) is a secondary view and is consistent with this. Over $T = 1$ to 8 the holdout PVE moves by 1.1 mm (116.0 to 117.1 mm), below the 2.5 mm by which the two runs with $T = 4$ differ, and the 3DPW PVE by 3.3 mm (106.8 to 110.1 mm); the formally lowest combined test error at $T = 2$ is not interpreted as an improvement. No temperature brings the student close to the error thresholds; at $T = 8$ the holdout recall (82.5 %) also falls just below its threshold of 82.7 %. The run with $T = 16$ completed training and has validation values, but no test evaluation for technical reasons (Appendix A).

Neither temperature nor training length changes the 3DPW result or brings the student within the retention thresholds; longer training lowers the synthetic holdout error by 11.9 mm. The 30-epoch student remains the reported model because the schedule was fixed in advance; preferring the 60-epoch run for its holdout error would be selection on the test set.

## 5.5. Multi-person detection

The evaluation sets differ in the number of people per image. The AnnyOne holdout has 222 people in 100 images; the 3DPW test set labels one or two people per image (11,854 people in 8,190 images); the Relative Human validation split has 2,130 labelled people in 635 images, many of them crowded. The teacher and the student find 99 % of the labelled people on 3DPW and 87.0 and 85.4 % on the holdout (Table 5.1). Precision on 3DPW is low for every model, because unlabelled bystanders count as false positives. These sets contain few people per image, so Relative Human is the only one that tests crowded scenes.

Table 5.7 splits the Relative Human recall by the number of labelled people per image. The teacher's recall is 78.5 % in images with two people and falls steadily with crowd size, to 56.6 % with five to eight and 45.3 % with nine or

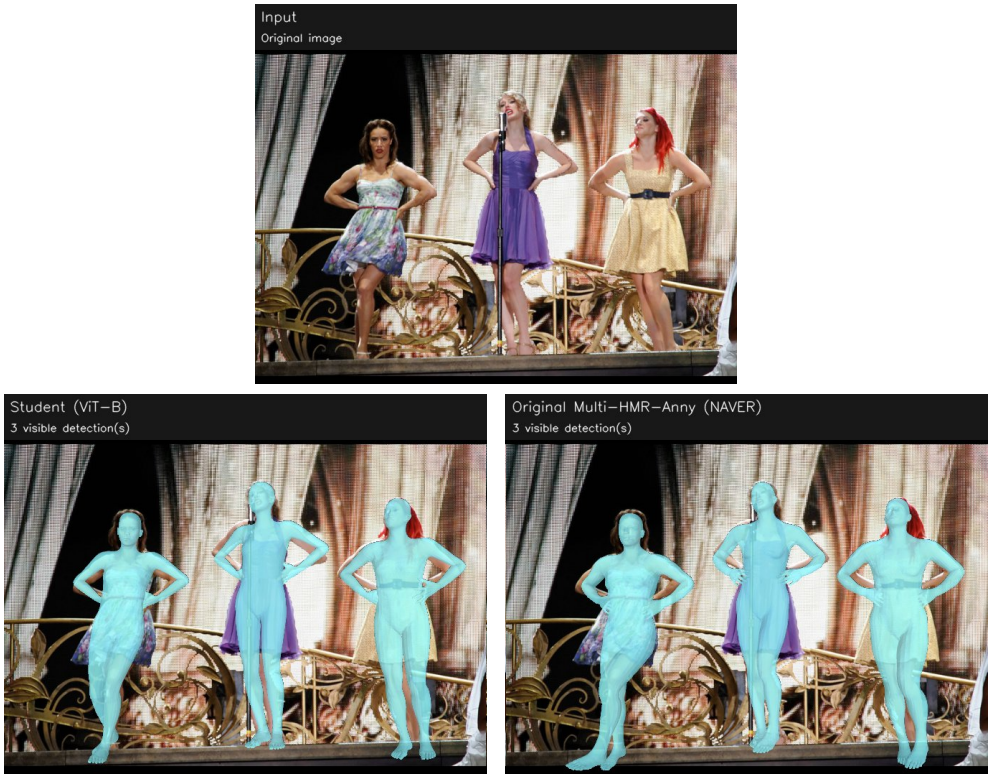



**Figure 5.4.:** *A real photograph of three people on a stage: input (top), the student (bottom left, labelled "Student (ViT-B)") and the released baseline (bottom right, labelled "Original Multi-HMR-Anny (NAVER)"). Both models detect all 3 people; the photograph has no ground truth.*

more. The initial model is lower in every bin (33.3 % with one person, 51.0 to 43.9 % from two people on), and the gap narrows in the most crowded images. The single-person bin holds only 21 images. Recall is counted at IoU $\geq 0.3$ against the labelled boxes; Relative Human has no 3D ground truth, so the reconstruction quality of the people found in crowds is not measured.

The dependence of recall on crowd size is a teacher result; the student was not evaluated on Relative Human. On the two sets where both were measured its recall is close to the teacher's (Table 5.1), but these sets hold few people per image. No occlusion-specific or crowd benchmark with 3D labels was used, and recall was not split by visibility or occlusion; the only occlusion case shown is the holdout image in which two people lying in the foreground are missed by both the initial model and the teacher (Figure 5.6, continued). The video of Section 5.6 shows one person, and the still image three.

| | Model inference[a] | | With preprocessing[b] | |
|---|---|---|---|---|
| Panel | FPS | included | ms/frame | FPS |
| Student, raw | 18.3 | network and mesh decoding | 64.9 | 15.4 |
| Student, filtered | 14.9 | plus filter and re-decoding | n/m | n/m |
| Student, aligned | 12.9 | plus filter and alignment fit | 87.8 | 11.4 |
| Released baseline | 8.5 | network and mesh decoding | 127.9 | 7.8 |

Batch-one sequential timings on one NVIDIA A40 (fp32 with TF32, 672-pixel input) over 346 frames after 10 warm-up frames. [a]The values printed in the video (Figure 5.5). [b]Including image preprocessing and output transfer. n/m: not measured.

**Table 5.8.:** *Speed of the student variants and the released baseline on the walking clip (356 frames), separated into model inference and per-frame processing including preprocessing. Neither column includes landmark extraction or video input and output, so neither is an end-to-end speed.*

## 5.6. Qualitative results and video

### 5.6.1. Still image

Figure 5.4 shows three singers on a stage, reconstructed by the student and by the released baseline. Both models detect all 3 people; each body is placed at its predicted depth, with predicted fingers. Both models place a body with the hands on the hips over each singer; the released baseline's bodies appear slightly larger. The photograph has no ground truth, so it shows that the student operates on a real photograph from outside 3DPW, not that it is as accurate as the released baseline there.

### 5.6.2. Video

The walking clip has 356 frames (29.97 FPS) of one man walking across a square. Both models were run on the same frames with the procedure of Section 4.9, and five panels were rendered (Figure 5.5): the input, the raw student, the student with temporal filtering, the student with landmark alignment to the MediaPipe shoulder, elbow, wrist, hip, knee and ankle landmarks (Section 4.9.2), and the released baseline. Neither step retrains the student. The student kept a matched person in 356 of 356 frames and the released baseline in 354; these counts come from the tracker and are not a detection recall against labels.

Under visual inspection the raw student mesh sometimes extends beyond the person (in Figure 5.5 it is drawn ahead of the man); filtering makes its motion steadier, and alignment improves its position and apparent size. The motion statistics of the run agree with this. The mean pelvis-relative joint acceleration falls from 38.0 to 10.5 $\mathrm{m/s^2}$ with the filter (released baseline, unfiltered: 29.6 $\mathrm{m/s^2}$), the mean speed of the head anchor from 1.69 to 0.63 m/s (0.53 m/s after alignment), and the standard deviation of the predicted gender over the clip from 0.058 to 0.024. These are smoothness measures, not accuracy: a filter can also flatten real motion, and the clip has no ground truth. Only this one clip, with one person and no occlusion, was evaluated.

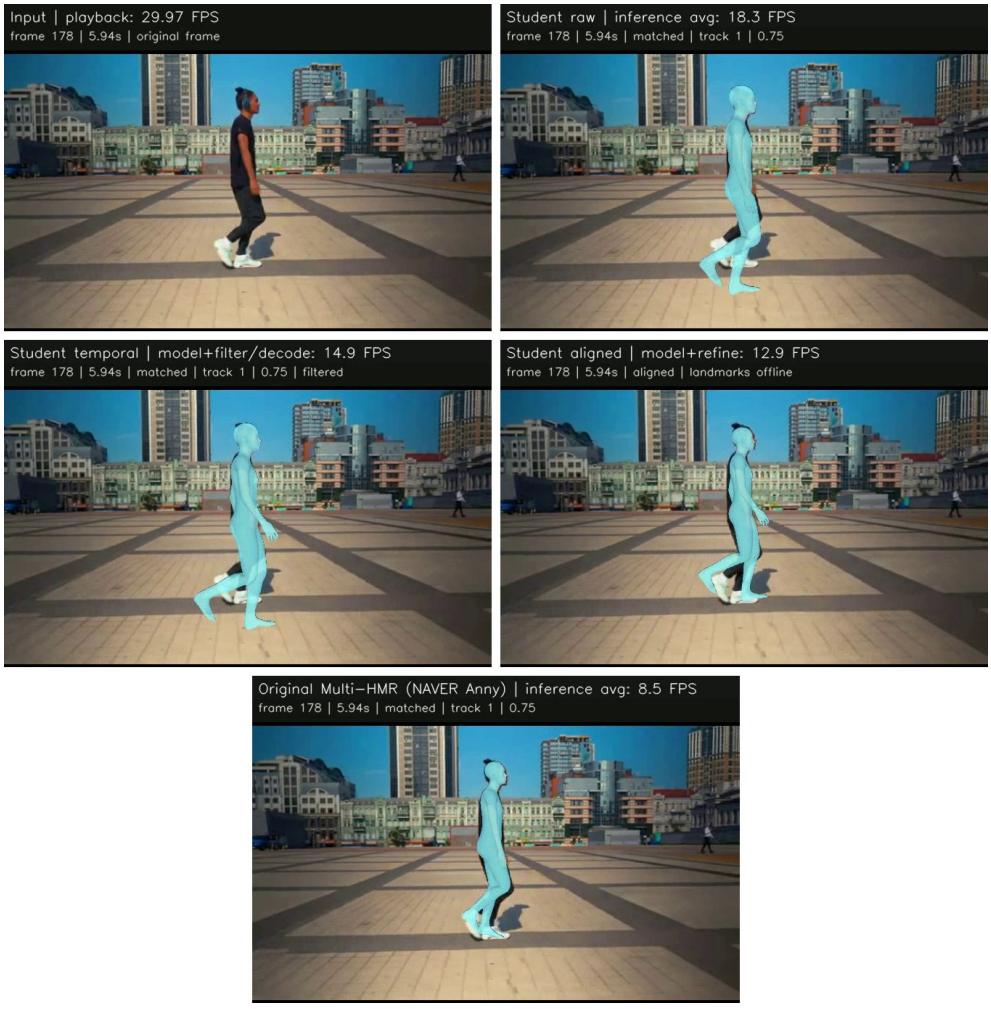



**Figure 5.5.:** *Frame 178 of 356 of the walking clip: input (top left), raw student (top right), student with temporal filtering ("Student temporal", middle left), student with landmark alignment ("Student aligned", middle right) and the released baseline ("Original Multi-HMR (NAVER Anny)", bottom). The FPS printed in each panel (playback for the input, model inference for the raw panels, model plus filter or alignment for the others) excludes preprocessing, landmark extraction and video input and output (Table 5.8).*

Table 5.8 separates model inference from per-frame processing. For model inference alone, the student runs at 18.3 and the released baseline at 8.5 FPS on the same NVIDIA A40 and frames (54.5 and 117.7 ms per frame), about 2.16 times faster; the heads also differ (8 against 2 cross-attention blocks). Including image preprocessing (10.1 ms) and output transfer, a frame takes 64.9 ms for the student (15.4 FPS) and 127.9 ms for the released baseline (7.8 FPS). The filter adds 12.7 ms and the alignment fit 10.2 ms; with both and preprocessing a frame takes 87.8 ms (11.4 FPS), and 9.3 FPS with rendering. Two parts of the complete pipeline were not timed: the offline landmark extraction (MediaPipe on the CPU, a separate pass over the video) and video decoding and encoding. These numbers therefore describe inference speed, not end-to-end speed, and no real-time claim is made for the complete pipeline. The A40 times do not compare with the L40S times of Table 5.2.

## 5.7. Analysis of the teacher

This section analyses the teacher, not the student, which is the deployed model (Section 5.5). The teacher differs from the uncorrected teacher only by the head-anchor supervision and the bone correction (Section 4.5). Pilot results and further details are in Appendix H.

### 5.7.1. Reconstruction on AnnyOne

On the holdout (Table 5.9a) the teacher has the lowest PA-PVE, 6.1 mm below the uncorrected teacher, and a 3.7 mm lower PVE, while the initial model has the lowest PVE. Recall is 2.0 points below the uncorrected teacher and 3.2 below the initial model, possibly because the detection cell moved from the pelvis to the head (untested). The PVE differences among these models are within run-to-run variation (Section 4.8). The recall loss of 2.0 points has the sign and size seen in every pilot with head-anchor targets (Section F.3) and is probably a cost of that correction. The two corrections thus do not raise the body error; PVE aligns the pelvis and cannot register the depth gain, and the neck and fingers are only a few vertices of the body (Section 6.2).

### 5.7.2. Real-image supervision

On 3DPW (Table 5.9b) the uncorrected teacher is already 30.3 mm PVE better than the initial model, so most of the gain is attributable to real-image training itself; the two corrections add 2.5 mm PVE and 2.1 mm PA-PVE. An earlier model trained on the uncorrected labels with a different schedule is 6.5 mm worse than the teacher (Table H.1), so the share of the label correction in this gain is unknown.

### 5.7.3. Camera-space placement

The uncorrected teacher places 99.1 % of the people too far from the camera (Figure 5.7); Figure D.1 shows the same error for the initial model and its

| | Initial model | Uncorrected teacher | Teacher |
|---|---|---|---|
| PVE (mm) | 70.5 | 77.4 | 73.7 |
| PA-PVE (mm) | 53.3 | 56.4 | 50.3 |
| Recall (%) | 90.2 | 89.0 | 87.0 |
| Root error (mm) | 1208 | 1402 | 728 |
| Depth slope | 0.33 | 0.28 | 0.42 |
| Depth intercept (m) | 2.36 | 2.66 | 1.75 |
| Size ratio | 0.63 | 0.59 | 0.78 |
| Neck girth pred/GT | 0.461 | 0.464 | 0.996 |
| $\mathrm{PVE}_{\mathrm{neck}}$ (mm) | 27.3 | 27.7 | 11.0 |
| Global head direction (deg) | 12.0 | 13.0 | 13.1 |
| Fingers (deg) | 17.3 | 17.4 | 8.3 |
| neck02 bone (deg) | 175.6 | 175.8 | 4.8 |
| head bone (deg) | 177.9 | 177.7 | 8.9 |

**(a)** *AnnyOne holdout (100 images, holdout protocol, matched people); placement metrics with each body at its model's predicted depth.*

| | Initial model | Uncorrected teacher | Teacher | Released baseline |
|---|---|---|---|---|
| PVE (mm) | 121.8 | 91.5 | 89.0 | 126.9 |
| PA-PVE (mm) | 72.0 | 59.7 | 57.6 | 68.8 |
| MPJPE (mm)[b] | 126.9 | 90.9 | 88.3 | 129.4 |
| Precision (%) | 54 | 62 | 64 | 75 |
| Recall (%) | 100 | 99 | 99 | 97 |
| Holdout PVE, 3DPW protocol[c] | 74.3 | 81.2 | 77.5 | (60.3)[a] |

[a]The released baseline masks the helper bones by design, so this value follows the holdout protocol and compares with the teacher's 73.7 in (a); it is potentially in-sample. [b]Mean over all 163 Anny joints, pelvis-centred. [c]The 100 images of (a) with the 3DPW protocol (helper bones predicted). Pilot columns: Table H.1.

**(b)** *3DPW test (corrected labels, all 8,190 images, 3DPW protocol), the same evaluation code for every model.*

**Table 5.9.:** *The teacher against the uncorrected teacher and the initial model on the AnnyOne holdout and the 3DPW test set. The PVE gap between the teacher and the uncorrected teacher lies within run-to-run variation (Section 4.8); the gaps in root error, neck and fingers do not.*

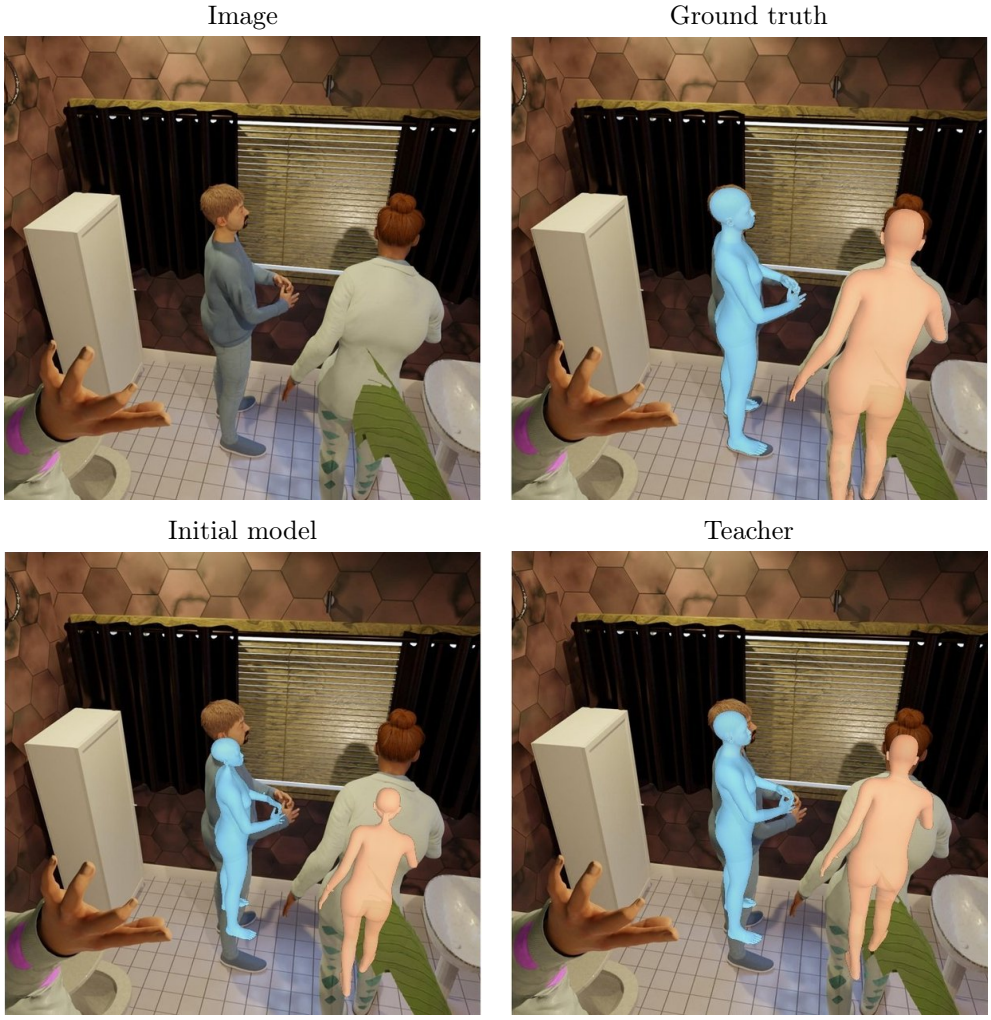



**Figure 5.6.:** *AnnyOne holdout image 579862 with the ground truth and the predictions of the initial model and the teacher, each body at its model's predicted depth. The initial model draws its bodies smaller and further back than the ground truth; the teacher's are close to its size and place.*

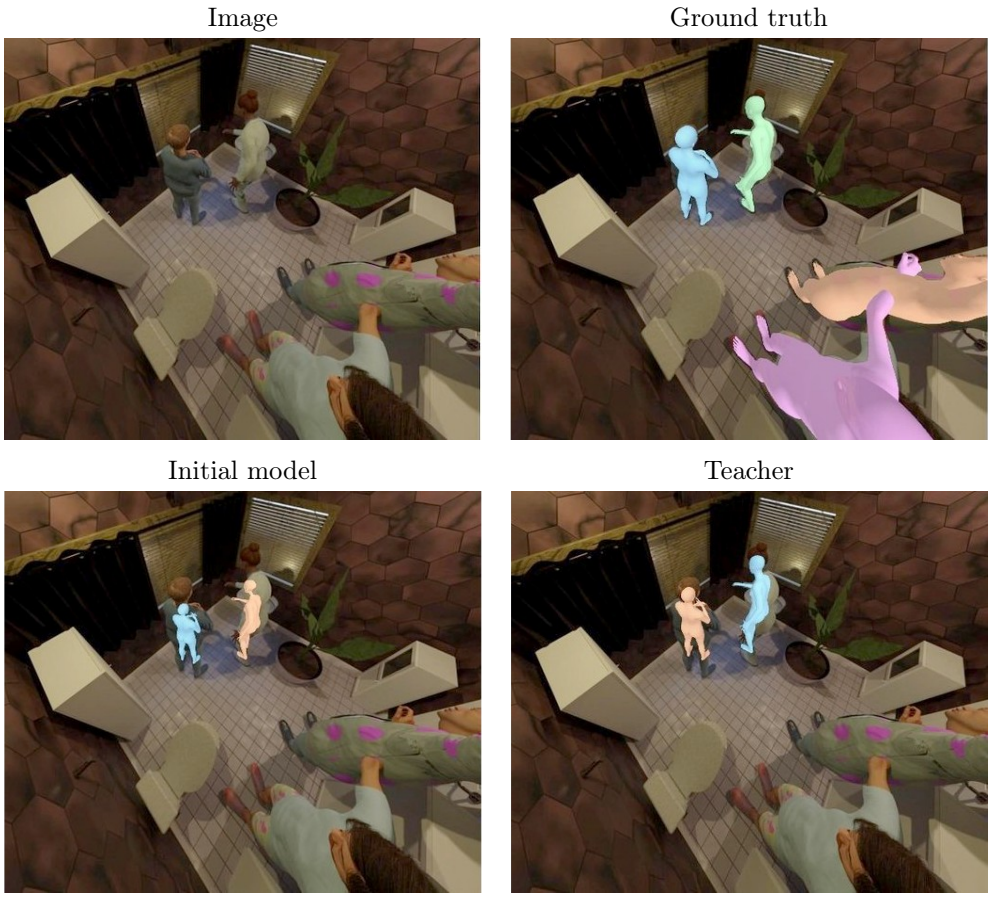



**Figure 5.6.:** *(continued) AnnyOne holdout image 579865, arranged as above, cropped and brightened; the two people lying in the foreground are missed by both models.*

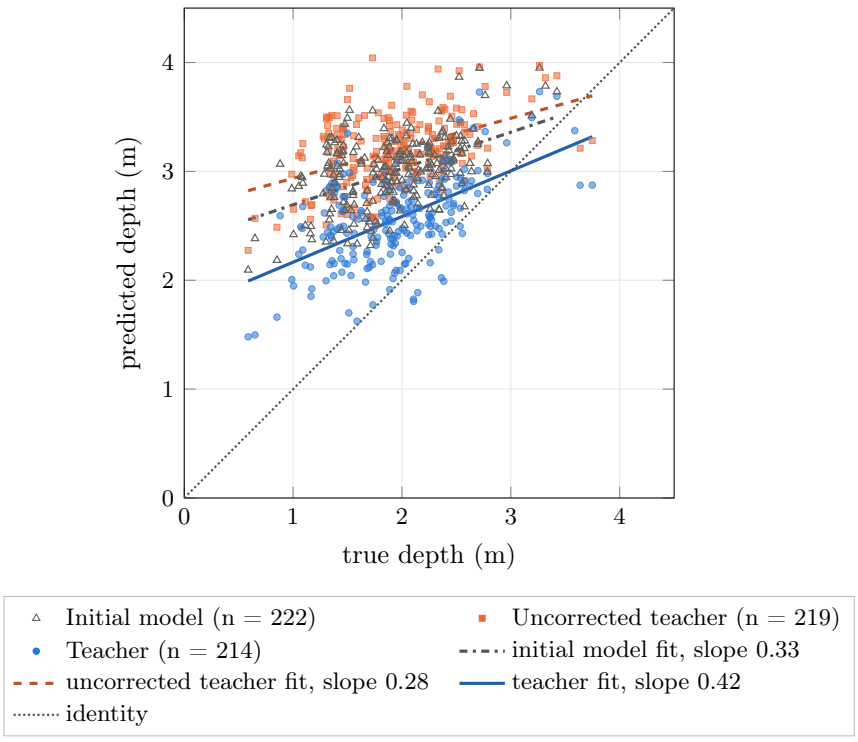



**Figure 5.7.:** *Predicted against true root depth for every matched person of the AnnyOne holdout, each body at its model's predicted depth, with least-squares fits and the identity line.*

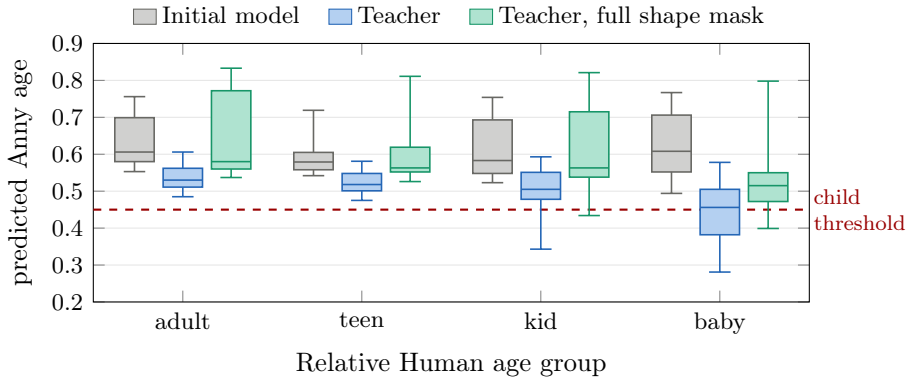



**Figure 5.8.:** *Predicted Anny age (phenotype value, not years) per Relative Human age group for the initial model, the teacher and the teacher with full shape mask, matched people of the validation split. Boxes show quartiles and median, whiskers the 5th and 95th percentiles; the dashed line marks the child threshold of 0.45.*

removal in a 10-epoch pilot with head-anchor targets (Appendix H). With the head-anchor targets the root error falls by 48 % (1,402 to 728 mm), the drawn size rises from 0.59 to 0.78 (Table 5.9a), and the mean 2D head error falls from 67.1 to 7.6 px; every run with these targets has a lower root error than every run without them (Table F.3). The corrections lower the intercept and raise the drawn size; the slope rises only from 0.28 to 0.42 and stays far from 1, so near people are still placed too far (94.4 % of all people). All depth numbers are measured on synthetic images (Appendices F.3 and H).

### 5.7.4. Bone correction

The teacher keeps the bone correction through ten epochs with real images (Table 5.9a): neck02 is 4.8 deg off against 175.8 for the uncorrected teacher, and the neck girth is 0.996 of the true girth against 0.464. The head direction, which worsened with the correction alone (19.1 deg), returns to the uncorrected teacher's level; fine-tuning thus preserves the correction, and PVE does not increase. The correction without training, a real-frame example and the open question of why the bones flip are in Appendix F.4.

### 5.7.5. Hand articulation

Predicted fingers are 17.3 deg off in the initial model and 17.4 deg in the uncorrected teacher, above the rest-pose error (15.1 deg; Table F.5). With the flipped bones metacarpal1.L and finger4-1.R corrected, the teacher reaches 8.3 deg, and the correction alone gave 8.2: about half of the finger error was due to two flipped bones, and the remainder is unexplained. On three real images the teacher's fingers appear flexed (Figure D.16).

| | Initial model | Teacher | Teacher, full mask | Teacher, partial mask |
|---|---|---|---|---|
| Recall (%) | 47.1 | 64.7 | 57.2 | 63.0 |
| Gender accuracy (%) | 46.5 | 64.2 | 49.3 | 47.4 |
| mean, men (male = 0) | 0.535 | 0.551 | 0.717 | 0.725 |
| mean, women (female = 1) | 0.551 | 0.861 | 0.860 | 0.888 |
| Child-vs-adult AUC | 0.606 | 0.719 | 0.708 | 0.727 |
| Mean age, adult | 0.631 | 0.536 | 0.638 | 0.532 |
| teen | 0.597 | 0.524 | 0.616 | 0.515 |
| kid | 0.613 | 0.500 | 0.611 | 0.506 |
| baby | 0.624 | 0.438 | 0.524 | 0.465 |
| Babies with age $< 0.45$ (%) | 2.6 | 49.3 | 16.9 | 34.3 |
| Mean height, adult | 0.507 | 0.500 | 0.505 | 0.527 |
| kid | 0.451 | 0.466 | 0.461 | 0.497 |
| baby | 0.394 | 0.464 | 0.478 | 0.504 |
| Matched children + adults | 948 | 1,305 | 1,157 | 1,270 |
| *Demo images: standard deviation of the prediction* | | | | |
| Muscle | – | 0.04 | 0.06 | n/m |
| Weight | – | 0.04 | 0.11 | n/m |
| Cup size | – | 0.05 | 0.11 | n/m |
| Firmness | – | 0.04 | 0.11 | n/m |
| Age | – | 0.03 | 0.08 | n/m |

Upper part: detected people matched to the labelled boxes at IoU $\geq 0.3$; ages and heights are Anny phenotype values in $[0, 1]$, and "child-shaped" means a predicted age below 0.45. Lower part: spread of the predicted phenotypes over all detected people of the 12 demo images; – not reported, n/m not measured. Masks as in Table 5.11.

**Table 5.10.:** *Body shape of the teacher variants on real photographs: Relative Human validation split (635 images, 2,130 labelled people) and the demo images. No value is a student result.*

### 5.7.6. Body shape on real photographs

On Relative Human (Table 5.10) the teacher assigns the correct gender to 64.2 % of the matched people, against 46.5 % for the initial model; the mean predicted gender is 0.861 for women and 0.551 for men (1 is female). The child-against-adult AUC, the probability that a random child receives a lower predicted age than a random adult, rises from 0.606 to 0.719, and 49.3 % of the babies are child-shaped against 2.6 %. In Figure 5.8 the teacher's median predicted age decreases from adults to babies, and only its baby box reaches below the child threshold; the initial model's median is nearly constant. A uniform shift cannot raise a rank measure, although part of the lower ages is likely a shift caused by the constant age of the converted 3DPW labels. The child–adult separation improves but remains unreliable, and the predicted gender of men stays near 0.551.

The gender AUC, which needs no cut-off, is 0.780 for the teacher against 0.562 (Table H.2, Appendix H); a likely reason for the intermediate gender of men is that the AnnyOne gender labels span only 0.38 to 0.82. On the AnnyOne holdout the teacher follows weight best (Pearson $r = 0.77$) and muscle worst (0.27), with every slope of prediction on label below 1 (Table H.3). The shape head can thus express shape on synthetic images, while on real photographs the teacher's bodies vary little (Figure D.17); since the initial model already shows part of this narrow spread, the gap between synthetic and real images is part of the cause, not the 3DPW training alone.

### 5.7.7. Shape-supervision masking

Two masked teachers test the shape mask of Section 4.4 (Table 5.11). The full mask removes most of the teacher's gain on Relative Human (gender 49.3 %, babies child-shaped 16.9 %) but raises the weight spread on the demo images from 0.04 to 0.11; the partial mask preserves the child-adult separation (AUC 0.727). Under both masks the mean predicted gender of both sexes exceeds 0.5, although the AUCs still rank women above men, and the body errors stay close to the teacher's. A replacement rule fixed before the first masked run required gender accuracy of at least 60 %, a child-against-adult AUC of at least 0.7 and a muscle and weight spread above 0.08. Both masked models fail on gender and pass on the AUC, and the full mask also fails the muscle-spread threshold (0.06), so the teacher without shape mask was retained as the teacher of the student (Appendix H).

| | Teacher | Teacher, full mask | Teacher, partial mask | Rule |
|---|---|---|---|---|
| Masked 3DPW phenotypes | none | 8 (age + 7) | 7 (age kept) | |
| *AnnyOne holdout* | | | | |
| PVE (mm) | 73.7 | 71.7 | 73.0 | |
| PA-PVE (mm) | 50.3 | 50.7 | 51.2 | |
| Recall (%) | 87.0 | 86.2 | 87.4 | |
| Root error (mm) | 728 | 682 | 680 | |
| *3DPW test* | | | | |
| PVE (mm) | 89.0 | 91.1 | 89.9 | |
| PA-PVE (mm) | 57.6 | 59.0 | 58.9 | |
| *Relative Human validation* | | | | |
| Recall (%) | 64.7 | 57.2 | 63.0 | |
| Gender accuracy (%) | 64.2 | 49.3 | 47.4 | $\geq 60$ |
| Child-vs-adult AUC | 0.719 | 0.708 | 0.727 | $\geq 0.7$ |
| Babies with age $< 0.45$ (%) | 49.3 | 16.9 | 34.3 | |
| *Demo images, standard deviation of the prediction* | | | | |
| Muscle | 0.04 | 0.06 | n/m | $> 0.08$ |
| Weight | 0.04 | 0.11 | n/m | $> 0.08$ |

Full mask: the 8 constants of the converted labels (age 0.55, seven values at 0.5); partial mask: the seven at 0.5; gender stays supervised in all three. Relative Human: matched people (Table 5.10); demo: 12 images, not matched to people; n/m: not measured. Rule: Section 5.7.7.

**Table 5.11.:** *The teacher without, with full and with partial shape mask on the AnnyOne holdout, the 3DPW test set, Relative Human and the demo images. Both masked models fail the gender threshold of the replacement rule.*

# 6. Discussion

This chapter answers the research questions (Section 1.2), interprets the metrics and states the limitations.

## 6.1. Answers to the research questions

**Real-image supervision.** The converted 3DPW labels, whose joints lie a median 14.1 mm from those of 3DPW, enable training and testing Anny models on real images, but 8 phenotypes are constant and head and hands remain at rest (Section 3.7). Most of the teacher's real-image gain comes from these labels: the teacher without corrections (uncorrected teacher) reaches 91.5 mm PVE on converted 3DPW against 121.8 mm for the initial model, the teacher 89.0 mm (Section 5.7.2). On real photographs the teacher separates children from adults better than the initial model (AUC 0.606 to 0.719) and ranks women above men (gender AUC 0.780), but men receive mid-range gender values, probably because the AnnyOne labels contain no fully male body (Sections 5.7.6 and 5.7.7).

**Compact reconstruction.** The student has 3.14 times fewer parameters and 3.19 times fewer FLOPs than its teacher and ran 2.10 times faster. Its PVE is 119.6 against 73.7 mm on the AnnyOne holdout and 110.1 against 89.0 mm on converted 3DPW, missing the pre-set 95 % retention thresholds on every error metric (Section 5.2.2). Its 3DPW PVE is lower than the released baseline's (126.9 mm) and its PA-PVE slightly higher (71.8 against 68.8 mm); this compares trained systems, not architectures (Section 5.2.1).

The teacher improves the compact model: a ViT-B model trained without it has a 3DPW test PVE 6.6 mm and an MPJPE 11.6 mm higher than a repeat run of the student (Section 5.3). This gain is obtained by head transfer alone; in the configuration tested, the distillation losses add no measurable further effect, although the feature loss brings the student's features closer to the teacher's. The answer to this question is therefore threefold: a model with 3.14 times fewer parameters was obtained; its gap to the teacher remains, outside the retention thresholds; and its advantage over a ViT-B model without teacher is obtained through the transferred heads, with no measurable further gain from the distillation losses in this configuration.

Training length and temperature were fixed in advance, and the sensitivity runs, judged on the 3DPW validation images, changed no setting (Section 5.1). Doubling the training length lowered the 3DPW validation PVE only from 87.7 to 86.0 mm and the holdout PVE to 107.7 mm, without improving the 3DPW test PVE (110.9 mm; Section 5.4.1). Across the five temperatures the 3DPW

validation PVE differed by at most 2.6 mm, against 1.5 mm between two runs with the same temperature, and the last checkpoint was at most 1.4 mm above the best epoch (Section 5.4.2). Untested causes of the gap are the smaller backbone, the 3DPW share per batch and the teacher's 3DPW error.

**Multi-person detection.** For the teacher, recall on Relative Human falls with the number of people per image: 76.2 % and 78.5 % with one and two labelled people, 67.8 % for three or four, 56.6 % for five to eight and 45.3 % for nine or more (Section 5.5).

**Video.** On the walking clip the student kept a tracked body in 356 of 356 frames and the released baseline in 354 (tracker matches, not detection recall; Section 5.6). Temporal filtering visibly steadied the motion and landmark alignment improved position and apparent size; without ground truth this shows steadier output, not better 3D accuracy. On one NVIDIA A40, model inference ran at 18.3 FPS for the student and 8.5 FPS for the released baseline, and the student with preprocessing, filtering and alignment at 11.4 FPS, excluding offline landmark extraction, rendering and video input and output; these are not end-to-end speeds.

## 6.2. Interpreting geometry and placement

PVE aligns the pelvises before measuring vertex distances and is therefore insensitive to placement: with the two corrections the root error fell from 1,402 to 728 mm, while the PVE changed only from 77.4 to 73.7 mm (Section 5.7.3). Local changes are diluted likewise; the neck-skin PVE fell from 27.7 to 11.0 mm, but the 171 neck vertices account for an estimated 0.2 mm of whole-body PVE, so the bone correction was judged by bone angles and neck geometry (Appendix F). Detection, accuracy, stability and speed are therefore reported separately.

## 6.3. Limitations

- **Evaluation data.** The synthetic test set is the last 100 AnnyOne images, without children, which also served as the in-training monitor; it supports no broad generalisation claim. The 3DPW labels are converted, and the 3DPW errors are not comparable with published results (Section 3.7).
- **Baseline fairness.** No baseline was trained on the same data, so a lower PVE does not show a better architecture (Section 5.2.1).
- **Distillation ablation.** Each ablation model was trained once. The head copy, its warm-up freeze and the teacher's bone correction in training were not separated from each other. The model without teacher was trained with the schedule and learning rate chosen for transferred heads,

which may disadvantage freshly initialised heads. No other loss weights or feature layers were tried for distillation, so the finding concerns this configuration, not distillation in general (Section 5.3).

- **Single runs.** Each model was trained once without a fixed seed (Section 4.8).
- **Shape and crowd recall.** Shape and crowd recall on real photographs were evaluated on the teacher only; the student, the deployed model, was not evaluated on Relative Human. The child–adult separation improves but remains unreliable, and the predicted gender of men stays near 0.551 (Section 5.7.6).
- **Depth.** The teacher's predicted depth grows by only about 0.42 m per metre of true depth, and a single image leaves size and distance ambiguous (Section 5.7.3).
- **Multi-person and video evaluation.** Multi-person evaluation is limited to detection recall, with no occlusion or crowd benchmark with 3D labels; the video is one clip of one person.
- **Timing.** The forward-pass benchmark and the video timings were measured differently, and latency depends on hardware, resolution and the number of people; no real-time claim is made.
- **Reproducibility.** The cluster scripts contain site-specific paths, and the terms of use of datasets, body model and checkpoints were not checked for a release of weights or labels.

# 7. Conclusion and future work

## 7.1. Conclusion

This thesis addressed multi-person whole-body mesh recovery from a single image in the Anny body model at reduced inference cost, for which no real images with Anny labels existed. The solution has four parts: 3DPW's SMPL labels were converted to Anny by joint-based fitting; a ViT-L teacher was trained on AnnyOne and converted 3DPW with corrected head-anchor targets and a fixed rotation correction of four output bones; a ViT-B student inherited the teacher's heads through a learned projector and was distilled from its features and outputs; and inference-time temporal filtering and landmark alignment were added for video.

The student has 3.14 times fewer parameters than its teacher and ran 2.10 times faster, at a higher error: 110.1 against 89.0 mm PVE on converted 3DPW, outside the 95 % retention thresholds set in advance. Its 3DPW PVE is lower than that of the released baseline, trained on different data, and neither longer training nor another distillation temperature closed the gap. The teacher improves the compact model: a ViT-B model trained without it was clearly less accurate. In the configuration tested, this gain was obtained through the transferred heads, and the distillation losses added no measurable further effect. For the teacher, detection recall on real photographs falls as the number of people per image grows. On video, filtering and alignment produced visibly steadier and better placed output.

Unresolved are the cause of the accuracy gap, the separate roles of the head copy, its freeze and the bone correction, the student's recall and shape on real photographs (not evaluated), the teacher's shape for children and men, the compression of predicted depth and a learned temporal model. The evaluation is bounded by a small synthetic holdout, single runs and the absence of a fairly trained baseline (Section 6.3).

## 7.2. Future work

**Temporal evaluation and modelling.** Running the filter and the alignment on the 3DPW test sequences, whose converted labels exist at 10 frames per second, would measure steadiness and per-frame PVE together; clips with occlusion, a moving camera or several people, timed end to end, would follow. A learned temporal model (Section 2.7) remains open.

**Accuracy at reduced cost.** Separating the head transfer from its warm-up freeze and from the teacher's bone correction would show which part carries

the teacher's benefit. Other distillation targets, such as other feature layers or loss weights, and more real images per batch could narrow the gap. A released baseline fine-tuned on the same converted 3DPW data would separate the effect of architecture and distillation from that of the training data.

**Real-image shape.** Age-group labels, body measurements or further Anny fits on real photographs could replace the constant phenotypes. Tests with children should separate the representability of child proportions in Anny from their recognition by the network.

**Reliability and reproducibility.** Repeated runs, an evaluation of the student on Relative Human and a larger, randomly drawn holdout that includes children would show which differences are real and how far the results generalise.

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# A. Model names, runs and reproduction

The main text refers to every model by a descriptive name, such as the teacher, the student or the initial model. This appendix collects the run identifiers behind these names, together with the Slurm jobs, logs, checkpoints and files that produced each number.

Every number can be traced in two steps. Each headline number is a macro in one of the files `numbers_*.tex` of the thesis source, and the comment next to it names the job and the line of its log, for example `3611336:191`. A job is one unit of work run by Slurm, the scheduler of the cluster, and the job id is the number Slurm assigned to it. Table A.1 maps every descriptive name to its run identifier and to the run name recorded during the project; Table A.2 maps each job to the model, the log and the checkpoint. Job ids and recorded run names appear only in this appendix.

## A.1. Model names

The ViT-L line ("L") comprises every model that keeps the ViT-L backbone. L-stage 1 to 4 are the steps of the transfer schedule that turned the released Multi-HMR-Anny checkpoint (the released baseline) into the initial model (Base), which is the end of that schedule and the start of every later fine-tune. The teacher (Final) and the teacher without corrections (Control) are the two arms of the last teacher experiment (exp7): the same 10-epoch fine-tune of Base, with and without the two corrections. The number after "Student-S/" is the old version number of a ViT-S student; the student (Student) is the ViT-B model distilled from the teacher. The 60-epoch student (Student-60) is the student's training configuration trained twice as long (Section 5.4.1); it is not the reported model. The sweep of Section 5.4.2 trains the same configuration at 5 temperatures. The two arms of the distillation ablation (Section 5.3), the student without distillation losses (Student-noKD) and the ViT-B model without teacher (ViT-B-noTeacher), are the last rows of the table. Two evaluation protocols are used (Section 4.8): the holdout protocol, with the helper bones set to rest pose before decoding, and the 3DPW protocol, with the helper bones scored as predicted. File and log names use V1 for the holdout protocol and V0 for the 3DPW protocol.

In Table A.1, "ep." is the epoch kept. Backbone: L, B, S = DINOv2 ViT-L, ViT-B, ViT-S with 14-pixel patches; "all" = every block trained, "last 4" = the last 4 of 24 blocks, "+p" = a linear projector to the heads' width 1024. Data: A = AnnyOne; A+P = 70 % AnnyOne and 30 % 3DPW train, with

the old labels (before their correction, Chapter 3) or the corrected v23 labels. Schedule: epochs × iterations per epoch × batch, then the learning rate, halved every 30 epochs unless noted (planned epochs where the run stopped early). Checkpoints lie in `logs/anny_model/<run>/checkpoints/` with copies in `models/multiHMR/`; Table A.2 gives the path of every model with a number in the thesis, and Appendix C the full flags of Final, Control and the student. "Not recorded" means that no script, log or document in the repository states the value.

**Table A.1.:** *Descriptive name used in the main text, run identifier (bold) and recorded run name of every model, with starting weights, backbone, training data and schedule. Key in Section A.1.*

| Main-text name, **identifier** | Recorded run / checkpoint name | Starting weights | Backb. | Data | Schedule; what it is |
|---|---|---|---|---|---|
| *Released starting point and baseline* | | | | | |
| the released baseline **Multi-HMR-Anny** | `multiHMR_672_L_anny` (released checkpoint) | trained by NAVER | L, own decoder | A, BEDLAM, COCO, MPII | not recorded; evaluated with its own model class |
| *ViT-L line ("L"): the transfer schedule and the initial model* | | | | | |
| earlier model **L-stage 1** | `anny_s1_pretrained_bb_v2` | Multi-HMR-Anny backbone, fresh heads | L, all | A | 150 × 1,000 × 4, $10^{-4}$, from epoch 177 at $3 \cdot 10^{-5}$; side branch, no checkpoint kept |
| earlier model **L-stage 2** | `anny_s2_frozen_bb_v2`, ep. 299 ("v2") | Multi-HMR-Anny backbone, fresh heads | L, frozen | A | 150 × 1,000 × 4, $10^{-4}$, from epoch 283 at $3 \cdot 10^{-5}$ |
| earlier model **L-stage 3** | `anny_s2_partialft_v3`, ep. 54 ("v3") | L-stage 2, epoch 299 | L, last 4 | A | 300 × 1,000 × 4, $10^{-5}$; helper bones held at rest |
| earlier model **L-stage 4** | `anny_s2_boosted_v4`, ep. 299 ("v4") | L-stage 3, epoch 54 | L, last 4 | A | 300 × 1,000 × 4, $10^{-5}$; neck and hand bones weighted ×4 |
| the initial model **Base** | `anny_s2_shape_v5`, ep. 99 ("v5") | L-stage 4, epoch 299 | L, last 4 | A | 100 × 1,000 × 4, $10^{-5}$; shape weight 20; AnnyOne only, start of every later fine-tune |
| earlier model **Mixed-L** | `anny_teacher_v9_mixed`, ep. 62 and 149 ("v9") | Base | L, last 4 | A+P, old | 150 × 1,000 × 4, $10^{-5}$, halved every 60 |
| earlier model **Fresh-heads-L** | `anny_v10_scratch_mixed`, ep. 290 ("v10", not from scratch) | Multi-HMR-Anny backbone, fresh heads | L, last 4 | A+P, old | 300 × 1,000 × 4, $10^{-5}$; 3DPW from the first step, no neck or hand weight |
| *Teacher experiment (exp7): two arms and two shape-mask variants* | | | | | |
| the teacher without corrections (uncorrected teacher) **Control** | exp7 `v5_root_dist1_nofix_v23_e10`, ep. 9; job 3611339 | Base | L, last 4 | A+P, v23 | 10 × 1,000 × 4, $10^{-5}$; 3DPW hands masked, pelvis targets |
| the teacher **Final** | exp7 `v5_pc_dist5_bonefix_v23_e10`, ep. 9; job 3611336 | Base and its bone correction | L, last 4 | A+P, v23 | as Control, with head targets, depth and translation weights 5, bone correction |

| Main-text name, **identifier** | Recorded run / checkpoint name | Starting weights | Backb. | Data | Schedule; what it is |
|---|---|---|---|---|---|
| the teacher with full shape mask **Final-masked** | exp7 `v5_pc_dist5_bonefix_v23_e10_shpmask`, ep. 9; job 3612325 | as Final | L, last 4 | A+P, v23 | as Final, 8 converter constants masked for 3DPW people |
| the teacher with partial shape mask **Final-partial-mask** | exp7 `v5_pc_dist5_bonefix_v23_e10_shpmask_noage`, ep. 9; job 3614042 | as Final | L, last 4 | A+P, v23 | as Final, the 7 constants other than age masked; not adopted |
| *ViT-S students ("Student-S"), earlier models; teacher Base unless noted* | | | | | |
| earlier model **Student-S/1** | distillation v1, `anny_distill_vits` | ImageNet DINOv2-S, random heads | S, all | A | 300 × 1,000 × 4, $10^{-4}$; feature distillation |
| earlier model **Student-S/2** | v2, `anny_distill_vits_v2_a` | Student-S/1, epoch 199 | S, all | A | 200 × 1,000 × 4, $2 \cdot 10^{-5}$, halved every 60 |
| earlier model **Student-S/3** | v3, `anny_distill_vits_v3` | ImageNet DINOv2-S; 54 of 67 head tensors from Base | S, all | A | 150 × 1,000 × 4, $5 \cdot 10^{-5}$, halved every 60; output distillation |
| earlier model **Student-S/4** | v4, `anny_distill_vits_v4` | NAVER's `multiHMR_672_S` backbone; heads as /3 | S, all | A | 150 × 1,000 × 8, $7 \cdot 10^{-5}$, halved every 60 |
| earlier model **Student-S/5** | v5, `anny_distill_vits_v5` | as Student-S/4 | S, all | A | as /4; heads frozen for 5 epochs |
| earlier model **Student-S/6** | v6, `anny_distill_vits_v6`, ep. 137 | NAVER's ViT-S backbone; all 67 head tensors from Base | S+p | A | as /4; stopped at epoch 138 by the time limit |
| earlier model **Student-S/6b** | v6b, `anny_distill_vits_v6b`, ep. 140 | Student-S/6, epoch 137 | S+p | A | 150 × 1,000 × 8, $2 \cdot 10^{-5}$, halved every 60; teacher in bfloat16 |
| earlier model **Student-S/6c** | v6c, `anny_distill_vits_v6c`, ep. 149 | Student-S/6b, epoch 140 | S+p | A | 150 × 1,000 × 16, $3 \cdot 10^{-5}$, halved every 60 |
| earlier model **Student-S/mixed** | v7, `anny_distill_vits_v7_mixed`, ep. 130 | Student-S/6b, epoch 140 | S+p | A+P, old | 150 × 1,000 × 8, $2 \cdot 10^{-5}$, halved every 60 |
| earlier model **Student-S/final** | v8, `anny_distill_vits_v8_final`, ep. 52 | Student-S/6c, epoch 149; teacher Mixed-L | S+p | A+P, old | 150 × 1,000 × 16, $3 \cdot 10^{-5}$, halved every 60 |
| *ViT-B student and its sensitivity runs* | | | | | |
| the student **Student** | exp8 `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30`; job 3614191 (done 19 Oct) | NAVER's `multiHMR_672_B` backbone; every head tensor from Final; teacher Final | B+p, all | A+P, v23 | 30 × 1,000 × 8, $3 \cdot 10^{-5}$, halved every 12; Final's targets and masks; holdout 119.6, 3DPW 110.1 mm, 95 % retention thresholds not met; the reported model |

| Main-text name, **identifier** | Recorded run / checkpoint name | Starting weights | Backb. | Data | Schedule; what it is |
|---|---|---|---|---|---|
| the 60-epoch student **Student-60** | exp8 `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e60`, ep. 59; job 3623342 | as Student | B+p, all | A+P, v23 | as Student, but 60 epochs, halved every 24; holdout 107.7, 3DPW 110.9 mm, 95 % retention thresholds not met; training-length analysis, not the reported model |
| the student at temperature $t$ **Student,** $T = t$ | exp8 (submitted by exp9) `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30_kdT<t>_sweep`; jobs in Table A.2 | as Student | B+p, all | A+P, v23 | as Student, with feature-distillation temperature $t \in \{1, 2, 4, 8, 16\}$; evaluation after every epoch and validation loss on (both only measure); temperature analysis |
| *Distillation ablation* | | | | | |
| the student without distillation losses **Student-noKD** | exp8 `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30_nokd`, ep. 29; job 3744297 | as Student (teacher heads copied) | B+p, all | A+P, v23 | as Student, with both distillation losses off; holdout 118.2, 3DPW 110.0 mm, 95 % retention thresholds not met; arm (a), results in Table 5.4 |
| the ViT-B model without teacher **ViT-B-noTeacher** | exp8 `student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30_supervised`, ep. 29; job 3744307 | NAVER's `multiHMR_672_B` backbone, fresh heads | B+p, all | A+P, v23 | as Student, with no teacher: no distillation losses, no head copy or freeze, no bone correction in training; holdout 122.6, 3DPW 116.7 mm, 95 % retention thresholds not met; arm (b), results in Table 5.4 |

## A.2. Environment

All training, evaluation and demo jobs ran on the DFKI Pegasus Slurm cluster, from the repository clone in `/netscratch/najib/multi-hmr`. The repository is public on GitHub (`saadnajib/Multi-Person-Whole-Body-Human-Mesh-Recovery-in-a-Single-Shot-with-Annyone-dataset-and-3dpw-dataset`). Checkpoints, logs and datasets are not in it. Each job script prints the short git hash of the clone into the run's `settings.txt`, so a run can be matched to a commit.

Every job runs inside the container image `ubuntu+20.04_v4.sqsh` (in `/netscratch/najib/ContainerImages/`), started by `srun` with the option `--container-image` and the mounts `/fscratch`, `/netscratch`, `/home/najib` and `/dev/shm`. Inside it, `slurm/env.sh` is sourced first. It activates the conda environment `multihmr` (Python 3.9, PyTorch 2.0.1 with CUDA 11.7, xformers 0.0.20, torchvision 0.15.2) and moves the Hugging Face, Torch and

Anny caches to `/netscratch/najib`. It also sets one thread per data-loader worker and strips the `X | None` type hints from the downloaded DINOv2 hub code, because Python 3.9 rejects them. `slurm/egl.sh` then installs the EGL libraries for the headless renderer. The video launcher also tests a real render and stops before loading a checkpoint if that test fails.

The landmark detector of the video pipeline, MediaPipe 0.10.14, runs in its own Python 3.10 environment, with the packages pinned in `requirements-video-landmarks.txt`. There, `scripts/video/extract_pose_landmarks.py` reads the video once on the CPU and writes a JSON file before the GPU job starts. `docs/video_inference.md` creates this environment at `/netscratch/najib/venvs/video-landmarks`.

That PyTorch build has GPU kernels up to compute capability 8.6. For this reason the training and evaluation scripts ask for A100, A40, L40S and RTX A6000 nodes and refuse an H100 or H200 (compute capability 9.0). The V100 nodes `gifu` and `glendale` are excluded: every converter task sent there failed within a minute with an argument-parser error whose cause was not identified. Only the 3DPW converter ran on H100 nodes, through a small virtual environment with PyTorch 2.1.2 for CUDA 12.1 on top of `multihmr` (`PY_VENV=/netscratch/najib/venv_cu121`). Steps that need only NumPy (building the 3DPW label files, the re-fit check) run on the login node. The model-cost script needs a GPU node, because `model.py` imports the renderer and the login node has no EGL.

## A.3. Runs, jobs, logs and checkpoints

Table A.2 lists every job behind a number, a table or a figure of the thesis. Paths are relative to the project folder on the cluster. The log of job $j$ is `logs/<logname>_`$j$`.out` (array jobs: `_`$j$`_<task>.out`; the video and image jobs: `logs/<logname>-`$j$`.out`, with a hyphen). `E7` stands for `results/experiments/exp7_final_model/`, `M` for `models/multiHMR/`.

Of the 5 temperature-sweep jobs (submitted 6 November), four completed; the $T = 16$ job (3717598) trained to the end, but its post-training evaluation failed on a network time-out, so it has no final table. The final-epoch validation values of all five runs, including the $T = 16$ run, were read from the training logs by `scripts/analysis/sweep_validation_table.py` (Table 5.5). The two runs of the distillation ablation (jobs 3744297 and 3744307) were submitted together on 12 November by `slurm/experiments/exp10_kd_ablation.sh` (record `logs/exp10_kd_ablation_20261112_225014.txt`, commit `7b7edf1`) and completed after about six hours each; their results are in Table 5.4. Their inference times are not reported, since they share the student's architecture and the second job was timed under a different load.

**Table A.2.:** *Slurm jobs behind the numbers, tables and figures of the thesis, with log name and checkpoint or output. Run identifiers as in Table A.1; path conventions in Section A.3.*

| Model or item: what the job did | Job(s) | Log name; checkpoint or output |
|---|---|---|
| *Models* | | |
| Base: holdout trace (PVE in the holdout protocol, placement) | 3570222; re-trace 3571596 | `diag_holdout`; `logs/anny_model/anny_s2_shape_v5/checkpoints/00099.pt` |
| Mixed-L: holdout trace | 3571963 | `diag_holdout`; `M/anny_teacher_v9_mixed_00062.pt` |
| Fresh-heads-L: holdout trace | 3571794 | `diag_holdout`; `logs/anny_model/anny_v10_scratch_mixed/checkpoints/00290.pt` |
| Final: training, holdout trace, 3DPW test | 3611336 (smoke test 3611333) | `exp7_final_model`; `E7/v5_pc_dist5_bonefix_v23_e10/train/checkpoints/00009.pt`, copy `M/exp7_v5_final_00009.pt` |
| Control: the same | 3611339 | `exp7_final_model`; `E7/v5_root_dist1_nofix_v23_e10/` |
| Final-masked: the same | 3612325 | `exp7_final_model`; `E7/v5_pc_dist5_bonefix_v23_e10_shpmask/` |
| Final-partial-mask: the same (done 19 Oct) | 3614042 | `exp7_final_model` (holdout table lines 191–215); `E7/v5_pc_dist5_bonefix_v23_e10_shpmask_noage/` |
| Student: distillation from Final, 30 epochs, own evaluation (done 19 Oct) | 3614191 (smoke test 3613926) | `exp8_distill_vitb`; `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30/train/checkpoints/00029.pt`; `final_table.json` in the same run folder |
| Multi-HMR-Anny: holdout and 3DPW test with the evaluation code of this thesis | 3614192 (3613928 crashed: FFN width) | `eval-test3dpw` (task 0); `M/multiHMR_672_L_anny.pt` |
| Model cost: parameters, GFLOPs, latency of Multi-HMR-Anny and Final (A100) | 3614947 | `model_cost`; `results/diagnostics/model_cost_naver_vs_final.json` |
| *Training length, distillation temperature and distillation ablation* | | |
| Student-60: the same training configuration trained twice as long, own evaluation | 3623342 | `exp8_distill_vitb`; `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e60/train/checkpoints/00059.pt`; `final_table.json` in the same run folder |
| Loss curves of the 30- and 60-epoch runs, from their tensorboard files (Figure 5.2) | none (login node) | `scripts/analysis/plot_loss_curves.py`; `results/experiments/exp9_kd_temperature/loss_curves/<run>_loss_curves.png`, `.pdf`, `.csv`; thesis figure drawn from the CSV copies `data/loss_curves_student_e30.csv`, `_e60.csv` by `figures/scripts/mk_f23_loss_curves.py` |
| Temperature sweep, 5 runs (submitted 6 Nov; $T = 1, 2, 4, 8$ finished; $T = 16$ trained to the end, but its post-training evaluation failed on a network time-out) | 3717507, 3717530, 3717538, 3717581, 3717598 ($T = 1, 2, 4, 8, 16$) | `exp8_distill_vitb` (one job per temperature, submitted by `slurm/experiments/exp9_kd_temperature.sh`; record `logs/exp9_kd_temperature_<date>.txt`); collector `scripts/analysis/kd_temperature_sweep.py` writes `results/experiments/exp9_kd_temperature/kd_temperature_sweep.csv`, `.png`, `.pdf`; final-epoch validation values of all five runs (Table 5.5) from `scripts/analysis/sweep_validation_table.py`, `results/experiments/exp10_kd_ablation/validation/sweep_validation_table.csv`, thesis copy `data/sweep_validation_table.csv` |

| Model or item: what the job did | Job(s) | Log name; checkpoint or output |
|---|---|---|
| Distillation ablation, the student without distillation losses (Student-noKD): both distillation losses off, teacher heads copied (submitted 12 Nov, 22:50, git 7b7edf1; completed) | 3744297 | `exp8_distill_vitb`, submitted by `slurm/experiments/exp10_kd_ablation.sh` (record `logs/exp10_kd_ablation_20261112_225014.txt`); `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30_nokd/`: `final_table.json`, `bone_fix.json`; collector `scripts/analysis/kd_ablation_table.py` writes `results/experiments/exp10_kd_ablation/kd_ablation_table.csv`; thesis copy `data/kd_ablation_table.csv` |
| Distillation ablation, the ViT-B model without teacher (ViT-B-noTeacher): same data and schedule, no teacher (submitted 12 Nov, 22:50, git 7b7edf1; completed) | 3744307 | as the row above, run folder `..._e30_supervised/` |
| *Evaluations and demos* | | |
| 3DPW test (v23): Base, Mixed-L, neck pilot (Final and Control in their own jobs) | 3612073, 3612074, 3612075 | `eval-test3dpw` (task 0), lines 117–125 |
| 3DPW test (v23): Student-S/6c, Student-S/final, also AnnyOne test-100 | 3615020_2, 3615020_4 | `eval-test3dpw`, line 119 (_2) and 118 (_4) |
| Relative Human: Base, neck pilot, Final, Final-masked, Final-partial-mask | 3611285_0, _1, 3612131, 3613616, 3616821 | `eval_rh`; `results/diagnostics/relative_human_val_<tag>/` |
| Demos: neck pilot, Final, Final-masked, each next to Base | 3581312, 3612078, 3613617 | `demo-compare`; `demo_montage_photos/`, `demo_montage_3dpwtest/` |
| Demos: predicted fingers (Base, Final); rest pose (Base, Final, Final-masked) | 3615018, 3615019 | `demo-compare`; scripts `slurm/demo/run_demo_compare_hands.sh`, `_rest.sh` (copies made on the cluster) |
| Video, raw Student and Multi-HMR-Anny on `man_walking.mp4` (frame and match counts) | 3642415 | `video-compare` (log `logs/video-compare-3642415.out`); `results/video_compare/video_3642415/comparison.mp4` |
| Video, five panels with temporal filter and landmark alignment (Figure 5.5) | 3657550 | `video-compare`; `slurm/demo/run_video_compare.sh` with `TEMPORAL=ema`, `ALIGNMENT=landmarks` and `LANDMARKS=<json>`; landmarks from `scripts/video/extract_pose_landmarks.py` (separate MediaPipe 0.10.14 environment); `results/video_compare/video_3657550/`: `comparison.mp4`, `timing.csv`, `student_temporal/`, `student_aligned/alignment.json`; thesis copy `figures/img/F25_video_panels.jpg` (frame 178 of the video in the final presentation) |
| Stage photo, Student and Multi-HMR-Anny (Figure 5.4) | 3657637 | `image-compare`; `slurm/demo/run_image_compare.sh`, which runs `demo_image_compare.py`; `results/image_compare/images_3657637/comparison/`; thesis copy `figures/img/F26_image_compare.jpg` (cut from the final presentation) |
| *3DPW labels* | | |
| Label check before and after the re-fit | 3575369, 3611284 (v23) | `check_3dpw_conversion`; `results/diagnostics/check_3dpw_conversion_v23/` |

| Model or item: what the job did | Job(s) | Log name; checkpoint or output |
|---|---|---|
| Re-fit with converter v2.3, 87 person-sequences | 3590385, 3602813 | `smpl2anny-all`; `threedpw_anny_v23/labels/`; `logs/check_refit_all_v23.log` |
| Label rebuilds from the old fits | 3579266, 3581320 | `3dpw-labels`; `threedpw_anny_full/labels_rebuilt/` |
| *Pilots and diagnostics (outputs in `results/experiments/` and `results/diagnostics/`)* | | |
| Phenotypes; youngest 50 training people | 3569948; 3569950 | `diag_pheno`; `diag_young` |
| Real-photo ages with the crop camera: Base, sampler pilot, its control | 3569949, 3572419, 3572420 | `diag_cropK` |
| Depth pilots, 2 epochs (×1, control, ×5, ×20) and 10 epochs (×1, control, ×5) | 3571598, 3571599, 3572418, 3572483; 3571933, 3571934, 3572482 | `exp1_depth_anchor` |
| Sampler pilot and control; batch ages only | 3571955, 3571956; 3571601 | `exp2_children` |
| Head-loss pilot; two identical controls (noise) | 3571957; 3571958, 3571960 | `exp3_neck_global_rot`; `exp4_fingers` (3571960) |
| Finger pilot (control 3571960: crop trace, hand size) | 3571959 | `exp4_fingers` |
| Geodesic pilot and its L1 control | 3573520, 3573521 | `exp5_neck_fix` |
| Per-bone dumps of eight checkpoints | 3573518, 3573519 | `diag_neck_bones`; `results/diagnostics/neck_bones/` |
| Bone correction: Base (with the neck pilot), Fresh-heads-L | 3579014, 3579015 | `exp6_bone_fix`; `results/experiments/exp6_bone_fix/anny_s2_shape_v5_00099_n200_s0/bone_fix.json`; neck pilot M/`exp6_v5_bonefix_00001.pt` |

The scripts behind these jobs and the exact commands that produced the thesis numbers are listed in the file `docs/thesis/REPRODUCE.md` of the repository. Each command runs from the login node in the repository clone. The flags each training run passes are in Appendix C.

## A.4. What cannot be reproduced exactly

The training code sets no random seed. The `--seed` of the holdout trace only selects the training images on which the bone correction is estimated. The student's script passes its seed to that trace and nowhere else (`exp8_distill_vitb.sh`, lines 257 and 1142). The sweep runs through the same script, so this also holds for the 60-epoch student and for every temperature. The order in which training images are drawn (with replacement) therefore differs between two runs of the same command, as does the number of optimiser steps skipped for non-finite gradients (16 in Final, 10 in Control, out of 10,000). Two identical 2-epoch pilots differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points (Table F.7); they do not establish an interval for a repeat of Final or Control (Section 4.8).

Final and Control also ran on different GPU types (Final on an A100, Control on an A40, both with TF32). This changes the speed, and it can change the last digits of the arithmetic. Evaluating a fixed checkpoint involves no sampling; whether it repeats exactly across GPU types was not tested.

Apart from the temperature, the sweep runs evaluate after every epoch instead of every fifth and also compute the training loss on the validation images (`--val_loss 1`). Both additions only read values out of the model and change no weight. The sweep's own $T = 4$ run therefore repeats the reported training configuration and gives a second estimate of run-to-run variation (Section 4.8).

Three further items cannot be repeated. The history scripts of the L line and of Student-S were edited after their runs, so their training configurations in Table A.1 are as recorded there, not re-run. `train.py` keeps only the last two or three checkpoints of a run, so intermediate epochs are lost. The checkpoints, the AnnyOne and 3DPW data and the Relative Human annotations are not in the repository, and 3DPW and Relative Human come with their own licences.

## A.5. Records of the video comparison

The video comparison of job 3642415 is documented by the log `video-compare-3642415.out` and the matching error log. It used the 30-epoch checkpoint of the student (Student) `results/experiments/exp8_distill_vitb/student_vitb_from_v5_pc_dist5_bonefix_v23_e10_00009_e30/train/checkpoints/00029.pt` with the correction selected by that run's `student_fix_arg.txt`. The released baseline (Multi-HMR-Anny) was `models/multiHMR/multiHMR_672_L_anny.pt` and the input was `man_walking.mp4`. Both checkpoints loaded with 100 % state coverage, meaning every saved weight found its place in the model. The renderer reported an NVIDIA L40S and passed its test mesh.

The final log lines report 356 processed frames for each model, 356 student matches and 354 baseline matches. The output is `results/video_compare/video_3642415/comparison.mp4`. The job ran before the temporal filter and the later timing code existed. Its log therefore gives neither the inference speed of the two models on the same frames nor any measure of frame-to-frame change in pose or body shape.

The five-panel video of Section 5.6 came from a later run of the same launcher with `TEMPORAL=ema` and `ALIGNMENT=landmarks`. It adds the filtered and the aligned student panels (Section 4.9.2) and a frame-rate label on every panel. This run is job 3657550 (`results/video_compare/video_3657550/`). It ran on one NVIDIA A40 in fp32 with TF32 at 672 pixels, timed 346 frames after 10 warm-up frames, and used the filter constants of Section 4.9.2. Its own summaries give the same matched-frame counts as job 3642415: 356 of 356 for the student and 354 for the baseline.

The frame-rate label on each panel counts a different amount of work (Table 5.8; definitions in `docs/video_inference.md`, lines 387–440). `timing.csv` has one row per model panel, and the input panel's playback rate is a column of those rows. The raw student and the baseline show `inference_fps`, model inference alone. The filtered panel shows `processing_fps` from `student_temporal/summary.json`, which also counts the filter and the Anny re-decode. The aligned panel shows the same quantity from `student_aligned/summary.json`, which counts the filter work as well as the alignment fit, but not the

offline landmark extraction.

The video implementation and the EGL correction are in commits `1d5621b` and `7b0b5e1`, and the revised FPS implementation is `a0fa59f`. The video comparison is launched through `slurm/demo/run_video_compare.sh`, and `docs/video_inference.md` describes how to run it, where each timing starts and stops, and which files it writes. The stage photo of Section 5.6.1 was made by `demo_image_compare.py`, normally launched through `slurm/demo/run_image_compare.sh`, whose defaults are the same student, correction and baseline (`docs/image_comparison.md`). It is job 3657637, with its output in `results/image_compare/images_3657637/`.

# B. SMPL–Anny joint correspondence

The converter requires, for each SMPL joint, the Anny bone that represents it. The two skeletons are the ones in Table 2.1: SMPL has 24 joints and Anny has 163 bones, of which the root and 18 rotate in the fit. Some joints of the two rigs sit at different places, so names alone are not enough. The converter compares the two rest skeletons (the default standing pose) before it fits the posed frames.

For each hip, Anny has two candidate bones, the upper-leg bone and the pelvis bone. The converter selects the one whose head (the point where the bone starts) lies about as far from the pelvis as SMPL's hip joint; for the v23 fits this is the upper-leg bone. For the spine, the Anny spine bones are ordered by their height, normalised so that the hip midpoint is 0 and the neck is 1, and the ordered triple of bones whose heights are jointly closest to the heights of SMPL's three spine joints is kept. For the limbs and the head, the converter takes the Anny bone of the same body part from a fixed list of names: the shoulder, for example, goes to the head of the upper-arm bone. Each run saves the bones it picked together with the fits.

For the torso joints, the converter fits a virtual joint (Equation (3.2)): a point that sits at a fixed offset from the Anny bone head. The converter measures this offset once, in the rest pose. Let $\mathbf{j}_k^S$ be the position of SMPL joint $k$, $\mathbf{o}_A$ and $\mathbf{o}_S$ the hip midpoints of the Anny and the SMPL rest skeleton, $s$ the ratio of Anny's to SMPL's hip-midpoint-to-neck length, $\mathbf{Q}_{S\to A}$ the matrix that lines up the axes of the two rest skeletons, and $\mathbf{h}_b^0$ and $\mathbf{Q}_b^0$ the rest head and orientation of the bone $b$ assigned to joint $k$:

$$\mathbf{j}_k^{S\to A} = \mathbf{o}_A + s\mathbf{Q}_{S\to A}(\mathbf{j}_k^S - \mathbf{o}_S), \qquad \boldsymbol{\delta}_k = (\mathbf{Q}_b^0)^\top(\mathbf{j}_k^{S\to A} - \mathbf{h}_b^0). \tag{B.1}$$

The first formula moves the SMPL joint into Anny's rest skeleton. It shifts the joint from SMPL's hip midpoint to Anny's, scales it by $s$ and turns it with $\mathbf{Q}_{S\to A}$. The second formula then measures the offset from the Anny bone head, in that bone's own frame. The offset stays fixed and turns with the bone, so it is not an extra joint fitted in each frame.

Each fit stores its own correspondence in three arrays: `anny_bone_names`, `joint_offsets_local` and `offset_joints`. A particular fit is reproduced from these arrays, not from this table. The code is in `scripts/data/smpl_to_anny.py`. Appendix G shows how geometric matching and virtual joints changed the fit error, and why matching by name failed in the first version of the converter.

| SMPL target | Anny correspondence and fitting point |
|---|---|
| Pelvis | root bone with a measured torso offset |
| Left/right hip | upper-leg or pelvis bone, chosen by its distance from the pelvis (the upper-leg bone in v23), with an offset |
| Spine1, spine2, spine3 | spine bones matched by normalised rest height, in order, with offsets |
| Neck | neck candidate with a measured rest-frame offset |
| Left/right collar | clavicle bone with an offset |
| Shoulder, elbow, wrist | heads of the matching upper-arm, lower-arm and wrist bones |
| Knee, ankle, foot | heads of the matching lower-leg, foot and toe bones |
| Head and hand targets | heads of the matching bones; their pose follows from the parent bones that the fit is allowed to rotate |

**Table B.1.:** *Correspondence used by the converter for the v23 labels. The exact bone names and offsets are saved with each fit and can differ between Anny rig versions. The fit rotates the root and 18 body bones; the others, including most fingers and the finer head and neck bones, keep their rest rotation.*

# C. Flags and weights

This appendix lists the command-line flags passed to the training script, so that each run can be repeated; run identifiers are those of Table A.1. Table C.1 covers the teacher (Final) and the teacher without corrections (Control). The student (Student) of Section 4.7 and the 60-epoch student (Student-60) of Section 5.4.1 share Table C.2. The temperature sweep of Section 5.4.2 has its own, Table C.3. The two distillation-ablation arms (Section 5.3) use the flags of Table C.2 with `--eval_freq 1 --val_loss 1`. The student without distillation losses sets `--lambda_kd 0 --lambda_kd_out 0`; the ViT-B model without teacher passes no distillation flag (no `--distill_teacher_ckpt`, hence no head copy or head freeze) and trains with `--bone_fix none`.

A flag is listed when its value differs from the default in the code, or when it keeps the default but matters for the result; every other flag keeps its default. The loss weights are in Table 4.1 of Chapter 4. The commands that start Final, Control and the student are in the file `docs/thesis/REPRODUCE.md` of the repository (Appendix A). The 60-epoch run was started with `EPOCHS=60` in front of the same student command, and the sweep with `slurm/experiments/exp9_kd_temperature.sh`.

Both teacher runs start from the initial model (Base) and train for 10 epochs of 1,000 iterations at batch 4, that is 40,000 sampled image presentations, at a constant learning rate. In Table C.1, "Value" is Final's; where Control differs, its value follows in brackets. Paths are relative to the project folder on the cluster.

**Table C.1.:** *Training flags of the teacher (Final) and the teacher without corrections (Control; values in brackets where they differ).*

| Flag | Default | Value (Control) | Meaning |
|---|---|---|---|
| *Start and network* | | | |
| `--pretrained` | none | Base checkpoint | weights the run starts from, heads included |
| `--pretrained_remap` | 1 | 0 | no renaming of weight names; Base was saved by this code |
| `--backbone`, `--img_size` | ViT-S, 336 | `dinov2_vitl14`, 672 | DINOv2 ViT-L, 48 × 48 tokens of 14 pixels |
| `--freeze_backbone`, `--unfreeze_last_n_blocks` | 0, 0 | 1, 4 | backbone frozen except its last 4 of 24 blocks and final norm |
| `--bone_fix` | off | Base's `bone_fix.json` (`none`) | constant output rotation on neck02, head, metacarpal1.L and finger4-1.R, stored in the checkpoint (Section 4.5.2) |
| *Targets and losses* | | | |
| `--gt_center_joint` | root | `person_center` (`root`) | the joint the depth, translation, cell and offset targets describe: the head, which the network anchors (Control: the pelvis) |

| Flag | Default | Value (Control) | Meaning |
|---|---|---|---|
| `--alpha_dist`, `--alpha_transl` | 1, 1 | 5, 5 (1, 1) | weights of the depth and translation terms |
| `--alpha_shape`, `--alpha_j2d`, `--alpha_v2d` | 1, 1, 1 | 20, 0.1, 0.1 | shape and 2D loss weights |
| `--boost_neck_weight`, `--boost_hand_weight` | 1, 1 | 4, 4 | extra weight on neck, head and hand bones, renormalised to mean 1 |
| `--start_2d_epoch` | 10 | 0 | 2D terms from the first epoch |
| `--mask_helper_joints_train` | 0 | 1 | hold the 88 helper bones at identity, no rotation targets |
| `--use_anny_shape` | 0 | 1 | phenotype targets from the dataset |
| `--mask_3dpw_hands` | 0 | 1 | no rotation targets for the 38 hand bones of 3DPW people |
| `--mask_3dpw_shape` | 0 | not passed | shape mask for 3DPW people; 1 only in Final-masked and Final-partial-mask |
| *Data* | | | |
| `--train_data`, `--val_anny_n` | BEDLAM, 0 | AnnyOne, 100 | AnnyOne for training, its last 100 images as the holdout |
| `--exclude_indices_file` | none | the 55 manifest images | never sample the diagnostic-manifest images |
| `--mix_3dpw_labels` | none | `threedpw_anny_v23/labels` | corrected 3DPW labels mixed in, 30 % of a batch on average (default ratio 0.3) |
| `--val_3dpw_n` | 0 | 300 | evaluate on 300 evenly spaced 3DPW validation images |
| *Optimisation, bookkeeping, job* | | | |
| `--batch_size`, `--learning_rate` | 2, $5 \cdot 10^{-6}$ | 4, $10^{-5}$ | images per step; Adam learning rate |
| `--n_iters_per_epoch`, `--max_iter` | 100, 10,000 | 1,000, 10,000 | 10 epochs of 1,000 steps |
| `--amp` | 1 | 0 | no mixed precision (it produced NaNs with the Anny layer); TF32 stays on |
| `--num_workers`, `--nb_max_ckpt`, `--log_freq` | 0, 10, 100 | 12, 2, 50 | bookkeeping; output in `results/experiments/exp7_final_model/<run>/train` |
| `TIME`, `EVAL_TEST` (script) | | 5 h, 1 | Slurm limit on A100, L40S, RTX A6000; after training the holdout trace and the 3DPW test (v23) |

Every flag of the student not listed in Table C.2 is as in Table C.1. This includes the targets, the masks, the loss weights and the 3DPW mixing, so the student also learns the corrected depth; there is no shape mask. The student trains for 30 epochs of 1,000 iterations at batch 8, that is 240,000 sampled image presentations, with a 20-hour Slurm limit. The script passes the colour-jitter flags (`-brightness 0.2` and so on), but the code never reads them, so there was no augmentation. Of the listed flags, the 60-epoch student differs only in the two values in brackets. Its job settings also differ: it has its own run name, ending in `_e60`, and the script computes its time limit from the epoch count instead of using 20 hours. That run's settings file is not in the repository; the two values follow from the script's rule (the learning rate halves every 2/5 of the epochs) and from the recorded launch command, which set only the epoch count. The loss curves of Figure 5.2 drop at the expected epochs.

**Table C.2.:** *Training flags of the student (Student) where they differ from Final's, with the 60-epoch student's values in brackets.*

| Flag | Default | Value (60 epochs) | Meaning |
|---|---|---|---|
| `--backbone` | ViT-S | `dinov2_vitb14` | DINOv2 ViT-B |
| `--pretrained` | none | `models/multiHMR/multiHMR_672_B.pt` | NAVER's ViT-B as the backbone start |
| `--pretrained_remap`, `--load_only_backbone`, `--freeze_backbone` | 1, 0, 0 | 1, 1, 0 | rename NAVER's weight names; load the backbone only and train all of it |
| `--head_dim` | 0 | 1024 | heads at the teacher's width, fed by a 768 to 1024 projector |
| `--distill_teacher_ckpt`, `--teacher_bf16` | none, 0 | Final, 1 | frozen teacher, its backbone in bfloat16; turns the run into distillation |
| `--init_heads_from_teacher`, `--freeze_heads_epochs` | 0, 0 | 1, 5 | copy every matching head tensor from the teacher; keep them frozen for 5 epochs |
| `--lambda_kd`, `--lambda_kd_out` | 1.0, 0 | 1.0, 1.0 | feature distillation (KL, temperature 4) and output distillation (L1) |
| `--bone_fix` | off | the teacher's | Final's correction in training; the Student's own, estimated on 200 training images, for evaluation |
| `--batch_size`, `--learning_rate` | 2, $5 \cdot 10^{-6}$ | 8, $3 \cdot 10^{-5}$ | images per step; Adam learning rate |
| `--max_iter`, `--lr_decay_every` | 10,000, 30 | 30,000, 12 (60,000, 24) | 30 epochs; learning rate halved after epochs 12 and 24 (60-epoch run: after 24 and 48) |
| `--eval_freq`, `--nb_max_ckpt` | 1, 10 | 5, 3 | evaluate every 5 epochs, keep 3 checkpoints |

Each temperature $T$ of the sweep is one full run of the student's training configuration. The repeated $T = 4$ run measures how much two runs with the same training settings differ (Section 4.8). The 30- and 60-epoch students were trained before `--val_loss` existed, so they have no validation loss.

**Table C.3.:** *Training flags of the distillation-temperature sweep that differ from the student's (Table C.2).*

| Flag | Default | Value | Meaning |
|---|---|---|---|
| `--kd_temperature` | 4 | 1, 2, 4, 8, 16 | softens the channel softmax of the feature distillation; the output L1 has no temperature |
| `--eval_freq` | 1 | 1 | evaluate after every epoch (Student: every 5); this changes nothing in training |
| `--val_loss` | 0 | 1 | a new flag that computes the training loss on the validation images in evaluation mode, without gradients, so it does not change the training either |
| `RUN` (script) | | `..._kdT<t>_sweep` | own output folder per temperature |
| `TIME` (script) | | 16 h | Slurm limit per temperature (Student: 20 h); each temperature is one job |

# D. Extra figures

This appendix collects the figures to which the main text refers without showing them. The models are the initial model (Base), the teacher (Final), the teacher with full shape mask (Final-masked) and two pilots of Appendix E: the depth pilot and the neck pilot. Several images print the run names recorded during the project: "v5" and `anny_s2_shape_v5_00099` denote the initial model, "final model (exp7)" and `exp7_v5_final_00009` the teacher (Table A.1). All image panels were judged visually and none is a measurement. Where a body is placed from the detector box (the demo setting), a wrong depth is not visible in the picture.

## D.1. Depth, bone correction and in-training curves

Figures D.1 and D.2 render two synthetic holdout images at each model's own predicted depth. The depth pilot uses head-joint targets and a depth weight of $\times 5$ for 10 epochs on AnnyOne only. In image 579862 the predicted body grows from 0.76 to 0.95 of its true height; in image 579860 the initial model draws a shrunken body too far away, so the ground-truth head shows above it. Both are single cases and not typical ones.

Figure D.3 adds two holdout pairs to Figure F.1. In each image the top row is the initial model as trained and the bottom row the same checkpoint with the bone correction applied to its output, with no further training (Section F.4). The columns are the input, the ground truth and the prediction rendered in three evaluation settings, labelled in the image. The bodies are drawn at the model's own predicted depth, so the depth defect still draws them too small; no metric exists for the necks.

Figure D.4 shows the in-training PVE of the teacher and the uncorrected teacher, taken from the training logs of one run per model; Appendix H reads it.

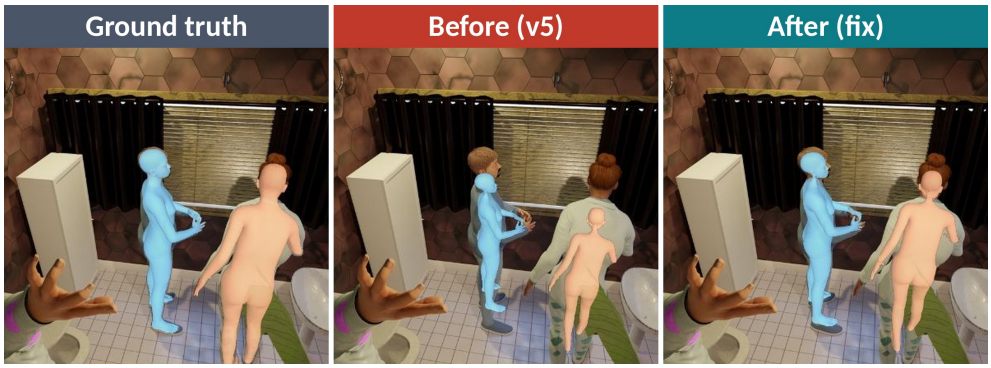



**Figure D.1.:** *Holdout image 579862 at each model's predicted depth: ground truth; initial model (labelled "Before (v5)" in the image); depth pilot (labelled "After (fix)"). Synthetic image; one of the better cases.*




**Figure D.2.:** *Holdout image 579860, same models and labels as Figure D.1. The initial model places a shrunken body too far away.*

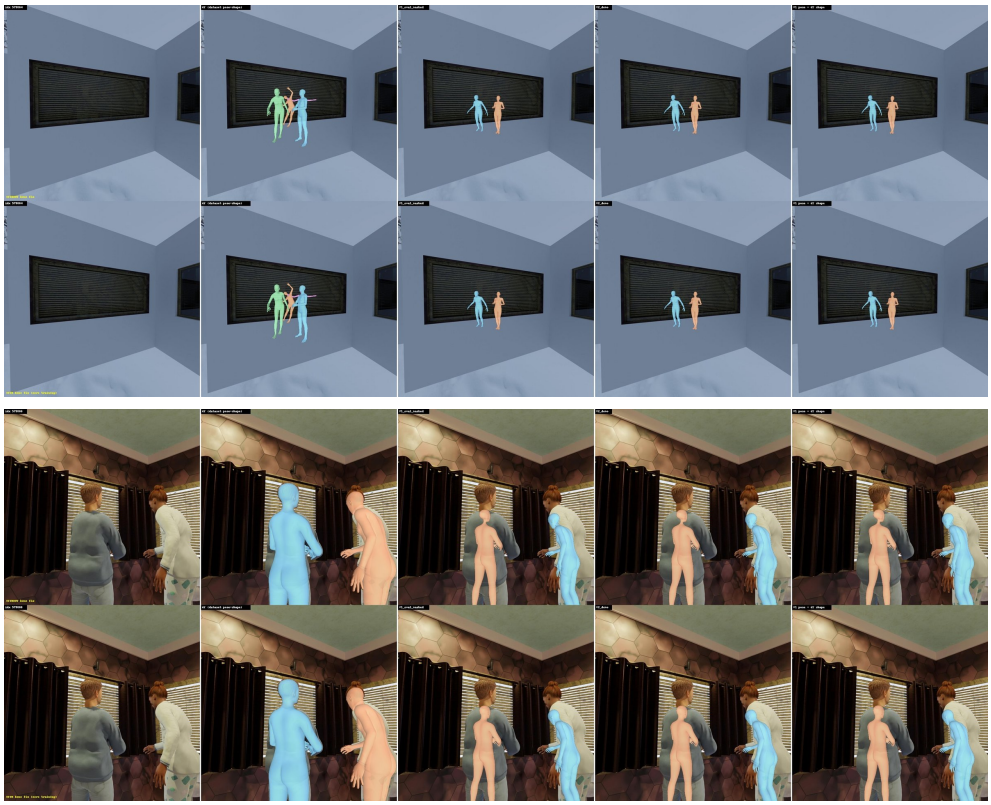


**Figure D.3.:** *Holdout images 579864 (top) and 579866 (bottom): in each, the initial model (Base) without (upper row) and with (lower row) the bone correction, without retraining. Synthetic images at the predicted depth.*

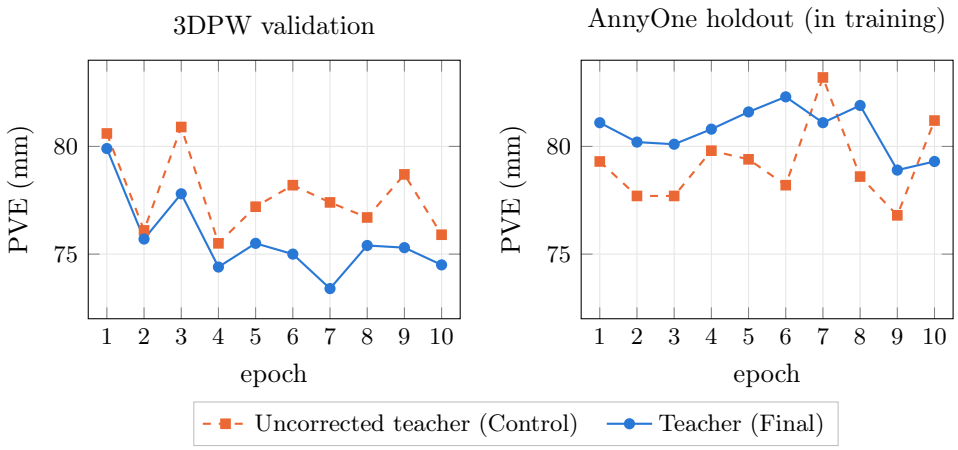



**Figure D.4.:** *PVE of the teacher (Final) and the uncorrected teacher (Control) after each training epoch. Left: 300 corrected 3DPW validation images; right: the AnnyOne holdout as evaluated during training (3DPW protocol), hence above the holdout-protocol values of Table 5.9a (73.7 mm and 77.4 mm).*

## D.2. Demo images: initial model and teacher

The demo images below compare the initial model with the teacher or with the neck pilot (the bone correction followed by a 2-epoch fine-tune of the initial model). Each montage shows the photo and the first model's prediction, then the photo and the second model's prediction. Bodies are placed from the detector box and, unless stated otherwise, the fingers are set to rest pose. In the neck-pilot figures the two columns differ by the correction and also by the fine-tune. The first four 3DPW test frames of the demo are in Figure D.13; the tennis photo also appears in Figure D.18.

Figures D.8 to D.10 show all twelve demo images with the predicted fingers (not set to rest pose). The initial model still has the flipped base bones of the left index finger (metacarpal1.L) and of the right ring finger; the teacher has the correction. In most images the hands are only a few pixels wide and cannot be judged, so Figure D.16 enlarges the three images where they can.

Figures D.11 and D.12 show the other nine demo images in rest pose: every detected person stands in the rest pose, so only the body shape differs between the models. Only the three render panels of each montage are shown, enlarged; the photo is visible behind the bodies. Only large differences in build can be seen. The other three images are in Figure D.17.




**Figure D.5.:** *3DPW test frames* `downtown_bar_00` *701 (top) and* `office_phoneCall_00` *440 (bottom). Left: initial model (labelled "Before: v5" in the image); right: neck pilot (labelled "After: v5 + neck fix").*



**Figure D.6.:** *Four 3DPW test frames, from top to bottom* `downtown_walking`*, stairs,* `downtown_runForBus` *and fencing. Left half: initial model (labelled* `anny_s2_shape_v5_00099` *in the image); right half: teacher (labelled* `exp7_v5_final_00009`*).*




**Figure D.7.:** *The four real photographs of the demo (garden, tennis, stage, party), none used in training. Left half: initial model (labelled* `anny_s2_shape_v5_00099` *in the image); right half: teacher (labelled* `exp7_v5_final_00009`*).*



**Figure D.8.:** *Predicted fingers on seven demo images (garden, tennis, stage, fencing, stairs,* `downtown_walking`*,* `downtown_runForBus`*). Left half: initial model (labelled "Base" in the image); right half: teacher (labelled "Final").*

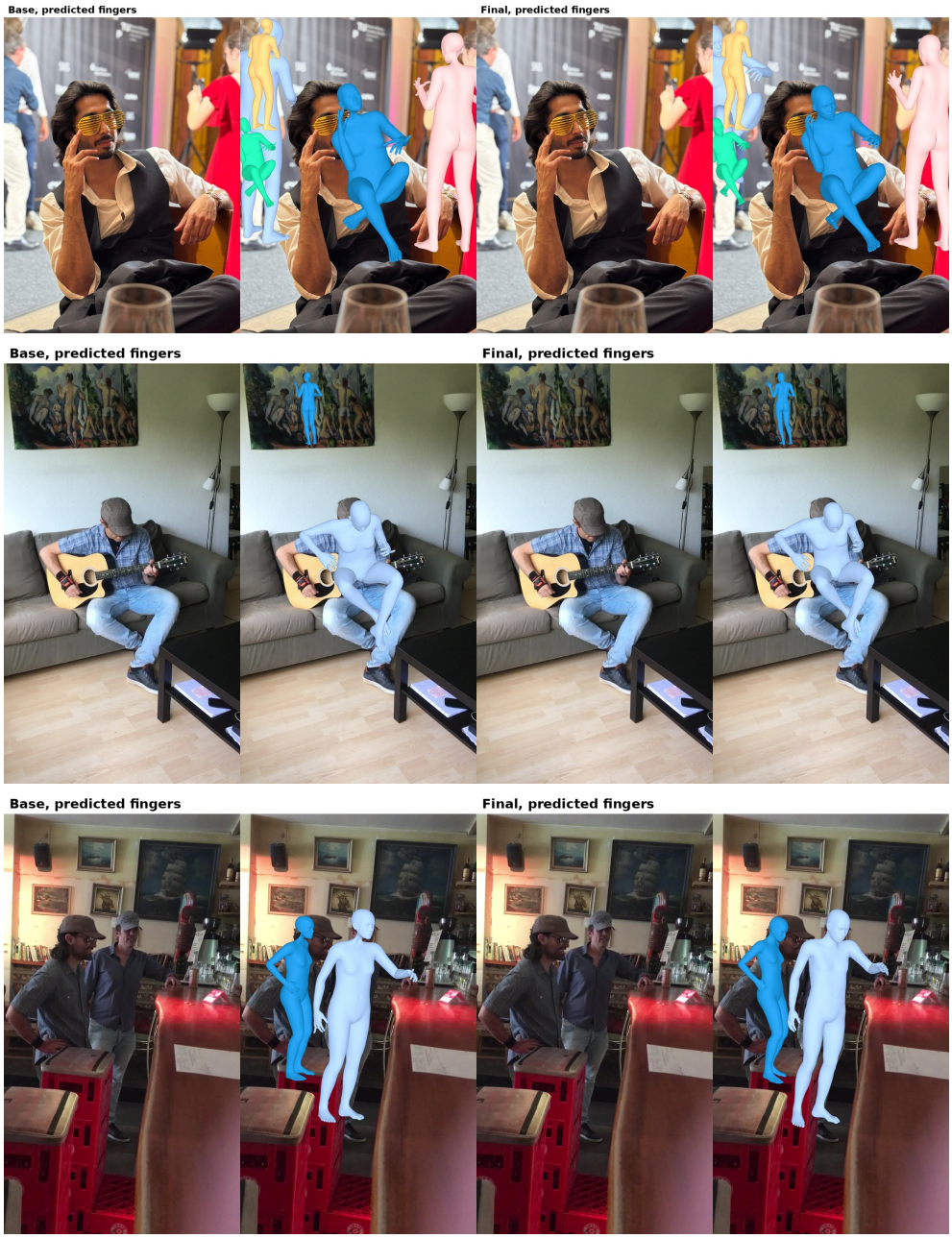



**Figure D.9.:** *Predicted fingers on the party photo and the 3DPW test frames guitar and* `downtown_bar`*. Left half: initial model (labelled "Base"); right half: teacher (labelled "Final").*




**Figure D.10.:** *Predicted fingers on the 3DPW test frames* `office_phoneCall` *and weekly market. Left half: initial model (labelled "Base"); right half: teacher (labelled "Final").*

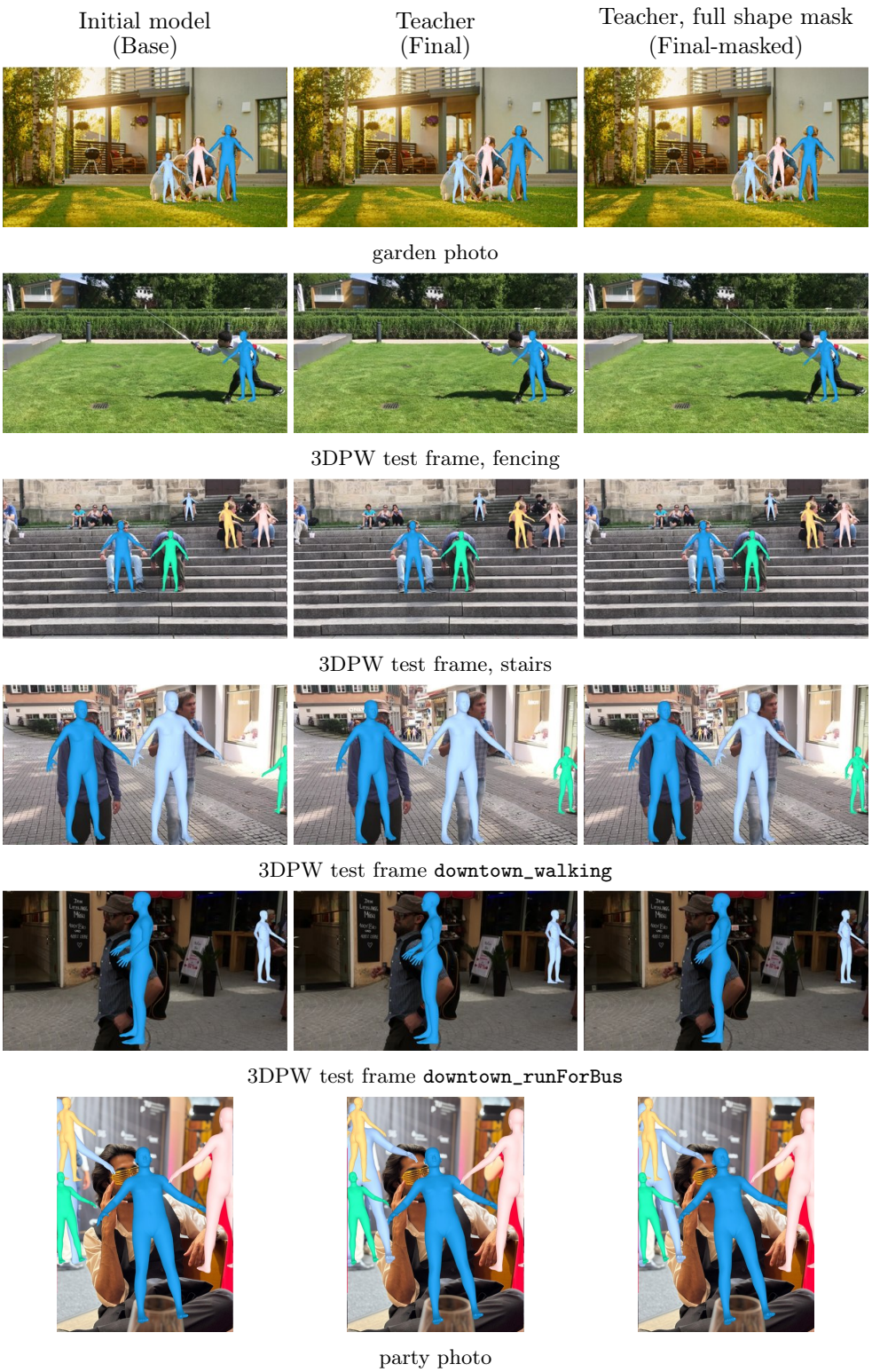



**Figure D.11.:** *Rest-pose line-ups on six demo images: initial model, teacher and teacher with full shape mask, from left to right. Demo setting, render panels only.*




**Figure D.12.:** *Rest-pose line-ups on three 3DPW test frames, same models and setting as Figure D.11.*

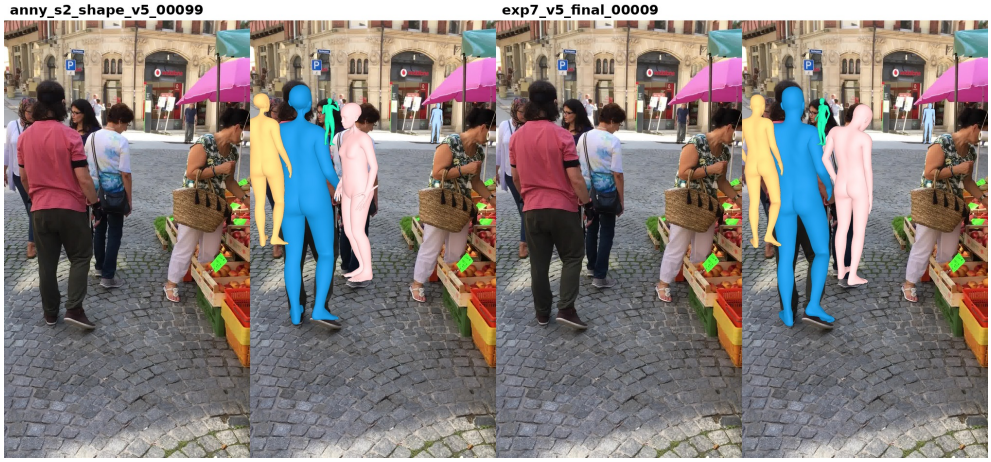
## D.3. Figures of the teacher analysis

These figures belong to the teacher analysis (Section 5.7) and the shape analysis (Section 5.7.6); they use the demo setting of Section D.2, with one run per model.

Figures D.13 and D.14 show four 3DPW test frames for the initial model and the teacher. The other four frames of the demo are in Figure D.6.




**(a)** *downtown_bar*




**(b)** *weekly market*

**Figure D.13.:** *Two 3DPW test frames. Left half: initial model (labelled* `anny_s2_shape_v5_00099` *in the image); right half: teacher (labelled* `exp7_v5_final_00009`*).*




(a) *office_phoneCall*




(b) *guitar*

**Figure D.14.:** *Two more 3DPW test frames, same models and labels as Figure D.13.*

Figure D.15 shows the bone correction on a real frame. A second frame is in Figure D.5.

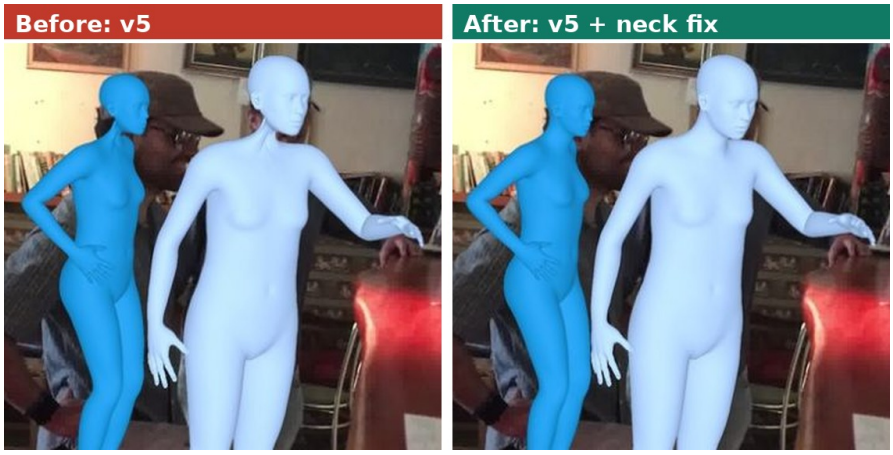
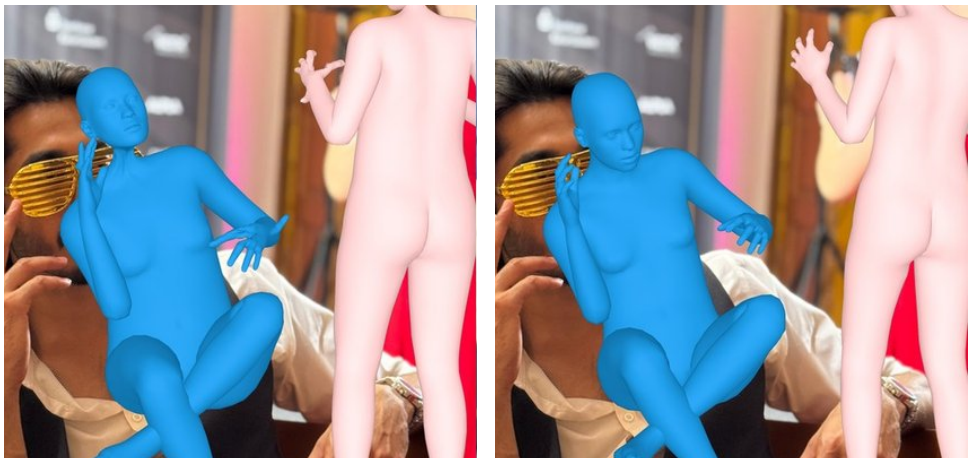



**Figure D.15.:** *3DPW test frame* `downtown_bar_00` *701. Left: initial model (labelled "Before: v5" in the image); right: neck pilot, fine-tuned for 2 epochs with the correction (labelled "After: v5 + neck fix").*

Figure D.16 enlarges the hands in three demo images with predicted fingers. Because the bodies are placed from the detector box, the meshes sit beside their people. The initial model still carries the two flipped hand bones and the teacher has the correction; neither model puts the fingers on the real ones. The columns differ by the correction and the fine-tune. All twelve montages are in Figures D.8 to D.10.

Initial model (Base) Teacher (Final)



(a) party photo: the hand at the face and the hand over the armrest



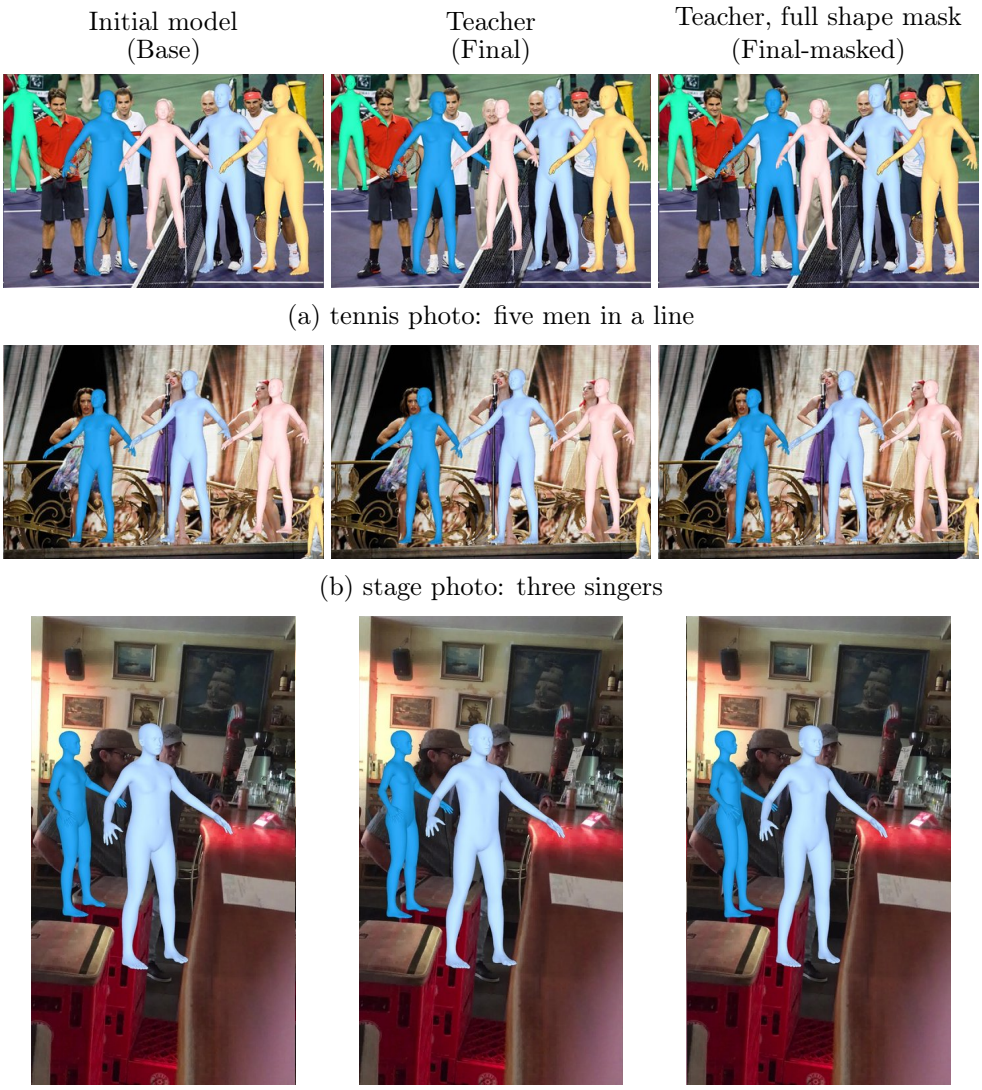
(b) 3DPW test frame, guitar: strumming hand (left) and fretting hand (right)



(c) 3DPW test frame `downtown_bar`: the hand on the bar and the hanging hand

**Figure D.16.:** *Predicted fingers (not set to rest pose) on three demo images, enlarged on the hands. Left: initial model; right: teacher.*

Figure D.17 draws every detected person in rest pose over the photo, so that only the body shape differs between the models; the other nine images are in Figures D.11 and D.12.




(a) tennis photo: five men in a line

(b) stage photo: three singers

(c) 3DPW test frame `downtown_bar`: two men

**Figure D.17.:** *Rest-pose line-up on three demo images: initial model, teacher and teacher with full shape mask, from left to right. Demo setting, render panels only.*

Figure D.18 shows three real images that no model was trained on. Most men receive male bodies, women stay female, and the boy keeps an adult body; no metric exists for these images.

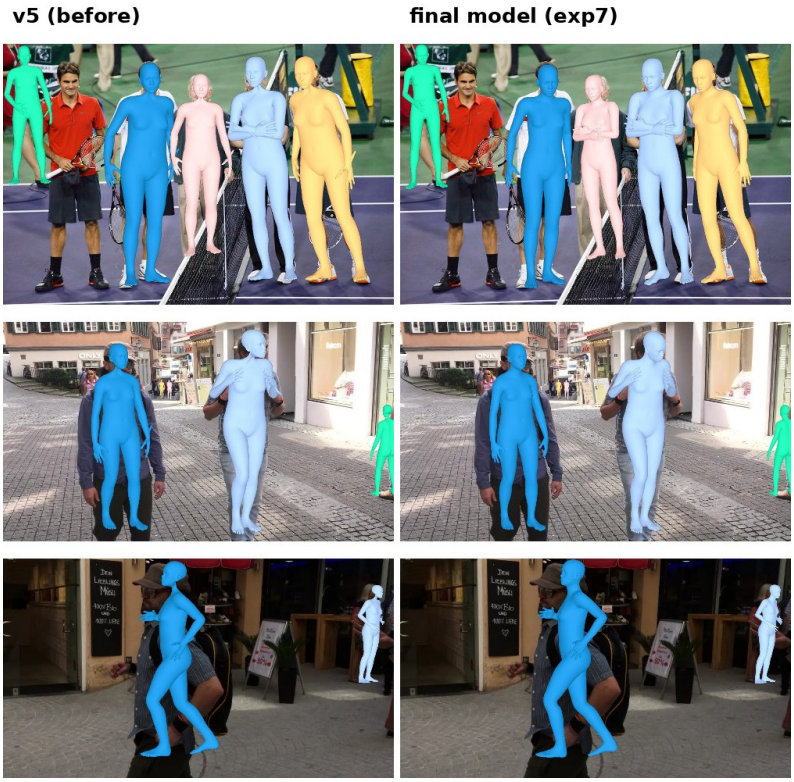



**Figure D.18.:** *A tennis photograph and the 3DPW test frames* `downtown_walking_00` *and* `downtown_runForBus_00`*. Left: initial model (labelled "v5 (before)" in the image); right: teacher (labelled "final model (exp7)").*

# E. Pilot results in full

Six candidate changes were tested in pilots before the final runs. A pilot is a short run, usually a 2-epoch fine-tune of the initial model (Base), that tests one change against a control, the same training without the change (Table E.1). Appendix F interprets the results; Section E.2 records the earlier stages that produced the initial model and the ViT-S students.

## E.1. The six pilots

Unless noted, each pilot fine-tuned the initial model on AnnyOne only (learning rate $10^{-5}$, batch 4, 2 epochs or 8,000 images) and was evaluated on the 100-image AnnyOne holdout in the holdout protocol, in which the helper bones (face, toes, breasts) are set to rest pose (Section 4.8). The success rule of each pilot was written into its script before the run; the full depth runs are in Table F.3. The geodesic angle is the angle of the single rotation that maps the predicted bone onto the true one, whereas the standard loss compares the rotation matrices element-wise with L1. A flagged bone is one that the per-bone dump of the initial model (Section F.4) marks as flipped or as rotated by a constant offset. Age bias is the mean of predicted minus true age phenotype.

**Table E.1.:** *The six pilot experiments, each a single fine-tune of the initial model (Base) without a seed, on the 100-image AnnyOne holdout (holdout protocol, matched people). The success rule was written into each script before the run. Head direction is the global head orientation error.*

| Pilot | What changed | Success rule | Result | Control |
|---|---|---|---|---|
| Depth pilots (2 and 10 epochs) | training targets at the head joint instead of the pelvis; depth and translation weights ×1, ×5, ×20 | slope toward 1 and intercept toward 0 m, with PVE ≤ 71.5 mm and recall ≥ 89.2 % | rule not met (PVE or recall missed in every run, controls too); root error 1.21 → 0.54 m, size 0.63 → 0.84, slope 0.33 → 0.38 (×5, 10 epochs) | same schedule, targets at the pelvis, weight ×1 (2 and 10 epochs) |
| Sampler pilot | age-balanced sampling: children (age < 0.3) per batch 2.1 → 9.7–11.1 % | 10 manifest children: PVE < 47.8 mm and \|age bias\| < 0.078; holdout PVE ≤ 71.5, recall ≥ 89.2 | inconclusive: children 48.5 vs. 53.1 mm, age bias 0.078 vs. 0.085; baby age floor 0.100 → 0.102 | sampler off (age ×3 and height ×2 shape weights in both) |

| Pilot | What changed | Success rule | Result | Control |
|---|---|---|---|---|
| Head-loss pilot | L1 loss on the camera-frame orientation of head, neck03 and both wrists, weight 5 | head direction $<$ 12.0 deg; PVE $\leq$ 71.5 mm, recall $\geq$ 89.2 % | head direction 11.3 deg, within the controls' 11.7–13.6; safety limit missed (71.8 mm, 89.0 %) | two identical runs at weight 0 |
| Finger pilot | region-normalised rotation loss (body, hands, head-neck averaged separately); a test-time person crop | fingers below the rest-pose floor 15.1 deg without a PVE cost | no effect: fingers 17.5 vs. 17.4 deg; crop 17.2 deg at body PVE 124.8 mm | loss off (one of the two identical runs above) |
| Geodesic pilot | geodesic angle instead of L1 on the rotation matrices | flagged bones down by $\geq$ 3 deg more than the control, or neck and head under 15 deg | no movement: neck02 175.6 $\rightarrow$ 175.0, head 177.9 $\rightarrow$ 177.1 deg | the same run with the L1 loss |
| Bone-correction pilot | constant output rotation on neck02, head, metacarpal1.L, finger4-1.R; no training, then a 2-epoch fine-tune | bones $<$ 15 deg, girth $\geq$ 0.9; no training: head direction $\leq$ initial model + 3 deg; fine-tuned: guards against the control | no training: fails (head direction 19.1 vs. limit 15.0 deg); fine-tuned: passes (neck02 4.7 deg, girth 0.997, fingers 8.3 deg) | the geodesic pilot's L1 control |

Two identical pilot configurations differed by 7.6 mm PVE (Section F.7); the placement and bone corrections were therefore adopted on their targeted diagnostics against matched controls, not on a small PVE gain.

## E.2. The ViT-S students and the stages that produced the initial model

The initial model was trained in stages (names in Table A.1, numbers in Table E.2a). L-stage 1 trained new Anny heads together with every block of NAVER's ViT-L backbone; it was discontinued, and no checkpoint is kept. L-stage 2 restarted from NAVER's backbone, kept it frozen and trained only new Anny heads; each later stage started from its predecessor. L-stage 3 unfroze the last 4 of the 24 blocks and added the helper mask, L-stage 4 weighted the loss on the neck and hand bones four times, and the initial model added the shape-loss weight of 20.

Before the ViT-B student, ViT-S students (Student-S/1 to Student-S/final; ViT-S is one size below ViT-B) were trained with the initial model as teacher to reduce the model size. All runs of Table E.2 predate the depth fix, the bone correction and the corrected 3DPW labels and report the values measured at

| Run | Change against the previous run | PVE | PA-PVE | MPJPE | Rec. |
|---|---|---|---|---|---|
| L-stage 1 | fresh heads, all blocks trained (side branch) | – | – | – | – |
| L-stage 2 | fresh heads, backbone frozen | 145 | – | – | – |
| L-stage 3 | + last 4 of 24 blocks, helper mask | 73.6 | – | – | – |
| L-stage 4 | + neck and hand loss ×4 | 73.6 | 56.5 | 73.2 | – |
| Base | + shape loss weight 20 | 74.3 | 57.5 | 74.4 | 90 |

**(a)** *The transfer schedule that produced the initial model, Base (Table A.1); recall (Rec.) in %. L-stages 2–4 are evaluated during training, the initial model with the standalone script.*

| | | | AnnyOne holdout, 3DPW protocol | | | | | 3DPW test, v23[c] | |
|---|---|---|---|---|---|---|---|---|---|
| Model | Teacher | Params | PVE | PA-PVE | MPJPE | Prec. | Rec. | PVE | PA-PVE |
| Base (ViT-L) | – | 318.4 | 74.3 | 57.5 | 74.4 | 81 | 90 | 121.8 | 72.0 |
| Mixed-L | – | 318.4 | 78.5 | 57.4 | 78.6 | 83 | 89 | 95.5 | 62.9 |
| Student-S/2 | Base | 32.8 | 193.4 | 128.2 | 210.7 | – | 86 | – | – |
| Student-S/6b[a] | Base | 37 | 126.9 | 92.1 | 132.0 | 77 | 88 | – | – |
| Student-S/6c | Base | 37 | **114.5** | **84.6** | **118.4** | 81 | 89 | 142.6 | 91.0 |
| Student-S/mixed[b] | Base | 37 | 120.6 | 88.2 | 125.2 | 78 | 87 | – | – |
| Student-S/final[b] | Mixed-L | 37 | 120.6 | 86.9 | 125.0 | 76 | 87 | 121.6 | 83.7 |

Student-S/2: feature distillation only; Student-S/6b: heads fed through a projector at the teacher's width; Student-S/6c: Student-S/6b at batch 16; Student-S/mixed and Student-S/final: continued with 3DPW images. [a]Evaluation during training. [b]30 % 3DPW images per batch, old labels. [c]Corrected labels, the script of Table 5.9b; the teacher (Final) reaches 89.0 / 57.6 mm.

**(b)** *The ViT-S students against their ViT-L teachers; parameters in millions, precision (Prec.) and recall (Rec.) in %.*

**Table E.2.:** *Earlier models, measured before the depth fix, the bone correction and the corrected 3DPW labels, on the AnnyOne holdout (100 images) in the 3DPW protocol (helper bones scored as predicted); errors in mm, single runs without a seed. "–": not measured with the same script; the students' old-label 3DPW numbers are retired.*

the time, in the 3DPW protocol (helper bones scored as predicted). The best of them, Student-S/6c, reached 1.54 times the PVE of its teacher. All ViT-S students used the feature-distillation temperature $T = 4$ and weight 1.0, and from Student-S/3 on an output-distillation weight of 1.0; the ViT-B student took these over unchanged.

Four observations from these runs shaped the student's training. With feature distillation alone nothing trains the heads, and Student-S/2 has by far the largest error (Table E.2b); the student therefore starts with a copy of the teacher's heads. The copied heads worked only when fed by the projector, not behind a newly initialised input layer, and heads at the teacher's width outperformed narrower ones; the student therefore keeps every head of the teacher (Final) at full width. The starting point mattered most: a backbone already trained by NAVER for mesh recovery gave the largest gain, so the student starts from NAVER's ViT-B. Finally, the data budget was increased, with corrected 3DPW labels alongside AnnyOne; the trials do not show that data, rather than capacity, was the only limitation.

Matrix products in TF32, a reduced-precision mode of the A100, and the teacher's backbone in bfloat16, a 16-bit format, reduced the ViT-S step time from 1.24 to 0.47 s on an A100 without changing the evaluation numbers. The student's run keeps both (TF32 as in the teacher, Table C.1; the teacher in bfloat16, Table C.2). The training log of these step times was not kept.

# F. Supporting diagnostic experiments

The initial model (Base), trained on AnnyOne only, showed three visible failures: thin and twisted necks, fingers bent into claws, and adult bodies for children. This appendix reports the experiments that isolated their causes; the two resulting corrections are defined in Sections 4.5.1 and 4.5.2. Unless noted, numbers are for the initial model on the 100-image AnnyOne holdout in the holdout protocol, in which Anny's helper bones (face, toes, breasts) are set to rest pose; the 3DPW protocol scores them as predicted (the holdout and 3DPW protocols of Section 4.8). Every variant is a single run without a fixed seed, and differences are read against the pilot variation of Section F.7.

## F.1. The three visible failures

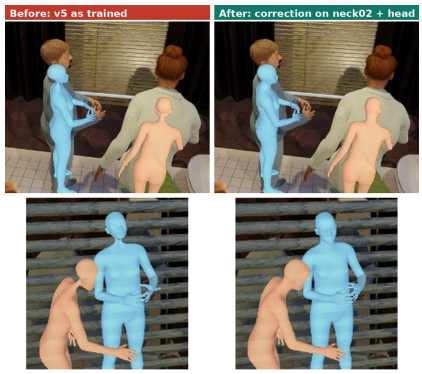



**Figure F.1.:** *Holdout images 579862 (top) and 579863 (bottom); left: the initial model (Base) as trained, right: the same checkpoint with the output correction of Section F.4, without training. Bodies are drawn at the predicted depth, so the depth error of Section F.3 still renders them too small. Neck shape is judged visually.*

Figure F.1 (left) shows the thin necks; the curled fingers were set to rest pose in every demo image, and a standing boy received an adult body (Figure D.18). Three untested explanations were proposed initially: the skinning of the stacked neck bones as a property of the rig (the skeleton that deforms the mesh), the small image size of the hands, and an unlearned age range, the last citing five small people in the demo images predicted at adult ages (0.57–0.67). The rig explanation was refuted (Section F.4), and hand size explains only a small part of the finger error (Section F.5). Four of the five small people were

adults who crouched, sat or were truncated (height phenotype 0.47–0.56); only the boy was a child, and he received an adult body (age 0.60, height 0.48). The thin necks are quantified by the neck girth of Table F.4, predicted over true neck circumference.

## F.2. Six structural checks

| Check | Sample | Result | Verdict |
|---|---|---|---|
| 1 labels decode onto the people (no network) | manifest, 55 people | 84.5 % of joints inside the image, 84.9 % inside the box (the rest is truncation); dataset camera correct on 100 % of images | labels correct |
| 2 body model, age sweep | body model | skeletal height 0.53 m at age 0, 1.55 m at age 0.8; head/height 0.200 to 0.106 | child shape exists |
| 3 single-joint moves | body model | all 16 pass (96–100 % of moved vertices in the intended part) | rig correct |
| 4 gradients | 4 holdout images | zero gradient only for the 88 helpers; all 75 active bones and 11 shape outputs receive gradient (598 at the root to 4.7 at the fingers); two of three camera outputs receive none | path correct |
| 5 prediction vs. truth, shape swap | holdout, manifest | neck length 99.3 % of truth; true shape lowers the 10 manifest children's PVE 47.8 → 41.6 mm | shape head is the gap |
| 6 capped overfit | 40 manifest training images | training PVE −36 % (80.7 → 51.8 mm), holdout +45 % | learns; two warnings |

**Table F.1.:** *Six structural checks on the initial model (Base) before any training change, on synthetic AnnyOne images. The manifest is a fixed image sample excluded from every later fine-tune (Section F.2). Check 6 trains a discarded copy for 400 steps and is scored in the 3DPW protocol.*



**(a)** *holdout 579888, a close-up with hands* **(b)** *training image 462901, low camera*

**Figure F.2.:** *Check 1 on two synthetic manifest images (two of four rendered cases): image, ground-truth mesh, and joints with box, decoded through the training body model and drawn with the dataset camera, without a network. On visual inspection the mesh lies on the person down to the fingers. In (a) the frame truncates head and legs; the black corner is part of the dataset image.*

**Hypothesis and test.** The failures could originate outside the network, in the labels, the body model, its rig or the gradient path (Table F.1). Checks 1, 5 and 6 use a manifest of 40 training images (10 adults, 10 children, 10 with visible hands, 10 hard neck poses) and 15 holdout images without children. The entries were drawn from the AnnyOne labels by a script with a fixed random seed. The initial model saw the training entries during its own training, so the manifest children show whether a model fits a child it has seen, not whether it generalises. Check 1 draws the decoded labels over the image without a network, which would expose a wrong label or camera (Figure F.2). *Result:* labels, rig and gradient path are correct, so the cause lies in what the training targets and losses require of the network.

**Loss composition.** At the end of training, the shape term made up 25.7 % of the initial model's loss, the translation term 1.9 % and the distance term 0.6 %. The 3D terms align both pelvises before comparing bodies and cannot see a person's distance from the camera; only the distance and translation terms, together 2.5 % before the depth fix, constrain it. Two log entries are easily misread: the detection term logged as `bce` is the focal loss of CornerNet [LD18], not a binary cross-entropy, and the neck and hand boost (Table 4.1) redistributes rotation-loss weight between bones without adding a loss.

**Overfit warnings.** In check 6 the rotation loss barely decreased during memorisation (44.6 → 40.8), and the translation and 2D terms rose (1.67 → 1.89, 58 → 75, first against last five epochs), possibly an early sign of the depth fault. This evidence is weak: the values are single-batch prints, and the half-run averages decrease (2.10 → 1.96, 69 → 63).

**Faults outside the checks' reach.** Two faults escaped: a depth target built at a joint other than the one placing the mesh, invisible in pelvis-centred comparisons, and a bone rotated by the same wrong amount in every person, hidden in group means of bone errors.

## F.3. Depth placement and the anchor targets

| Quantity | Ground truth | Initial model (Base) |
|---|---|---|
| pelvis depth, mean (sd) | 1.92 m (0.49) | 3.00 m (0.32) |
| people placed farther than truth | – | 100 % |
| corr. of true depth with depth error | – | −0.76 |
| skeletal height (3D) | 1.409 m | 1.405 m ($r = 0.89$) |
| head / pelvis 2D error, median | 0 | 46 / 14 px |
| corr. of PVE with depth error | – | −0.03 |

**Table F.2.:** *Placement of people by the initial model (Base) on the AnnyOne holdout, holdout protocol, all 222 matched people; slope, intercept and drawn size are in Table F.3. The bodies have the correct metric size but stand at the wrong distance, which PVE does not register.*

| Run (depth weight) | Root (m) | Size | Slope | Icpt. (m) | Head px | Pelvis px | PVE | Recall |
|---|---|---|---|---|---|---|---|---|
| Initial (Base) | 1.21 | 0.63 | 0.33 | 2.36 | 64.5 | 23.4 | 70.5 | 90.2 |
| *2-epoch fine-tunes of the initial model, AnnyOne only* | | | | | | | | |
| control ×1 | 1.11 | 0.66 | 0.33 | 2.28 | 58.1 | 23.3 | 72.3 | 89.8 |
| fix ×1 | 0.69 | 0.78 | 0.30 | 1.93 | 9.6 | 37.3 | 77.3 | 87.8 |
| fix ×5 | 0.66 | 0.80 | 0.44 | 1.62 | 8.5 | 38.1 | 76.8 | 87.8 |
| fix ×20 | 0.53 | 0.87 | 0.43 | 1.44 | 8.8 | 31.4 | 75.7 | 87.4 |
| *10-epoch fine-tunes of the initial model, AnnyOne only* | | | | | | | | |
| control ×1 | 1.44 | 0.59 | 0.34 | 2.59 | 64.0 | 24.3 | 72.2 | 89.4 |
| fix ×1 | 0.73 | 0.78 | 0.30 | 1.98 | 7.6 | 38.5 | 75.5 | 87.0 |
| fix ×5 | 0.54 | 0.84 | 0.38 | 1.59 | 7.2 | 33.1 | 74.8 | 87.8 |
| *uncorrected teacher (Control), teacher (Final): 10 epochs, 30% 3DPW* | | | | | | | | |
| Control ×1 | 1.40 | 0.59 | 0.28 | 2.66 | 67.1 | 26.7 | 77.4 | 89.0 |
| Final ×5 | 0.73 | 0.78 | 0.42 | 1.75 | 7.6 | 41.1 | 73.7 | 87.0 |
| *other checkpoints trained with pelvis targets* | | | | | | | | |
| Mixed-L | 1.39 | 0.59 | 0.31 | 2.60 | – | 25.1 | 75.0 | 88.6 |
| Fresh-heads-L | 1.49 | 0.57 | 0.28 | 2.71 | – | 24.0 | 74.1 | 89.0 |

**Table F.3.:** *Placement with training targets built at the pelvis (control rows, run only at weight ×1) or at the head joint (fix rows); ×1/5/20 is the weight of the distance and translation terms, and the teacher also carries the bone correction. AnnyOne holdout (100 images), holdout protocol, matched people; head and pelvis px are mean 2D errors, PVE in mm, recall in %, other metrics as in Table 4.4. Single runs without a seed; each fix row is read against the control of the same length.*

**Observation.** The initial model places every person too far from the camera, the nearest people most (Table F.2, Figure 5.7). With a skeleton of correct height, each body is therefore drawn at 0.63 of its true image size. PVE, which moves the predicted pelvis onto the true pelvis before measuring, does not register the error.

**Cause.** The code places the mesh in the image by its head joint, but the anchor targets had been built at the pelvis (Section 4.5.1), so the network was trained to put the head where the pelvis is. A debug print on one batch showed anchor offsets of $-2$ to $-19$ cells of the token grid, against a valid range of $\pm 0.5$ cells. A plausible but untested mechanism: with the mesh suspended from a misplaced head, the 2D terms reduce the mismatch only by drawing the body smaller, that is, farther away; the weak direct depth terms do not counteract this, and the mismatch is largest near the camera.

**Test and result.** The fix builds the anchor targets at the head; Table F.3 compares it with controls of the same schedule that keep the pelvis targets. The five fix runs reach a root error (3D distance between predicted and true pelvis) of 0.53–0.73 m against 1.11–1.44 m, a drawn size of 0.78–0.87 against 0.59–0.66, and a head 2D error of 7.2–9.6 against 58.1–64.0 px; repeated-run uncertainty was not estimated. Two further checkpoints trained with pelvis

targets carry the same fault, which remained undetected because PVE aligns the pelvis and the demo images place each body from its detection box.

**Remaining error.** The fix moves the intercept and the drawn size but barely the slope of predicted against true depth (0.30 with the target change alone, 0.38–0.44 with a stronger depth loss, against 0.33 for the initial model). Predictions remain pulled toward 2.5–2.9 m; the cause was not identified. The pelvis 2D error rose to 31.4–41.1 px, plausibly because the head is now pinned while the body is still drawn too small (not checked). Every fix run is worse than its control, by +2.6 to +4.9 mm PVE and −1.6 to −2.4 recall points. Each gap lies within the pilot variation, but their common direction indicates that the fix most likely costs some accuracy. The success rule fixed before the runs (PVE ≤ 71.5 mm, recall ≥ 89.2 %) was met by no run, controls included.

## F.4. A per-bone dump of the neck

| | Initial model (Base) | Corr., no train. | Corr. + 2-ep. FT | 2-ep. control | Uncorr. teacher (Control) | Teacher (Final) |
|---|---|---|---|---|---|---|
| neck02 (deg) | 175.6 | 4.5 | 4.7 | 175.7 | 175.8 | 4.8 |
| head bone (deg) | 177.9 | 12.4 | 11.2 | 177.9 | 177.7 | 8.9 |
| metacarpal1.L (deg) | 180.0 | 0.0 | 0.0 | – | 180.0 | 0.0 |
| finger4-1.R (deg) | n.d. | 10.2 | 10.3 | – | 180.0 | 9.6 |
| neck girth, pred./true | 0.461 | 1.004 | 0.997 | 0.461 | 0.464 | 0.996 |
| $PVE_{neck}$ (mm) | 27.3 | 10.8 | 10.7 | 27.9 | 27.7 | 11.0 |
| body PVE (mm) | 70.5 | 71.3 | 68.8 | 76.9 | 77.4 | 73.7 |
| PA-PVE (mm) | 53.3 | 52.2 | 50.9 | 55.2 | 56.4 | 50.3 |
| recall (%) | 90.2 | 90.2 | 89.4 | 91.1 | 89.0 | 87.0 |
| head direction (deg) | 12.0 | 19.1 | 14.4 | 13.8 | 13.0 | 13.1 |
| fingers (deg) | 17.3 | 8.2 | 8.3 | 17.4 | 17.4 | 8.3 |

**Table F.4.:** *Median local bone errors (about 180 deg is half a turn), neck girth and neck-skin error $PVE_{neck}$ (171 vertices) before and after the bone correction (Corr.; four constants from 200 training images). AnnyOne holdout, holdout protocol, matched people; the 2-epoch columns are AnnyOne-only fine-tunes (FT) of the initial model, the two teachers 10-epoch fine-tunes with 30 % 3DPW, and n.d. means not dumped. Single runs; PVE gaps below about 10 mm lie within the pilot variation of Table F.7.*

**Observation.** Group means (62.7 and 89.5 deg) did not localise the neck error. A dump of every bone of the neck chain (Table F.4, first column) shows neck02 175.6 deg off, a half-turn about its own axis (twist 175.7, swing 3.3 deg), and the head bone 177.9 deg off. neck03 is correct relative to its parent (5.7 deg) but 173.3 deg off in the camera frame. The two half-turns cancel, so the head points correctly (12.0 deg), but the skin between them is twisted thin, hence the low girth.

**Hypothesis: a training setting causes the flip.** All eight dumped checkpoints of the ViT-L line leading to and continuing from the initial model (L-stages 2 to 4, Base, Mixed-L, three short fine-tunes of Base; Table A.1) show the flip: neck02 is off by 175.0–176.0 deg and the girth is 0.46–0.47. It already exists in L-stage 2, before the helper mask (helper bones held at rest), the neck boost and the shape weight of 20, which excludes these. The 10-epoch depth fix with $\times 5$ weight (Table F.3), one of the three fine-tunes, keeps the flip, which excludes the depth fault. Fresh-heads-L, a separate ViT-L run from the Multi-HMR-Anny backbone with new heads, 3DPW from the first step and no neck and hand weight, has a normal neck (girth 1.00) but other flipped bones: two bones of the left arm whose errors cancel, and three metacarpals.

**Hypotheses: rig or loss.** The rig is excluded: it passes all 16 single-joint moves, and the rest pose, in which every rotation is the identity, cannot exhibit a flip. Loss changes did not remove the flip: the neck boost ($\times 2.41$ for the boosted bones after renormalisation) left it unchanged, and a geodesic loss, which penalises the angle between predicted and true rotation, moved neck02 only from 175.6 to 175.0 deg, plausibly because neck02 and the head would have to turn back jointly (not tested). The output is corrected instead, with one constant rotation per flipped bone (Section 4.5.2): 178.8, 179.2, 179.8 and 179.7 deg for neck02, head, metacarpal1.L and finger4-1.R.

**Result.** Without training, the correction restores neck02, girth and neck-skin error. The head direction, however, worsens to 19.1 deg, above the limit of 15 deg set beforehand, presumably because the head's error differs between people; a 2-epoch fine-tune reduces it to 14.4 deg (control 13.8). The teacher was therefore fine-tuned with the correction in place: a constant removes the half-turn shared by all people but cannot follow a per-person error. Body PVE differences lie within the pilot variation; Figure D.3 shows two further holdout pairs. The origin of the flips is unknown: the network outputs each rotation in the 6D representation, whose singular initial value would explain the tilted metacarpal1.L but not the twist of neck02. Necks on real photos lack ground truth, so Figure D.15 rests on visual inspection.

## F.5. The fingers

**Observation.** The initial model's predicted fingers are 17.3 deg off, fingers at rest pose 15.1 deg (Table F.5): the prediction is worse than none.

**Hypothesis 1: the hands are too small.** A hand spans a median of 2.25 tokens (image patches of the encoder), and 41 % span fewer than two. A test-time crop around each person enlarged the hands, but the finger error stayed at 17.2 deg while body PVE nearly doubled. A region-normalised rotation loss had no effect; it moves the hands' share of the rotation loss from about 70 to 33 %, which reweights the hands without strengthening them.

*(a) finger rotation error on the holdout*

| Setting | Fingers (deg) | Reference (deg) |
|---|---|---|
| Initial model (Base), predicted fingers | 17.3 | – |
| Base, fingers set to rest pose | 15.1 | – |
| Base, test-time person crop (PVE 124.8 mm) | 17.2 | 17.3 (paired) |
| region-normalised rotation loss, 2 ep. | 17.5 | 17.4 (control) |
| Base + bone correction, no training | 8.2 | 17.3 (Base) |
| Base + bone correction, 2-ep. fine-tune | 8.3 | 17.4 (control) |
| Teacher (Final), 10 ep., 30% 3DPW | 8.3 | 17.4 (Control) |
| Fresh-heads-L, three metacarpals corrected | 17.8 | 32.0 (uncorr.) |

*(b) hand size in tokens (14-px patches of the 672-px input)*

| | Median length | Under 2 tokens |
|---|---|---|
| full image, all hands | 2.25 | 41% |
| person crop at test time | 4.2 | 0% |
| children's hands, full image | 0.7 | 80% |

**Table F.5.:** *(a) Finger rotation error on the AnnyOne holdout (100 images, holdout protocol, matched people), each row against the matching run without the change; the rest-pose row keeps the flipped metacarpal1.L. (b) Hand length in input tokens. Single runs without a seed.*

**Hypothesis 2: flipped bones.** A per-bone search on training images found metacarpal1.L and finger4-1.R half-turned like neck02; correcting them roughly halves the finger error, with and without training.

**Result.** About half of the finger error came from two flipped bones, much less from hand size. A separate hand-crop branch was not tested, and the rest-pose floor was not remeasured with the correction in place. Figure D.16 shows the fingers on real photos.

## F.6. Children and body shape

**Observation on synthetic data.** On 10 holdout images (Table F.6(a)), raising the shape weight from 1 to 20 moved the age bias from +0.199 to −0.002 and the height bias from −0.154 to −0.026. Part (b) shows that this holds only near the mode: in both tails predictions move toward the training median. The holdout contains no children (Section 3.7), so the young tail is measured on training images.

**Observation on real photos.** On Relative Human (Table 5.10) children are read as short adults: kids receive a mean age of 0.61, babies 0.62 and adults 0.63 (age AUC 0.61, chance 0.5). None of 89 kids and 1 of 38 babies fall below the child threshold; their smallness goes into height. Gender carries no signal (accuracy 46.5 %). On the demo images the medians over all 299 predicted people (one per detection, about 30 distinct people, nearly all adults) equal the training medians: gender 0.59, age 0.61, height 0.50.

*(a) predicted minus true phenotype, 10 holdout images*

| Phenotype | True mean | True std | Weight 1 | Weight 20 (Base) | Reading |
|---|---|---|---|---|---|
| age | 0.553 | 0.036 | +0.199 | −0.002 | near the mode |
| height | 0.579 | 0.026 | −0.154 | −0.026 | near the mode |
| proportions | 0.763 | 0.070 | −0.017 | −0.001 | near the mode |
| firmness | 0.757 | 0.039 | −0.167 | −0.190 | unchanged |
| gender | 0.694 | – | −0.042 | −0.063 | ≈ |
| weight | 0.382 | – | +0.056 | +0.064 | ≈ |

*(b) initial model (Base): full holdout and young tail*

| | |
|---|---|
| holdout, all 222 people (nobody younger than 0.513) | age bias −0.015 (abs. 0.037); gender −0.008 (0.068) |
| holdout, true age 0.6–0.8 ($n = 53$, mean 0.67) | predicted age 0.57 |
| 50 youngest training people (in-sample, true age 0.015) | predicted age 0.10; height −0.25 |
| training medians (AnnyOne) | gender 0.588, age 0.601, height 0.500 |

**Table F.6.:** *Shape-head errors in phenotype units* $[0, 1]$*: (a) the stage before the initial model (shape weight 1) against the initial model (weight 20) on the same 10 holdout images; (b) the initial model, holdout protocol, on the full holdout and the 50 youngest training people (in-sample). In both tails predictions move toward the training median.*

**Interpretation.** Children and men predicted with female-looking bodies are the same failure: under weak image cues the shape head reverts to the training median. The evidence cannot separate two causes: the synthetic-to-real gap and the loss. The shape outputs pass a sigmoid and are trained with L1, which is minimised at the median, and the sigmoid's gradient is 0.25 at an output of 0.5 but only 0.015 at an output of 0.015, so outputs near the edge of the range barely move during training. Only 2.53 % of the training people are younger than age 0.3. The result is a short adult because Anny's child shape appears only below an age of about 0.4–0.5; the shoulders, for example, are 2.24 head widths wide at age 0.1 and 3.00 at 0.6. A low height at age 0.6 thus gives a short adult (at least 1.22 m), and proportions barely matter (leg over stature −0.016).

**Further views.** Phenotype agreement on the holdout (Table H.3) and gender ranking on real photos (Table H.2) are in Appendix H. For the initial model, the uncorrected teacher (Control) and the teacher (Final), the slope of predicted against true value is below 1 for every phenotype, the same pull toward the middle. The teacher follows weight (Pearson $r = 0.77$) far better than muscle ($r = 0.27$). Open are which cause dominates and ages in years, which the code's three mutually inconsistent age scales do not permit.

| Quantity | Observed values | Interpretation |
|---|---|---|
| Two identical controls: PVE | 76.3 vs. 68.7 mm | 7.6 mm difference between these runs |
| Head direction | 13.6 vs. 11.7 deg | 1.9 deg difference |
| Recall | 0.8 points apart | different matched sets |
| Six arms, initial-model configuration | PVE 68.7–76.3 mm; recall 88.6–90.2 %; head 11.3–13.7 deg | observed range, not a confidence interval |
| Restart comparison | 72.9 vs. 70.5 mm for the initial model | 2.4 mm higher mean PVE after restart |
| Root error | fix 0.53–0.73 m; controls 1.11–1.44 m | lower in every tested fix run |

**Table F.7.:** *Observed variation in the 2-epoch pilots, fine-tuned from the initial model (Base) at learning rate $10^{-5}$ without a fixed seed, on the AnnyOne holdout (100 images, holdout protocol, matched people). The restart comparison uses the unchanged controls. Two repeated pilots establish no general significance threshold and do not predict the variation of the final models.*

## F.7. Pilot noise and restart cost

Two 2-epoch pilots with identical settings differed by 7.6 mm PVE, 1.9 deg head direction and 0.8 recall points (Table F.7); the training code sets no seed. Two further epochs without any change raised the initial model's PVE by about 2.4 mm (restart cost), so each pilot is compared with its own control, never with the initial model.

Two runs establish no general threshold, and no bootstrap intervals were computed; the differences are a rough yardstick, not a significance test, and do not predict the variation of the final models (Section 4.8). Every change this appendix builds on exceeds it by far: the root error of the depth fix, neck02 from 175.6 to 4.5 deg, the girth from 0.46 to 1.004, and the fingers from 17.3 to 8.3 deg. PVE changes of a few millimetres are smaller, so whether the depth fix and the neck pilot cost PVE remains undetermined.

## F.8. Training changes without effect

Six training changes fixed none of the three failures (Table E.1): an age-balanced sampler (inconclusive on the 10 training-image children available), a global head orientation loss (within the spread of two identical controls), the region-normalised hand loss, the geodesic rotation loss and the neck and hand boost (none removed the flipped bones), and a $\times 20$ depth loss (no gain over $\times 5$). None addressed the actual faults, a wrong anchor target and flipped bones, which the loss could not detect; the two effective changes, head-anchor targets and the bone correction, came from the checks in this appendix.

# G. Supporting conversion experiments

Chapter 3 describes the final converter, version 2.3; this appendix documents its development and the history of the 3DPW label sets. No earlier version produced labels used for the main results. The corrected labels of Chapter 3 are those built from the version 2.3 fits; "v23" in file names and in the appendix tables denotes these corrected labels.

## G.1. Development of the converter

The first converter (v1) matched SMPL joints to Anny bones by name. Its mean joint error on `downtown_bar_00` was 34.5 mm; on four test scenes the error stayed at 30 to 37 mm in every frame, and all four actors received the same body. An error independent of pose and a body independent of the person indicate that the shape parameters absorbed a fixed mismatch between the two skeletons instead of fitting each actor.

A rest-pose comparison of both models exposed two wrong correspondences invisible in the bone names. Anny numbers the spine from the top down and SMPL from the bottom up, so SMPL's lowest spine joint, `spine1`, had been mapped to a bone 25 cm above it; SMPL's hips had been mapped to `pelvis.L/R`, 4 cm from the hip joint, instead of `upperleg01.L/R`. Version 2 selects the spine bones by height along the body axis and the hip bones by distance from the pelvis (Appendix B). Limb and head errors fell to 1 to 10 mm, but the torso remained at 43 to 73 mm (Table G.1).

The residual torso error arises because the rigs place their torso joints differently: SMPL's pelvis joint lies 9 cm above its hips, Anny's root at hip height and 7 cm in front. For nine torso joints the offset $\boldsymbol{\delta}_k$ from the Anny bone head $b$ to the SMPL joint $k$ is therefore measured once in the rest pose, in the bone's own frame, and the fit moves the *virtual joint* $\mathbf{p}_b = \mathbf{h}_b + \mathbf{Q}_b\boldsymbol{\delta}_k$ onto the SMPL joint, with $\mathbf{h}_b$ and $\mathbf{Q}_b$ the posed bone head and orientation (Equation 3.2). The fitted point thus keeps a fixed offset from its bone and rotates with it. The offsets are 11.6 cm (pelvis), 4.3 cm (hips), 3 to 7 cm (spine) and 5 cm (collars). With them (v2.1) the torso error fell to 13 to 22 mm, and four varied test scenes reached 15.8 to 20.1 mm (one occluded frame 43.5 mm).

Age remained wrong: up to v2.1 every actor received exactly 0.67, because joints do not distinguish adult ages. Version 2.2 fixes age at 0.55. On all 87 person-sequences it fitted 22,915 frames with a median fit error of 18.2 mm. The converter's own fit statistics were unremarkable, yet these fits carried two of the three defects of the next section.

| Converter version | Mean (mm) | Max | Worst joints (mm) |
|---|---|---|---|
| v1: correspondence by name, neutral SMPL targets | 34.3 | 37 | all joints, common floor |
| v2: correspondence by geometry, 3DPW joints, gender from metadata (reversed until v2.3) | 24.6 | 30 | spine1 73, collars 55–59, spine3 58, pelvis 53 |
| v2.1: + virtual torso joints, bone mask, warm start | 15.8 | 18.8 | shoulder 36, wrists 25–27 |

**Table G.1.:** *Converter fit error against SMPL joint targets (v1: joints of the neutral SMPL model; v2 and v2.1: 3DPW's own joints), averaged over all 24 joints on* `outdoors_fencing_01` *(38 frames, including full lunges); these are not network predictions. The geometric correspondence is worth about 10 mm, the virtual joints a further 9 mm.*

## G.2. Three label defects and the v2.3 re-fit

This section gives the history behind the check of Table 3.2. Because the converter grades only its own fit, it cannot detect errors in its targets or in the later processing of its fits; an independent check therefore reads the finished label files. On sampled images from every sequence it compares the labels with 3DPW's joints, surface and 2D keypoints, confirms that the data loader passes them on unchanged, and tests that the gender is not reversed, that every valid actor has a label and that the head turn is preserved. On the first label set (old labels, v2.2 fits, about 2,000 images) it returned FAIL with three defects (Table G.3). Table G.2 lists the three label sets.

**Defect 1: gender reversed.** The converter wrote 1.0 for men and 0.0 for women, MakeHuman's convention and the reverse of Anny's, so every 3DPW man was labelled with a female body. A refit with the flipped value lowered the joint error for 72 of 73 men (median 19.2 to 15.3 mm). For women the joint error rose (13.7 to 16.3 mm), but the body surface came closer to 3DPW's for 27 of 31.

**Defect 2: missing actor.** In 407 of 960 sampled two-person images one valid actor had no label. The converter fits every third valid frame of each person, counted from that person's first valid frame, so each person had their own frame grid; after a gap in which one actor was not valid, the grids diverged and the two actors were fitted on different frames. The detection loss then treats a visible person as background.

**Defect 3: missing labels.** The training split had 4,451 label images where the fits implied 6,437 (31% missing). The image files of 8 sequences were incomplete on the cluster (`courtyard_captureSelfies_00`: 480 of 750 files), and the label builder skipped every frame without an image file without counting the skips. A rebuild after re-extracting the images recovered every label

| | Old labels | Rebuilt labels | Corrected labels |
|---|---|---|---|
| Converter fits | first fit | first fit | re-fit |
| Image files of 8 sequences | missing | re-extracted | re-extracted |
| Training, images / people | 4,451 / 5,183 | 6,437 / 7,589 | 5,882 / 7,584 |
| Validation | 2,458 / 2,995 | 2,777 / 3,314 | 2,622 / 3,311 |
| Test | 9,722 / 11,850 | 9,722 / 11,850 | 8,190 / 11,854 |
| Total | 16,631 / 20,028 | 18,936 / 22,753 | 16,694 / 22,749 |
| Gender reversed | yes | yes | no |
| Actor missing in two-person frames | yes | yes | no |
| Labels missing (no image file) | yes | no | no |
| Used for training | earlier models (App. E) | never | teacher, uncorrected teacher, student |

**Table G.2.:** *The three converted 3DPW label sets, all built at every third frame with the same drop rules. The corrected labels have fewer images than the rebuilt set but nearly the same number of people, because both actors of a two-person sequence share each frame. Converter versions are described in Section G.1.*

| Check | Old labels | Corrected labels |
|---|---|---|
| Overall grade | FAIL | WARN |
| Body joints vs 3DPW, median / p95 (mm) | 18.7 / 24.5 | 14.1 / 21.1 |
| Surface, Anny→SMPL / SMPL→Anny (mm) | 18.9 / 18.3 | 18.9 / 16.1 |
| 2D error over raw 3DPW (px) | +1.1 | +0.4 |
| Gender labels | reversed | men 0, women 1 |
| refit with gender flipped | better for 72 of 73 men | worse: men 13.9 → 20.0 mm |
| Two-person images, an actor unlabelled | 407 of 960 | 0 of 1,080 |
| Fitted frames; median frame error (mm) | 22,915; 18.2 | 22,912; 14.0 |
| Head turn vs chest, labels / SMPL (°) | 9.5 / 20.2 | 19.6 / 19.3 |
| *Post-check of the re-fit, all fitted frames, 87 people* | | |
| Mean joint error, men / women (mm) | 19.0 / 15.0 | 14.7 / 15.2 |
| Two-person frames, a valid actor unfitted | 4,457 of 8,274 | 0 of 6,055 |
| *Remaining warnings on the corrected labels* | | |
| Limb ratio Anny / SMPL, forearm / thigh | | 0.948 / 1.064 |
| People with a joint more than 100 mm off | | 3.2% |

**Table G.3.:** *Independent check of the converted 3DPW labels before and after the re-fit, on images sampled from every sequence and compared with 3DPW's joints, surface and 2D keypoints. The middle block is the re-fit's own post-check over all fitted frames. Old and corrected fits cover different frames, so this block is not a paired comparison.*

(Table G.2, "rebuilt"); as it still had defects 1 and 2, it was never used for training.

**The v2.3 re-fit.** Version 2.3 fixes defects 1 and 2 with two switches, `-gender_convention anny` and `-frame_grid shared`, whose defaults reproduce v2.2 exactly. All 87 person-sequences were refitted with both switches on, and the independent check passed on all 60 sequences (Table G.3, middle block). For 9 single-person outdoor sequences the error rose by 1 to 3 mm, plausibly because of a different warm start (each frame's fit starts from the previous one); this was not tested.

The labels built from the v2.3 fits (v23 labels) comprise 16,694 images and 22,749 people (Table G.2): fewer images than the rebuilt set but nearly the same number of people (22,753), because both actors now share each image. No person was dropped for a high fit error or a small bounding box. A further 163 frames, all from `outdoors_crosscountry_00`, were dropped because the person cannot be visible: with 3DPW's own camera the ground-truth joints lie at a depth of $-0.3$ m, behind the camera. The 3D fit is correct, but image labels cannot exist with this camera, and no label set ever contained this sequence. Figure G.1 shows three further v23 frames.

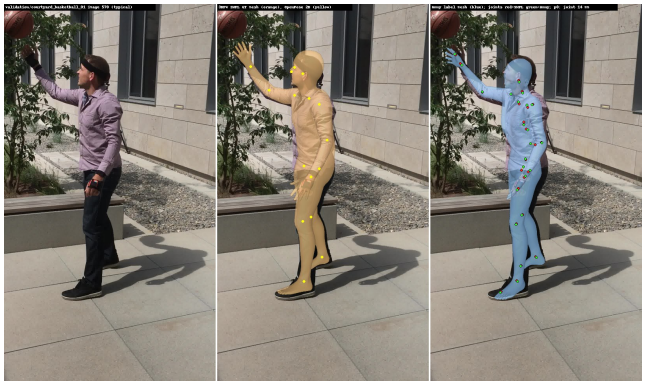
On the v23 labels (3,314 people in 2,360 images, 59 sequences) the grade moved from FAIL to WARN (Table G.3), and the median body-joint discrepancy fell from 18.7 to 14.1 mm. Flipping the gender now worsens the fit, which confirms the convention, and no sampled two-person image misses an actor. The head turn relative to the chest is now preserved; in the old test labels it was mostly lost (6.3° against 20.0° in SMPL). Two rows remain at WARN, the limb-length ratios and the people with a joint more than 100 mm off, mostly at the hip, plausibly because Anny and SMPL differ in limb proportions and hip-joint placement. Figure G.1(b) shows one of the three people the check renders as its worst cases, selected by the largest single-joint discrepancy; the value printed in the panel is that person's mean joint discrepancy.



**(a)** `courtyard_arguing_00`, *image 24 (training split), two people*



**(b)** `outdoors_climbing_02`, *image 633 (training split), one of the check's worst cases*



**(c)** `courtyard_basketball_01`, *image 570 (validation split), a typical frame*

**Figure G.1.:** *Three further converted labels from the v23 set (labels, not network predictions), judged visually. Each row shows the image, 3DPW's SMPL label (orange) and the converted Anny label (blue; red dots SMPL joints, green dots Anny joints), with each person's joint discrepancy printed above the right panel. In (a) both actors are labelled, which the old labels missed in 407 of 960 sampled two-person images (defect 2); in (b) the Anny body appears thinner than the climber.*

# H. Additional teacher results

This appendix collects observations and open points of the teacher analysis (Section 5.7) that are not needed for its findings. All statements refer to single runs on the sets of Section 5.1.

## H.1. Holdout examples and reconstruction

Figure H.1 shows two further holdout images; the label printed on the prediction panels ("V1_eval_masked") is the identifier of the holdout protocol. In all four images mesh colours mark detections, not identities; people without a ground-truth mesh (the foreground arm in 579862) are not scored, and people a model misses (the two lying in the foreground of 579865) have no mesh. During training the teacher stayed below the uncorrected teacher at all ten epochs on 3DPW validation, whereas on the holdout the two alternated (Figure D.4).

Table H.1 adds the pilot columns to Table 5.9. The depth pilot (depth fix, depth weight ×5, 10 epochs) has the lowest root error, and the neck pilot (bone correction and a 2-epoch fine-tune) has neck errors close to the teacher's; the earlier model was trained on the 3DPW labels before their correction and is tested on the corrected ones like every model here.

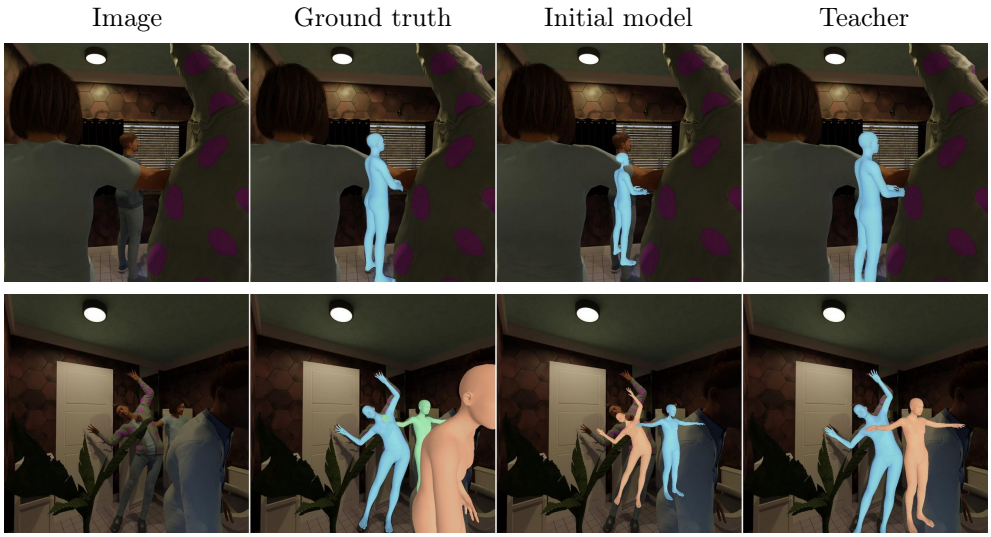



**Figure H.1.:** *Two further AnnyOne holdout images (579860 top, 579871 bottom), arranged as in Figure 5.6. The ground truth is drawn with the dataset camera, the predictions of the initial model and the teacher with the holdout protocol, each body at its model's predicted depth.*

| | Initial model | Depth pilot | Neck pilot | Uncorrected teacher | Teacher |
|---|---|---|---|---|---|
| PVE (mm) | 70.5 | 74.8 | 68.8 | 77.4 | 73.7 |
| PA-PVE (mm) | 53.3 | 54.0 | 50.9 | 56.4 | 50.3 |
| Recall (%) | 90.2 | 87.8 | 89.4 | 89.0 | 87.0 |
| Root error (mm) | 1208 | 544 | 1429 | 1402 | 728 |
| Depth slope | 0.33 | 0.38 | 0.34 | 0.28 | 0.42 |
| Depth intercept (m) | 2.36 | 1.59 | 2.56 | 2.66 | 1.75 |
| Size ratio | 0.63 | 0.84 | 0.58 | 0.59 | 0.78 |
| Neck girth pred/GT | 0.461 | 0.457 | 0.997 | 0.464 | 0.996 |
| $\text{PVE}_{\text{neck}}$ (mm) | 27.3 | 27.7 | 10.7 | 27.7 | 11.0 |
| Global head direction (deg) | 12.0 | 12.7 | 14.4 | 13.0 | 13.1 |
| Fingers (deg) | 17.3 | 17.4 | 8.3 | 17.4 | 8.3 |
| neck02 bone (deg) | 175.6 | 175.5 | 4.7 | 175.8 | 4.8 |
| head bone (deg) | 177.9 | 178.0 | 11.2 | 177.7 | 8.9 |

**(a)** *AnnyOne holdout (100 images, holdout protocol, matched people); placement metrics with each body at its model's predicted depth.*

| | Initial model | Earlier model[d] | Neck pilot | Uncorrected teacher | Teacher | Released baseline |
|---|---|---|---|---|---|---|
| PVE (mm) | 121.8 | 95.5 | 120.2 | 91.5 | 89.0 | 126.9 |
| PA-PVE (mm) | 72.0 | 62.9 | 68.7 | 59.7 | 57.6 | 68.8 |
| MPJPE (mm)[b] | 126.9 | 94.8 | 124.8 | 90.9 | 88.3 | 129.4 |
| Precision (%) | 54 | 73 | 53 | 62 | 64 | 75 |
| Recall (%) | 100 | 94 | 100 | 99 | 99 | 97 |
| Holdout PVE, 3DPW protocol[c] | 74.3 | 78.5 | 72.7 | 81.2 | 77.5 | (60.3)[a] |

[a]The released baseline masks the helper bones by design, so this value follows the holdout protocol and compares with the teacher's 73.7 in (a); it is potentially in-sample. [b]Mean over all 163 Anny joints, pelvis-centred. [c]The 100 images of (a) with the 3DPW protocol (helper bones predicted). [d]Earlier ViT-L model trained on the labels before their correction (Appendix E), tested on the corrected ones like every model here. Depth pilot: depth fix, depth weight ×5, 10 epochs; neck pilot: bone correction and a 2-epoch fine-tune.

**(b)** *3DPW test (corrected labels, all 8,190 images, 3DPW protocol), the same evaluation code for every model.*

**Table H.1.:** *Full version of Table 5.9, with the depth pilot, the neck pilot and the earlier model (Appendix E) on the AnnyOne holdout and the 3DPW test set.*

The comparison of the teacher with the uncorrected teacher has four gaps. The matched sets differ in size (214, 219 and 222 people) and PVE was not compared on the same people, recall was not computed separately for people with a visible head (which would test the explanation via the moved detection cell), no bootstrap intervals were computed, and there is no second run of the teacher.

On the 3DPW overlays (Figure D.13) most men receive male bodies from the teacher and female-looking ones from the initial model, the teacher's bodies are slightly larger, and the seated guitar player is reconstructed poorly by both. The low 3DPW precision arises because every detected bystander and every ghost body on a half-visible person counts as a false positive. On 3DPW the only repeated setting is the pair of $T = 4$ students (Section 4.8); the teacher was not repeated. The 3DPW result was not broken down by sequence, and only 8 frames were inspected visually.

## H.2. Placement

The Pearson correlation of predicted with true root depth rises from 0.46 for the uncorrected teacher to 0.56 for the teacher, computed over the matched people of Figure 5.7; the fits are $0.42 \times \text{true} + 1.75\,\text{m}$ (teacher), $0.28 \times \text{true} + 2.66\,\text{m}$ (uncorrected teacher) and $0.33 \times \text{true} + 2.36\,\text{m}$ (initial model), and the teacher's fit crosses the identity at 3.01 m.

The teacher's pelvis lands farther from its true pixel than the uncorrected teacher's (41.1 against 26.7 pixels), probably because the head is pinned while the body is still drawn too small. Its root error is also above that of the depth pilot (544 mm), possibly because of the real images or the bone correction. Neither explanation was tested, and why the slope of predicted against true depth stays far from 1 is unknown. The single-image figure of the depth fix (Figure D.1), in which the initial model draws the man far behind his true position and too small and the depth pilot close to his true size, shows the depth pilot, not the teacher.

## H.3. Bone correction

The neck-region joint error falls from 62.6 deg for the uncorrected teacher to 5.5 deg for the teacher (per-region evaluation of the same holdout runs). On the real frame of Figure D.15, which no model trained on, the initial model draws both necks as thin stalks under the head and the neck pilot draws them at normal width. The frame shows the neck pilot, not the teacher, and its columns also differ by the fine-tune. On the holdout, however, the same fine-tune without the correction keeps the girth at 0.461 (Table F.4), so the correction is the likely cause of the wider necks. Flips that cancel each other along a chain (Section F.4) would need a two-sided correction, which was not implemented.

| | Initial model | Teacher | Teacher, full mask | Teacher, partial mask |
|---|---|---|---|---|
| People with a gender tag | 932 | 1,308 | 1,163 | 1,271 |
| Gender AUC | 0.562 | 0.780 | 0.705 | 0.752 |
| Accuracy at 0.5 (%) | 46.5 | 64.2 | 49.3 | 47.4 |
| Mean prediction, women | 0.551 | 0.861 | 0.860 | 0.888 |
| Mean prediction, men | 0.535 | 0.551 | 0.717 | 0.725 |
| Best accuracy[a] | 0.636 | 0.723 | 0.685 | 0.707 |
| at cut-off | 0.62 | 0.89 | 0.95 | 0.93 |
| *Spread (sd) of the prediction over all matched people* | | | | |
| Weight | 0.101 | 0.058 | 0.146 | 0.120 |
| Muscle | 0.084 | 0.053 | 0.084 | 0.055 |

[a] Cut-off chosen on Relative Human itself, so this row only shows how large the threshold effect is; it is not a usable result. A real cut-off cannot come from the AnnyOne holdout either: its gender labels only span 0.38 to 0.82, with 85 % above 0.5, so it has no fully male body.

**Table H.2.:** *Gender on the Relative Human validation split (matched people with a male or female tag; Anny's gender 0 is the male end): the AUC needs no cut-off (0.5 = chance), accuracy uses the cut-off 0.5. The masked models still rank women above men but move both sexes up, so at 0.5 most men are classified as women.*

## H.4. Fingers on real images

In the crops of Figure D.16, which are the first demos with predicted instead of rest-pose fingers, the initial model's fingers are splayed and stiff, and one finger of the left hand sticks out as the flipped metacarpal1.L would make it. The teacher's hands lie over the armrest in the party photo, are curled as if fretting in the guitar frame, and bend over the edge of the bar in the bar frame. Small hands (Figure D.8) cannot be judged at print size.

## H.5. Body shape

Table H.2 gives the gender measures on Relative Human; the masked models still rank women above men but move both sexes above 0.5. Table H.3 gives the phenotype agreement on the AnnyOne holdout: the teacher follows weight (0.77), gender (0.73) and cup size (0.69) best and muscle worst (0.27, down from 0.41 for the initial model, likely because the converter fixes muscle for every 3DPW person), and the largest slope (0.61, for weight) is still below 1, so all three models regress towards the mean. Ages are not converted to years, because the code uses three age scales that disagree, and the holdout contains no children to check them against.

On the demo photos (Figures D.18 and D.7) most men look male with the teacher, and the boy keeps an adult body. The gender gain is consistent with the corrected 3DPW gender labels, which supervise the shape head on real people for the first time. Only detected people are scored on Relative Human, so the matched sets differ between models. The neck pilot (bone correction and a 2-epoch fine-tune) equals the initial model on Relative Human within

| | Initial model | | Uncorrected teacher | | Teacher | | Teacher: spread (sd) | |
|---|---|---|---|---|---|---|---|---|
| Phenotype | $r$ | slope | $r$ | slope | $r$ | slope | prediction | label |
| weight | 0.80 | 0.67 | 0.80 | 0.65 | 0.77 | 0.61 | 0.194 | 0.247 |
| gender | 0.77 | 0.52 | 0.69 | 0.54 | 0.73 | 0.48 | 0.087 | 0.132 |
| cupsize | 0.70 | 0.52 | 0.68 | 0.50 | 0.69 | 0.47 | 0.164 | 0.243 |
| proportions | 0.74 | 0.57 | 0.66 | 0.42 | 0.67 | 0.50 | 0.056 | 0.074 |
| asian | 0.65 | 0.26 | 0.62 | 0.26 | 0.62 | 0.25 | 0.036 | 0.087 |
| height | 0.52 | 0.36 | 0.55 | 0.38 | 0.48 | 0.37 | 0.043 | 0.055 |
| firmness | 0.45 | 0.36 | 0.45 | 0.35 | 0.46 | 0.34 | 0.149 | 0.198 |
| caucasian | 0.50 | 0.25 | 0.48 | 0.23 | 0.46 | 0.22 | 0.050 | 0.104 |
| age | 0.44 | 0.16 | 0.31 | 0.14 | 0.31 | 0.16 | 0.036 | 0.068 |
| african | 0.40 | 0.23 | 0.35 | 0.20 | 0.30 | 0.17 | 0.040 | 0.071 |
| muscle | 0.41 | 0.21 | 0.35 | 0.16 | 0.27 | 0.13 | 0.052 | 0.108 |
| People | 222 | | 219 | | 214 | | | |

**Table H.3.:** *Agreement of predicted and labelled phenotypes on the AnnyOne hold-out (matched people): Pearson $r$, least-squares slope of prediction on label (1 = no shrinkage towards the middle), and the spread of the teacher's predictions and of the labels. Every slope is below 1.*

run-to-run variation (child-vs-adult AUC 0.613 against 0.606).

At the cut-off 0.5 the gender accuracy of the two masked models is near chance only because both sexes moved up, so most men are classified as women. A cut-off for real photos would have to be chosen on 3DPW validation, which was not done; the best accuracy in Table H.2 uses a cut-off chosen on Relative Human itself and therefore only shows the size of the threshold effect. On the AnnyOne holdout only 85 % of the gender labels lie above 0.5. The partial mask restores weight spread on real photos, but not muscle spread.

In the rest-pose line-up (Figure D.17) only the body differs. The initial model gives nearly everyone the same slim, flat body. The teacher gives the singers a fuller chest and hips and the man in the bar a broader torso. The tennis players stay slim in all three models, in line with the intermediate gender score of men, and the teacher with full shape mask resembles the teacher except for a slimmer man in the bar. The explanation of the weakened muscle correlation by the converter's constant muscle value of 0.5 was not tested.

## H.6. Shape-supervision masking

The full mask covers age (0.55) and the seven values at 0.5; on the demo images it raises the spread of weight, cup size and firmness from 0.04–0.05 to 0.11. The partial mask leaves 34.3 % of babies child-shaped and doubles the weight spread on Relative Human (0.058 to 0.120). Without a mask, the teacher learned the seven constants and predicts nearly the same muscle, weight, cup size and firmness for everyone on real photographs.

The 3DPW mesh targets still hold the neutral body under both masks, which is why the body errors barely change. With age supervised (partial mask) the age-group improvement remains in this run; this suggests an effect of the age

| | AnnyOne holdout | | | | 3DPW test | | | |
|---|---|---|---|---|---|---|---|---|
| $T$ | PVE | PA-PVE | MPJPE | Rec. | PVE | PA-PVE | MPJPE | Rec. |
| 1 | 116.0 | 77.8 | 123.1 | 85.0 | 109.9 | 71.0 | 111.6 | 100.0 |
| 2 | 116.7 | 80.2 | 124.1 | 85.0 | 106.8 | 71.3 | 108.0 | 100.0 |
| 4 (sweep) | 117.1 | 81.4 | 124.2 | 85.8 | 110.1 | 72.0 | 111.5 | 100.0 |
| 8 | 117.1 | 77.8 | 124.8 | 82.5 | 107.1 | 71.0 | 108.4 | 99.0 |
| 16 | no test evaluation: the post-training step failed on a network time-out | | | | | | | |
| 4 (reported) | 119.6 | 80.8 | 127.2 | 85.4 | 110.1 | 71.8 | 112.0 | 99 |
| Teacher | 73.7 | 50.3 | 77.9 | 87.0 | 89.0 | 57.6 | 88.3 | 99 |
| 95 % threshold | 77.6 | | 82.0 | | 93.7 | | 92.9 | |

Only the temperature of the feature distillation changes (it softens the channel softmax of the projected backbone tokens before the KL term, gradient scaled by $T^2$); the output distillation is an L1 term without temperature. Everything else is the reported training configuration (30 epochs).

**Table H.4.:** *Test-set evaluation of the student for each distillation temperature $T$ (errors in mm, recall in %; $T = 16$ without a test evaluation). These values are reported for completeness and were not used to select $T$, which was fixed in advance (Section 5.1).*

target but does not establish the mechanism. Gender accuracy falls under both masks although the gender label stays supervised. One untested reading is that the teacher's gender gain depends on the seven constants on the 3DPW men as much as on the gender label. The partial-mask model had no demo run, so it is not known whether it restores variety on the demo images, and the reason why gender falls remains open.

## H.7. Distillation temperature: test-set values

Table H.4 lists the final test-set evaluation of each temperature of the sweep, plotted in Figure 5.3. Only the feature-distillation temperature changes; the values were not used to select $T$ (Section 5.1); the sensitivity analysis itself rests on the validation values of Table 5.5.

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